



CARIBBEAN
EXAMINATIONS
COUNCIL

Caribbean Secondary
Education Certificate®

SYLLABUS

INTEGRATED SCIENCE

CXC 23/G/SYLL 23

Effective for examinations from May–June 2027



CSEC®

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NOTE TO TEACHERS AND LEARNERS

Please note that the syllabus was revised and amendments are indicated by italics.

Issued 1983

Revised in 1993, 2000, 2009, *2015, 2021, 2023*

Amended 2025

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For access to short courses, training opportunities and teacher orientation webinars and workshops go to our Learning Institute at <https://pli.cxc.org/>

PLEASE NOTE



This icon is used throughout the syllabus to represent key features which teachers and learners may find useful.

Integrated Science Syllabus

◆ RATIONALE

The study of science provides the knowledge and skills which are intended to improve the quality of living. An understanding of science is central to young citizens' preparedness for life in modern society. It empowers individuals to engage in public discussions on issues related to science and technology; and to be critical consumers of scientific information related to everyday life. Integrated Science brings together the everyday context in which science operates. It encompasses biology, chemistry, physics, earth science, environmental science, and technology.

The **CSEC®** Integrated Science Syllabus is based on three themes, *Organisms and Life Processes, Energy, and Our Planet* which adequately reflect the common areas of human activity and experience. These themes form the unifying points of the syllabus which should therefore be seen as a coherent unit. The syllabus is redesigned with greater emphasis on the integration and application of scientific concepts and principles. *Integrated Science by the very nature of the subject facilitates* the inquiry-based approach and should be used to develop long-term transferable skills of ethical conduct, creativity, collaboration, problem-solving, critical thinking, innovation and communication. This *syllabus* also encourages the use of modern technology and *learner-centred teaching, learning and assessment* strategies to inculcate these skills that are useful in everyday life, while at the same time catering to multiple intelligences, different learning styles and needs. *Its flexible approach ensures that the diverse learning population can achieve their full potential. Additionally, the CSEC® Integrated Science Syllabus also prepares students to make relevant career choices in the pure and applied sciences and provides a foundation to build on as they pursue further studies.*

This syllabus will contribute to the development of the Ideal Caribbean Person, as articulated by the CARICOM Heads of Government, *that is*, someone who: has a positive work *ethic*; demonstrates respect for human life and awareness of the importance of living in harmony with the environment; *improves, maintains and promotes physical, mental, and social well-being and contributes to the health and welfare of the community, country and region*; demonstrates multiple literacies, independent and critical thinking; and questions the practices of past and present and brings this to bear on the innovative application of science and technology to problem solving. In keeping with the UNESCO Pillars of Learning, this course of study will also contribute to a person who will learn how to know, learn how to do, learn to live together, *learn to be*, and learn to transform themselves and society.

◆ AIMS

This syllabus aims to:

1. develop scientifically literate students who *should be able to* engage in discussions on issues related to this field;
2. *develop an understanding of living organisms and how they function;*
3. *Increase students' awareness about the factors that contribute to food production;*

4. *develop scientific knowledge, principles and practical applications for use in everyday life situations;*
5. *increase students' awareness of the importance of living in harmony with the environment including the importance and benefits of the use of sustainable energy (renewable energy, energy efficiency and energy conservation) in the preservation of the natural environment;*
6. *empower students to use scientific inquiry;*
7. *develop students' critical thinking, problem-solving and collaborative skills;*
8. *develop competencies that will enable students to make appropriate and ethical decisions regarding health, safety and other everyday life problems; and,*
9. *enable students to understand the importance of integrating information, communication and technological tools and skills in solving problems.*

◆ SUGGESTED TIMETABLE ALLOCATION

It is recommended that a minimum **of 40 hours per term or a minimum of five 40-minute periods per week over two academic years** be allocated to the study of the Integrated Science Syllabus. This should include at least one double period each week. A minimum of two periods per week should be devoted to practical activities.

CLASS SIZE

It is recommended that practical classes accommodate a maximum of **twenty-five** students.

◆ ORGANISATION OF THE SYLLABUS

The syllabus is arranged in three **MODULES**. Each module covers a distinct set of topics that are clearly defined by General Objectives, Specific Objectives, corresponding explanatory notes, Suggested Practical Activities and *Suggested Teaching and Learning Activities*. The *Specific Objectives* are indicative of the content on which the *assessment* will be based and should not be treated in isolation as they are related to the general objectives and syllabus aims.

MODULE 1 - ORGANISMS AND LIFE PROCESSES

MODULE 2 - ENERGY

MODULE 3 - OUR PLANET

◆ APPROACHES TO TEACHING THE SYLLABUS

The organisation of each *module* in the syllabus is designed to facilitate inquiry-based learning and to ensure that connections among concepts are established. *This modularised syllabus follows the complementary model design which allows modules to be taken in any order because of the horizontal*

relationships between content, making the outcome of one independent of another. Teachers are encouraged to introduce concepts familiar to the students and ensure that their lessons stimulate the use of all the senses and incorporate the use of technology where possible during the teaching and learning process. This will help students view science as a dynamic and exciting investigative process.

The general and specific objectives indicate the scope of the content including practical work that should be covered. However, unfamiliar situations may be presented as stimulus material in examination questions.

This syllabus caters to varying teaching and learning styles, with specific attention being drawn to the interrelatedness of concepts. Whenever possible, a practical approach should be employed, with special attention to the identification of variables and to the use of controls in investigations. The need for repeated investigation and observations to arrive at meaningful conclusions should be emphasised.

Greater emphasis should be placed on the integration and application of scientific concepts and principles and less on the factual materials which encourage memorisation and short-term recall. Every opportunity should be made to relate the study of scientific principles to the environment *and integrate concepts using the reference links in the syllabus. These are included throughout the document to highlight and reinforce related content.*

The role of the teacher is to facilitate students' learning of accurate and unbiased information that will contribute to a more scientifically literate society that is capable of making educated and ethical decisions regarding the world we live in.

◆ QUALIFICATION OVERVIEW

Credentialing is defined as the issuing of formal certification for successful achievement of a defined set of outcomes targeting specific knowledge, skills, and competencies. It includes the granting of an award or digital unit of measurement for learning. CXC® offers three categories of credentials across all levels of qualifications:

- 1. A micro-credential is awarded after successful completion of a module. The Caribbean Targeted Education Certificate® (CTEC®) is a stackable micro-credential that allows individuals to achieve an intermediate or macro qualification.*
- 2. An intermediate credential is awarded after successful completion of a set of defined modules that together, make up a CSEC® subject.*
- 3. A macro-credential is awarded after successful completion of a combination of intermediate (subjects) and micro-credentials (modules) that culminate in the award of credits for a certificate, diploma or associate degree.*

A total of three micro-credentials may be obtained from pursuing the discrete modules captured in CSEC® Integrated Science. Candidates who pursue the requirements of individual modules of the syllabus may qualify for micro-credentials while those who pursue all the modules in one sitting (the regular examination) may qualify for micro-credentials (if they earn Grades A-C) and the intermediate credential. The modules in this syllabus may be pursued in any order.

Module	Duration	Credit Weighting
Module 1: Organisms And Life Processes	45 hours	1
Module 2: Energy	45 hours	1
Module 3: Our Planet	45 hours	1

**The SBA is a requirement for the completion of the first micro-credential undertaken.*

◆ CERTIFICATION

The syllabus is offered for General Proficiency certification. A candidate's performance will be indicated on the certificate by an overall numerical grade on a six-point scale for the intermediate qualification as well as a letter grade for each Module.

◆ SKILLS AND ABILITIES TO BE ASSESSED

On completion of the syllabus, students are expected to develop skills under three dimensions:

1. Knowledge and Comprehension (KC);
2. Use of Knowledge (UK); and,
3. *Experimental Skills (XS).*

Knowledge and Comprehension (KC)

Remembering

The ability to:

identify, recall, state basic facts, concepts and principles; and,

Understanding

select appropriate ideas, match, and compare and cite examples of facts, concepts and principles in familiar situations.

Use of Knowledge (UK)

Applying

The ability to:

transform data accurately and appropriately; use common characteristics as a basis for classification; use formulae accurately;

Analysing and Interpreting

identify the component parts of a whole and interpret the relationships between those parts; identify causal factors and show how they interact with each other. *Use experimental data to infer, predict and draw conclusions; identify trends and patterns; make necessary and accurate calculations and recognise the limitations and assumptions of data;*

Evaluating	make reasoned judgments and recommendations based on the value of ideas and information and their implications; and,
Creating	<i>combine component parts to form a new meaningful whole; make predictions and solve problems.</i>

Experimental Skills (XS)

	The ability to:
Drawing	<i>make large, clear, labelled and/or line representations of specimens, apparatus, or models which include calculating magnification;</i>
Observation Recording and Reporting	use the senses to perceive objects and events accurately; present a written and oral report; <i>construct tables, drawings or other graphical representation which are clear, concise, accurate and pertinent to the investigation;</i> report and recheck unexpected results;
Manipulation and Measurement	set up and use carefully and competently simple laboratory apparatus and measuring instruments; appropriately prepare specimens and materials for observation/investigation; <i>and,</i>
Planning and Designing	develop hypotheses and devise means of carrying out investigations to test them; plan experimental procedures and operations within the time allotted in appropriate sequence of operations as a result of difficulties encountered in carrying out experiments or obtaining unexpected results.

Note: In addition to the *Experimental* skills, candidates are expected to utilise the skills listed under the Use of Knowledge profile dimension in their practical work.

◆ THE PRACTICAL APPROACH

The syllabus is designed to foster the use of inquiry-based learning through the application of the practical approach. Students will be guided to answer scientific questions by a process of making observations, asking questions, and doing experiments. The CSEC® Integrated Science syllabus focuses on the following skills.

1. Planning and Designing (PD)

Student's ability to:

- (a) Ask questions: how, what, which, why or where. (Students must be guided by their teachers to ask scientific questions).

Example: Will plants that are grown using organic fertilisers grow taller than those that are grown using inorganic fertilisers?

- (b) Construct a hypothesis; the hypothesis must be clear, concise, and testable.

Example: Plants grown using organic fertiliser will grow taller than those grown using inorganic fertiliser.

- (c) Design *experiments/investigations* to test the hypothesis; experimental procedure must include the following:

- (i) an appropriate aim related to the hypothesis;
- (ii) list of materials and apparatus to be used;
- (iii) observations to be made or measurements to be taken;
- (iv) precautions to be taken;
- (v) *identification of variables (controlled, manipulated, responding);*
- (vi) method of controlling variables;
- (vii) clear, concise and step by step (*sequential*) procedure *using instructional language (command verbs);*
- (viii) display of *expected* results *using appropriate method (table, prose graph, drawings);*
- (ix) *possible* use of *expected/predicted/possible* results; and,
- (x) possible limitations, *sources of errors and/or conclusions.*

- (d) *Suggest alternative methods or modification to existing methods.*

2. Measurement and Manipulation (MM)

- (a) Student's ability to handle scientific equipment competently.

The list of equipment is:

- (i) Bunsen burner;
- (ii) Tripod stand with wire gauze;
- (iii) clamp and stand/retort stand;
- (iv) binocular and monocular light microscope;
- (v) measuring cylinders (25-100cm³);
- (vi) beaker (50-500cm³);
- (vii) thermometer;
- (viii) ruler;
- (ix) stop watch/clock;
- (x) balance;
- (xi) boiling tube;
- (xii) test tubes and test tube holders;
- (xiii) *spatula*;
- (xiv) *funnel*;
- (xv) *conical flasks*;
- (xvi) hand lens/magnifying lens; and,
- (xvii) syringe.

- (b) Student's ability to take accurate measurements.

- (c) Student's ability to use appropriate units.

3. Drawing

The following guidelines should be used for drawing:

- (a) *The drawing should be placed in a position on the page which will allow for neat and clear labelling.*

- (b) *Drawings should be done in pencil. The use of coloured pencils is not recommended.*
- (c) *The drawing should be large enough so that all structures can be clearly seen.*
- (d) *The drawing should be correctly proportioned, and parts should be accurately positioned.*
- (e) *To get a smooth, unbroken line when drawing, lift the pencil from the paper as infrequently as possible until the line is completely drawn. This method will help to eliminate haphazard and sketchy lines.*
- (f) *When a large number of small structures are present in a specimen, draw only a few of them carefully, showing structural detail.*
- (g) *Write labels in pencil.*
- (h) *Labels should be annotated (that is, accompanied by brief explanatory notes).*
- (i) *Label lines should never cross each other and should be horizontal where possible.*
- (j) *In drawings where only a few structures (**less than FIVE**) are being labelled, all labels should be written on the right of the drawing.*
- (k) *Drawings must have a full title. This is usually written below the drawing and underlined. The title tells the name of the structure or organism and the view from which the drawing was made. Biological illustrations must include the magnification which is usually included in the title.*

4. Observation Reporting and Recording (ORR)

(a) Recording

Student's ability to record observations and to collect, organise and present data. Observations and data may be recorded in the following format:

(i) Prose:

Written description of observations in the correct tense.

(ii) Table (Neatly enclosed):

- Numerical: physical quantities in heading, units stated in heading, symbols, decimal points; and,
- Non-numerical: headings correct, details present.

(iii) Graph

Axes labelled, correct scales (*at least 75% of the graph paper*), correct plotting (*use of dots, circles and crosses*), smooth curves/best fit lines, key to explain symbols if more than one dependent variable is being plotted. *The graphs must be plotted using a pencil.*

(iv) *Drawing*

Candidates' ability to make large, clear, labelled and/or line representations of specimens, apparatus, or models which include calculating magnification. Candidates must adhere to the syllabus requirements for drawings.

(b) **Reporting**

Student's ability to prepare a comprehensive written report on their assignments using the following format:

- (i) **Date** (date of experiment and date of write-up).
- (ii) **Aim/Purpose** (what is the reason for doing the experiment).
- (iii) **Apparatus and Materials** (all equipment, chemicals and materials used in the experiment must be listed).
- (iv) **Method/Experimental Procedure** (logically sequenced, step-by-step procedure written in the past tense, passive voice).
- (v) **Results and Observations** (*representation of results using appropriate method [table, prose graph, drawings]*).
- (vi) **Discussion** (*presentation of background information, analysis of the results*).
- (vii) **Conclusion** (*statement of relevant inferences*).

5. Analysis and Interpretation

Student's ability to *analyse and interpret **results and observations** and make conclusions by:*

- (a) *presenting relevant background information;*
- (b) *identifying patterns and trends;*
- (c) *making accurate calculations;*
- (d) *identifying limitations and sources of error;*
- (e) *making a conclusion to either support or refute the hypothesis and/or in relation to the aim; and,*
- (f) *discussing results in relation to the background/theoretical knowledge.*

◆ FORMAT OF THE EXAMINATIONS

The examination will consist of three papers: Paper 01, an objective type paper, Paper 02, a structured type paper and Paper 03, the SBA (031) or its alternative (032). **CXC®** will offer candidates the following three options for sitting the Paper 01 and Paper 02.

Option A – Regular Sitting

For the regular sitting, candidates will be required to complete Paper 01, comprising of 60 multiple-choice questions from across all THREE modules with 20 questions on each module. Paper 02 consists of six compulsory structured type questions (two from each module) across three sections. EACH section has one question that has a practical/investigative component.

Option B – Modular Sitting (One Module)

Candidates attempting one module will be required to complete a modular Paper 01 and a modular Paper 02. Papers 01 comprises of 20 multiple-choice questions from the selected module and Paper 02 comprises of two compulsory structured type questions from the selected module.

Option C – Modular Sitting (Two Modules)

Candidates attempting two modules will be required to complete a modular Paper 01 and a modular Paper 02. Papers 01 comprises of 40 multiple-choice questions from the selected modules and Paper 02 comprises of four compulsory structured type questions, two from each of the modules selected.

All candidates will be required to attempt the SBA or the Paper 032 at the first sitting regardless of the option of sitting chosen.

Paper	Modular Requirement												
Paper 01 <i>Regular (1 hour 15 minutes)</i> Modular <i>One Module (25 minutes)</i> <i>Two Modules (50 minutes)</i>	<table><tr><th>Module</th><th>Number of Items</th><th>Marks</th></tr><tr><td>Module 1</td><td>20</td><td>20</td></tr><tr><td>Module 2</td><td>20</td><td>20</td></tr><tr><td>Module 3</td><td>20</td><td>20</td></tr></table> Each item will be allocated one mark.	Module	Number of Items	Marks	Module 1	20	20	Module 2	20	20	Module 3	20	20
Module	Number of Items	Marks											
Module 1	20	20											
Module 2	20	20											
Module 3	20	20											
Paper 02 Regular <i>(2 hours and 30 minutes)</i> Modular <i>One Module (50 minutes)</i> <i>Two Modules (1 hour 40 minutes)</i>	The paper consists of a maximum of six compulsory questions drawn from all areas of the syllabus organised under the three modules. There are two questions for each module. A question may require knowledge of several topics. However, all topics may not be given equal emphasis. <table><tr><th>Module</th><th>Number of Items</th><th>Marks</th></tr><tr><td>Module 1</td><td>2</td><td>35</td></tr><tr><td>Module 2</td><td>2</td><td>35</td></tr><tr><td>Module 3</td><td>2</td><td>35</td></tr></table> The first question in EACH module will be a practical/investigative type question worth 20 marks and the second question is worth 15 marks.	Module	Number of Items	Marks	Module 1	2	35	Module 2	2	35	Module 3	2	35
Module	Number of Items	Marks											
Module 1	2	35											
Module 2	2	35											
Module 3	2	35											

Paper 031	The School-Based Assessment will evaluate the achievement of the candidates in the Practical Skills in the laboratory and field work. Candidates will be required to keep a laboratory notebook.
Paper 032 (2 hours 10 minutes)	<i>This paper is an alternative for Paper 031, the School-Based Assessment, and is intended for private candidates. The paper consists of THREE compulsory questions.</i> The paper will examine the same skills as those assessed in Paper 031. The focus, therefore, will be on Experimental Skills and Use of Knowledge (Analysis and Interpretation). This is a practical examination. <i>Question 1 will be a practical question worth 25 marks. Questions 2 and 3 are worth 15 and 10 marks respectively.</i>

◆ ASSESSMENT GRID

The Assessment Grid shows the marks assigned to *Modules*, *Papers* and to *the Skills and Abilities to be Assessed*, and percentage contributions of each paper and *module* to the total scores. *Assessment Grid A shows the overall marks assigned when the regular sitting is chosen, and Assessment Grid B shows the marks assigned to each module.*

Assessment Grid A – Overall marks Assigned (Regular)

Papers	Module 1	Module 2	Module 3	Total Weighted marks	Total %
Paper 01	20	20	20	60	29
Paper 02	35	35	35	105	50
Paper 031 OR Paper 032	15	15	15	45	21
Total (Weighted)	70	70	70	210	100

Assessment Grid B – Marks Assigned for EACH Module

Papers	Knowledge and Comprehension (KC)	Use of knowledge (UK)	Experimental skills (XS)	Total Weighted marks	Total %
Paper 01	15	5	-	20	29
Paper 02	15	15	5	35	50
Paper 031 OR	-	5 (30 raw)	10 (70 raw)	15	21
Paper 032	-	5 (10 raw)	10 (40 raw)		
Total (Weighted)	30	25	15	70	100

◆ REGULATIONS FOR PRIVATE CANDIDATES

Private candidates must be entered for examination through the Local Registrar in their respective territories and will be required to sit Papers 01, 02, and 032.

Paper 032 is a practical examination. The Paper will be of 2 hours and 10 minutes duration and will consist of three questions. Questions will test the Experimental Skills and Use of Knowledge (Analysis and Interpretation) profiles and will incorporate written exercises and practical activities.

◆ REGULATIONS FOR RESIT CANDIDATES

Resit candidates must complete Papers 01 and 02 and Paper 03 of the examination for the year for which they reregister.

Candidates may reuse any moderated SBA score within a four-year period. In order to assist candidates in making decisions about whether or not to reuse a moderated SBA score, the Council will continue to indicate on the preliminary results if a candidate's moderated SBA score is less than 50 per cent in a particular subject.

Candidates reusing SBA scores should register as "Resit candidates" and must provide the previous candidate number when registering.

All resit candidates may enter through schools, recognised educational institutions, or the Local Registrar's Office. *A candidate who sat the linear examination may opt to resit only the Module/s in which they were unsuccessful. That is candidates may reuse their modular performance if they earned Grades A-C for any Module/s when they register to resit.*

◆ GUIDELINES FOR MODULAR CANDIDATES

Candidates opting to sit the modular examinations must complete a Modular Paper 01, a Modular Paper 02 and EITHER the Paper 031 OR Paper 032 (Private Candidates only) at their first sitting. Candidates who are using a modular approach must successfully complete all modules within a four-year period to be awarded the full CSEC® award.

Candidates using a modular approach may reuse their moderated SBA score or Paper 032 score from their first sitting within a four-year period when attempting the other modules. In order to assist candidates in making decisions about whether or not to reuse a moderated SBA score or the Paper 032 score, the Council will indicate on the preliminary results if a candidate's moderated SBA or Paper 032 score is less than 50 per cent in the particular subject.

Modular candidates may enter through schools or recognised educational institutions, or as private candidates through the Local Registrar's Office.

♦ **MODULE 1: ORGANISMS AND LIFE PROCESSES**

SKILLS AND ABILITIES: KC, UK and XS

DURATION: Minimum of 45 hours

CREDIT WEIGHTING: 1

GENERAL OBJECTIVES

On completion of this *module*, students should:

1. be aware that a *living organism* comprises of cells and particles moving *into and out of them*;
2. develop an appreciation for the interdependence of life processes;
3. *have an appreciation for the various factors that contribute to the production of food*;
4. *understand the need to control human population*;
5. understand *how humans respond* to environmental changes;
6. understand the relationship between *our sense organs* and the environment;
7. *be aware of communicable and non-communicable diseases and how they affect health*;
8. *be aware of the effect of lifestyle choices on health*;
9. *be cognisant of factors and techniques which reduce micro-organisms on the body and in food*;
10. *understand that food contaminants come from different sources; and*,
11. develop *collaborative, investigative and problem-solving skills*.

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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1. TOPIC 1: UNITS OF LIFE			
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Students should be able to:

- | | | | |
|-----|---|--|---|
| 1.1 | <i>analyse the processes of diffusion, osmosis and active transport; and,</i> | (a) <i>Definition of terms must include movement of particles and concentration gradient.</i> | <i>Conduct simple investigations to demonstrate the processes of diffusion and osmosis.</i> |
| | | (b) <i>Role of diffusion, osmosis and active transport in moving substances in and out of cells and from</i> | |

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 1: UNITS OF LIFE (cont'd)

Students should be able to:

one cell to another in all living organisms.

(c) *Reference to the cell membrane as a selectively permeable membrane.*

(d) *Examples of diffusion in artificial and natural environments to include but not limited to ash from volcanic eruptions, smog from car exhaust and industries, smoke from landfills.*

Refer to Module 1, Topic 4, SO 1 – transport systems.

Refer to Module 2, Topic 2, SO 7 – Gaseous exchange.

1.2	<i>examine animal and plant cells.</i>	(a) <i>Functions of the cell structures in animal and plant cells (Simple treatment only. For example, nucleus is responsible for cell division, making protein and contains genetic information).</i> (b) <i>Structures: cell wall, cell membrane, nucleus, cytoplasm, ribosomes, mitochondria, vacuoles, and chloroplast.</i>	<i>Examine prepared slides of cells under a microscope and make simple drawings.</i>
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Simple annotated diagrams are required (as seen under the light microscope).

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
2. TOPIC 2: REPRODUCTION AND GROWTH IN PLANTS			
Students should be able to:			
2.1 distinguish between asexual and sexual reproduction;	(a) <i>Role of cell division - meiosis and mitosis (stages are not required).</i> (b) <i>Advantages and disadvantages of asexual reproduction (variety, adaptation, livestock and crops).</i>		
2.2 examine various methods of asexual reproduction in plants;	<i>Methods of asexual reproduction:</i> (a) <i>Natural vegetative propagation (examples of perennating organs):</i> (i) <i>corms;</i> (ii) <i>bulbs;</i> (iii) <i>rhizomes; and,</i> (iv) <i>runners.</i> <i>Details of structures not required.</i> (b) <i>Brief description of artificial methods of propagation:</i> (i) <i>budding;</i> (ii) <i>cuttings;</i> (iii) <i>tissue culture;</i> (iv) <i>cloning; and,</i> (v) <i>grafting.</i> <i>Refer to Module 1, Topic 3, SO 1 – Asexual reproduction in animals.</i>	<i>Examine and draw storage organs including corms, bulbs, rhizomes, runners, and cuttings.</i>	

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 2: REPRODUCTION AND GROWTH IN PLANTS (cont'd)			
Students should be able to:			
2.3	examine the process of sexual reproduction in plants;	(a) <i>Structure and function of flowers.</i> (b) Pollination: (i) <i>definition;</i> (ii) <i>types – self and cross pollination (advantages of cross pollination); and,</i> (iii) <i>agent of pollination (wind, animal).</i> (c) <i>Fertilisation.</i> (d) <i>Development of seeds/fruits (outline).</i>	<i>Draw and label cross section of various types of flowers.</i> <i>Draw and label internal and external views of seeds and fruits.</i>
2.4	analyse growth patterns in plants;	(a) <i>Conditions necessary for the germination of seeds (temperature, oxygen and water).</i> (b) <i>Growth patterns of seeds of annual plants - bean and corn (maize). From germination to fruit formation.</i>	<i>Plot graph of plant growth at regular intervals of one week and extrapolate to predict height at future time.</i>
2.5	describe the methods used in the production of crops;	Brief description of: (a) <i>strip planting.</i> (b) <i>crop rotation.</i> (c) <i>greenhouse farming.</i> (d) <i>hydroponics.</i> (e) <i>tissue culture.</i> (f) <i>organic farming.</i> (g) <i>container gardening.</i>	

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 2: REPRODUCTION AND GROWTH IN PLANTS (cont'd)			
Students should be able to:			
2.6	<i>relate soil fertility to the physical and chemical properties of soil; and,</i> (a) <i>Characteristics of sandy, clay, loam soils. (Include soil profiles).</i> (i) <i>soil composition;</i> (ii) <i>particle size;</i> (iii) <i>air content;</i> (iv) <i>pH of soil; and,</i> (v) <i>water holding capacity (drainage and retention).</i> (b) <i>Presence of soil organisms (earthworms, nematodes).</i> (c) <i>Role of nitrifying bacteria, denitrifying bacteria, and nitrogen-fixing bacteria.</i> (d) <i>Role of Humus.</i> (e) <i>Role of decomposers.</i> (f) <i>Role of composting.</i> <i>Refer to Module 1, Topic 7, SO 1 – microbes.</i>	<i>Conduct sedimentation tests in order to make inferences about the amount of clay and sand in the soil. Draw a labelled diagram of a soil profile.</i> <i>Conduct soil tests to determine the percentage of air, pH of soils, drainage, and water retention.</i>	
2.7	<i>evaluate the impact of soil erosion on food production.</i> (a) <i>Cause and effect of soil erosion: wind, water. (Simple treatment).</i>		

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 2: REPRODUCTION AND GROWTH IN PLANTS (cont'd)

Students should be able to:

- (b) *Methods of prevention:*
 - (i) *contour farming (ploughing, planting, cultivating, harvesting);*
 - (ii) *terracing;*
 - (iii) *crop rotation;*
 - (iv) *wind breaks; and,*
 - (v) *crop cover.*
- (c) *Impact on food production.*

3. TOPIC 3: REPRODUCTION AND GROWTH IN ANIMALS

- | | | |
|-----|--|---|
| 3.1 | <i>outline various methods of asexual reproduction in animals;</i> | <i>Budding, binary fission, fragmentation, parthenogenesis.</i>

<i>Refer to Module 1, Topic 2, SO 2 – Budding in plants.</i> |
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MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 3: REPRODUCTION AND GROWTH IN ANIMALS (cont'd)

Students should be able to:

- | | | |
|-----|---|--|
| 3.2 | <i>describe the structure and function of the sexual reproductive organs in humans;</i> | <div>(a) <i>Functions of organs (structures) in the human reproductive system.</i></div> <div>(i) <i>Female</i></div> <div style="margin-left: 20px;"><ul style="list-style-type: none">- <i>Ovary;</i>- <i>Oviduct (fallopian tube);</i>- <i>Uterus;</i>- <i>Cervix; and,</i>- <i>Vagina.</i></div> <div>(ii) <i>Male</i></div> <div style="margin-left: 20px;"><ul style="list-style-type: none">- <i>Testes;</i>- <i>Scrotum;</i>- <i>Epididymis;</i>- <i>Vas deferens or sperm duct;</i>- <i>Seminal vesicle;</i>- <i>Prostate gland;</i>- <i>Cowper's gland;</i>- <i>Urethra; and,</i>- <i>Penis.</i></div> |
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MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 3: REPRODUCTION AND GROWTH IN ANIMALS (cont'd)

Students should be able to:

- 3.3 analyse the menstrual cycle in humans;
- (a) Ovulation (definition, and time of occurrence).
- (b) Roles and levels of oestrogen and progesterone (include changes in the uterine lining).
- (c) Menopause (definition, level of oestrogen).

Refer to Module 1, Topic 6, SO6 – The endocrine system.

Graphs showing changes of the uterine lining during a 28-day menstrual cycle are required.

- 3.4 discuss the stages of pregnancy;
- (a) Stages of pregnancy:
- (i) fertilisation;
- (ii) implantation; and,
- (iii) simple description of the development of the foetus and birth with a brief description of:
- contraction and dilation;
 - contraction, crowning and expulsion of foetus; and,
 - contraction and expulsion of the placenta.
- (b) Simplified diagrams to illustrate processes.

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 3: REPRODUCTION AND GROWTH IN ANIMALS (cont'd)			
Students should be able to:			
3.5	discuss the methods of birth control;	Advantages and disadvantages:	
	(a) Natural (rhythm, withdrawal, billings.		
	(b) Barrier (male and female condom, diaphragm, cervical cap, Intrauterine devices).		
	(c) Hormonal (pill, injections, patch, implants).		
	(d) Surgical (vasectomy, hysterectomy, tubal ligation).		
	(e) Mention abstinence.		
3.6	assess the importance of pre-natal and post-natal care of mothers and babies;	(a) The effects of nutrition, drugs, x-rays and diseases.	
		(b) Advantages of breast feeding, and immunisation.	
		(c) Use and importance of ultrasound.	
3.7	compare growth patterns of males and females; and,	Analysis of data of height and weight with increase in age of boys and girls.	
		Graphs are required.	
3.8	discuss the need for human population control.	(a) Problems arising from overpopulation (effects on quality of life):	
		(i) world food production; and,	
		(ii) management of natural resources and material resources.	

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 3: REPRODUCTION AND GROWTH IN ANIMALS (cont'd)

Students should be able to:

- (b) Effects of teenage pregnancy and birth control methods on population growth.

4. TOPIC 4: TRANSPORT SYSTEMS

4.1	<i>justify</i> the need for transport systems within a living organism;	(a) Surface area/volume ratio. (b) Movement of nutrients, gases, excretory products, metabolic products. (c) Role of transpiration in plants.	Conduct experiments to investigate diffusion using agar cubes of different sizes to show how surface area/volume ratio affects total diffusion. Conduct experiments to investigate factors which affect the rate of transpiration.
		Refer to Module 1, Topic 1, SO 1 – Diffusion and osmosis.	
4.2	relate the structures in transport systems to their functions; and,	(a) Composition of blood (plasma, blood proteins, blood cells, platelets). (b) Types of blood cells and their functions: (i) Red Blood Cells (mention blood doping); and, (ii) White Blood Cells (lymphocytes and phagocytes). (c) Internal structures of the heart (diagram required): (i) left and right atria; (ii) left and right ventricles; and, (iii) valves (tricuspid, bicuspid semi-lunar).	

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 4: TRANSPORT SYSTEMS (cont'd)

Students should be able to:

- (d) Heartbeat (*diastole, atrial systole, ventricular systole*).
- (e) Names of *major blood vessels (aorta, vena cava, pulmonary artery, and vein)*.
- (f) Function of *stem-xylem and phloem (simple explanation)*.

Refer to SO 1 above.

Refer to Module 1, Topic 5, SO 2 – The excretory system.

- | | | |
|-----|--|---|
| 4.3 | <i>distinguish among the different blood groups.</i> | <ul style="list-style-type: none"> (a) <i>Blood groups (A, B, AB and O).</i> (b) <i>Antigen and antibody for each group.</i> (c) <i>Precaution in transfusion and handling; include the term agglutination.</i> (d) <i>Rh factor – risk in pregnancy and precautions.</i> |
|-----|--|---|

Mention that blood type is an inheritable trait.

5. TOPIC 5: EXCRETION

- | | | |
|-----|--|--|
| 5.1 | <i>distinguish between excretion and egestion;</i> | <ul style="list-style-type: none"> (a) <i>Definition of terms excretion and egestion.</i> |
|-----|--|--|
- Refer to Module 2, Topic 2, SO 2 – Digestion in humans.*
- | | | |
|--|--|---|
| | | <ul style="list-style-type: none"> (b) <i>Difference between excretion and egestion.</i> |
|--|--|---|

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 5: EXCRETION (cont'd)			
Students should be able to:			
5.2	explain the mechanism of excretion by the lungs, skin and kidneys in humans; and, (a) Relationship to metabolism, excretory organs and products: (i) Lungs (carbon dioxide and water vapor); (ii) Skin (water and salt); and, (iii) Kidneys (water, salt, and urea). (b) Kidney – structure of tubule related to ultra-filtration and re-absorption. Cross-section and parts of the kidney. Refer to Module 1, Topic 7, SO 5 – The circulatory system. (c) Osmoregulatory function of kidneys (role of anti-diuretic hormone ADH). (d) Dialysis for malfunctioning Kidneys. (e) Label the diagram of the skin and relate the structure of skin to its functions – excretion, temperature control. Refer to Module 2, Topic 5, SO 4 – The cooling effects of evaporation. Refer to Module 2, Topic 2, SO 6 – The mechanism of breathing.	Examine the cross-section diagrams/models of the skin and kidneys of human beings. Draw, label and annotate the diagrams using annotations.	
5.3	identify the methods of excretion in flowering plants. Waste products of respiration and photosynthesis only.		

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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6. TOPIC 6: SENSE ORGANS AND COORDINATION

Students should be able to:

- | | | |
|-----|---|---|
| 6.1 | <i>describe the sense organs and their functions;</i> | <i>Stimulus associated with sense organs:</i>
(a) <i>eye (sight);</i>
(b) <i>ear (sound);</i>
(c) <i>tongue (taste);</i>
(d) <i>skin (touch); and,</i>
(e) <i>nose (smell).</i> |
| 6.2 | <i>relate the structures of the mammalian eye to their functions;</i> | (a) <i>Functions of the following Structures as it relates to sight:</i>
(i) <i>retina;</i>
(ii) <i>lens;</i>
(iii) <i>iris;</i>
(iv) <i>cornea;</i>
(v) <i>pupil;</i>
(vi) <i>choroid;</i>
(vii) <i>fovea;</i>
(viii) <i>sclera;</i>
(ix) <i>optic nerve; and,</i>
(x) <i>ciliary body (ciliary muscles and suspensory ligaments).</i> |

(Annotated diagrams are required).

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 6: SENSE ORGANS AND COORDINATION (cont'd)			
Students should be able to:			
	(b) <i>Formation of an image on the retina.</i>		
	(c) <i>Accommodation and control of amount of light entering the eye.</i>		
6.3	analyse sight defects;	(a) <i>Causes and correction/treatment for the following defects:</i>	Conduct simple investigations using convex and concave lenses.
		(i) <i>long and short-sightedness;</i>	
		(ii) <i>glaucoma;</i>	
		(iii) <i>cataracts; and,</i>	
		(iv) <i>astigmatism (mention colour blindness).</i>	
	(b) <i>Function of convex and concave lenses.</i>		
	<i>Mention the effects of bright light and physical injury.</i>		
6.4	relate the structures of the mammalian ear to their functions;	(a) <i>Functions of the following Structures of the ear:</i>	Conduct simple investigations on pitch and loudness.
		(i) <i>Pinna;</i>	
		(ii) <i>Ear canal;</i>	
		(iii) <i>Ear drum;</i>	
		(iv) <i>Ear bones (anvil, stirrup, hammer);</i>	
		(v) <i>Eustachian tube;</i>	

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 6: SENSE ORGANS AND COORDINATION (cont'd)

Students should be able to:

(vi) *Semi-circular canal;*

(vii) *Cochlea; and,*

(viii) *Auditory nerve.*

(b) *Functions: hearing and balance.*

(c) *Mention should be made of:*

(i) *the approximate audio frequency spectrum of the human ear; and,*

(ii) *the effects of loudness and pitch on human beings.*

(Annotated diagrams are required)

6.5 *relate the structures of the nervous system to their functions; and,*

Structures of the central nervous system.

(a) *Functions of the brain with specific reference to:*

(i) *cerebrum;*

(ii) *cerebellum;*

(iii) *medulla oblongata;*

(iv) *pituitary gland; and,*

(v) *hypothalamus.*

Brief description only.

Include simplified diagrams of the brain.

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 6: SENSE ORGANS AND COORDINATION (cont'd)

Students should be able to:

- (b) *Function of spinal cord.*

Brief description only.

- (c) *Function of neurones (sensory neurone, relay neurone motor neurone). Adaptations to function not required.*

- (d) *Examples of voluntary and involuntary actions.*

Mention malfunctioning of system, for example, paralysis; physical disabilities.

6.6 *relate the structures of the endocrine system to their functions.*

- (a) *Structural diagram – identifying the location of organs, hormones produced and their uses/effects.*

- (b) *Endocrine – hormones as messengers:*

(i) *thyroid (thyroxine);*

(ii) *pancreas (insulin);*

(iii) *sex organs (estrogen and testosterone);*

(iv) *adrenal glands (adrenaline); and,*

(v) *pituitary glands (Antidiuretic hormone ADH).*

Refer to Module 1, Topic 3, SO 3 – The menstrual cycle.

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
7. TOPIC 7: HEALTH			
Students should be able to:			
7.1 discuss selected microbes;	<i>Virus, bacteria, fungi.</i> <i>Positive and negative effects.</i> <i>Refer to Module 1, Topic 2, SO 6 – How organisms contribute to soil fertility; the role of decomposers in the soil.</i> <i>Refer to Module 1, Topic 7, SO 11 – Micro-organisms in food spoilage, production and processing.</i>		
7.2 discuss communicable/infectious disease;	(a) <i>Definitions of disease, and communicable disease (infectious).</i> (b) <i>The causes, signs and symptoms; and prevention, control and treatment of sexually transmitted infections to include:</i> (i) <i>Bacterial – Syphilis, Gonorrhoea, Chlamydia;</i> (ii) <i>Viral – Herpes, Hepatitis, AIDS; and,</i> (iii) <i>Fungal – Candida.</i> <i>Refer to Module 1, Topic 7, SO 1 – microbes.</i>		
7.3 outline how the principles of immunisation are used in the control of communicable/infectious diseases;	(a) <i>The immune system and how it is impacted by diseases.</i> (b) <i>Definition of immunity.</i> (c) <i>Role of white blood cells.</i>		

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 7: HEALTH (cont'd)			
Students should be able to:			
	(d) <i>Types of immunity:</i>		
	(i) <i>natural; and,</i>		
	(ii) <i>artificial (Covid, Polio vaccines).</i>		
7.4	<i>discuss non-communicable/infectious diseases;</i>	(a) <i>Definition of non-infectious disease (non-communicable).</i>	
		(b) <i>Examples of non-infectious Diseases (non-communicable diseases) to include:</i>	
		(i) <i>allergies; and,</i>	
		(ii) <i>autoimmune diseases (lupus, rheumatoid arthritis).</i>	
		(c) <i>Diabetes and Hypertension (causes and effects).</i>	
	<i>Refer to Module 3, Topic 7, SO1 – effect of air pollution.</i>		
7.5	<i>examine the physiological effects of exercise;</i>	(a) <i>Effects on circulatory and respiratory systems, lifestyle diseases, effects on balancing energy input and output.</i>	<i>Investigate the effect of exercise on the pulse rate.</i>
		(b) <i>Effects of exercise on muscle toning.</i>	

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 7: HEALTH (cont'd)			
Students should be able to:			
7.6 evaluate the effects of drug use;	(a) <i>Definition of drugs:</i> (i) <i>prescription drugs (steroids, diet pills and hormonal injections); and,</i> (ii) <i>non-prescription drugs (alcohol, illegal drugs and recreational drugs).</i> (b) <i>Effects of drugs (stimulants, depressants, hallucinogens, and narcotics) on the nervous system.</i> (c) <i>Economic and social effects of drug use and abuse.</i>		
7.7 discuss the importance of practising good personal hygiene;	(a) <i>Practices: Elimination of body odours and care of genitalia.</i> (b) <i>Benefits: Social acceptance and prevention of infections.</i>		
7.8 discuss conditions that encourage the breeding of household pests and parasites;	(a) <i>Definition of pests, parasites, pathogens and vectors.</i> (b) <i>Conditions that encourage the breeding of: Cockroaches, flies, rats, mosquitoes.</i> (i) <i>impact of improper disposal of waste (industrial and domestic); and,</i> (ii) <i>improper household hygiene (household surfaces, storage of food).</i>		

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 7: HEALTH (cont'd)			
Students should be able to:			
7.9 recommend appropriate methods of pest control;	(a) <i>Methods of pest control: Biological, chemical, sanitary and mechanical.</i> (b) <i>The relevant methods of control for each developmental stage in the life cycles of mosquitoes and houseflies.</i>		
7.10 describe various types of food contaminants;	Types of food contaminants: (a) <i>Pathogens (virus, fungus, bacteria).</i> (b) <i>Chemical contaminant.</i> (c) <i>Physical contaminants.</i>		
7.11 determine conditions which promote the growth of microorganisms; and,	(a) <i>Conditions to include:</i> (i) <i>moisture;</i> (ii) <i>optimal temperature; and,</i> (iii) <i>nutrients.</i> (b) <i>The effects of microorganisms in food spoilage, production and processing.</i> (c) <i>Procedures for retarding and preventing the growth of bread mould.</i> <i>Refer to Module 1, Topic 7, SO 1 – microbes.</i>	<i>Investigate growth of mould on bread under different conditions.</i>	

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 7: HEALTH (cont'd)

Students should be able to:

7.12	<i>apply the principles used in food preservation.</i>	(a) <i>Definition of food preservation.</i> (b) <i>Principles – altering condition to delay the growth of microorganisms or to delay microbial decomposition of the food; for example, removal of moisture, inactivating enzymes.</i> <i>Methods: salting, drying, pickling, heating, canning, curing, refrigeration, adding sugar and treating with other preservatives.</i> (i) <i>brief description of the methods; and,</i> (ii) <i>Application of principles to the methods.</i> (c) <i>Mention solar drying, energy efficient/inverter type refrigeration.</i>	<i>Investigate one of the methods for preserving food.</i>
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MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

Suggested Teaching and Learning Activities

To facilitate students' attainment of the objectives of this Syllabus, teachers are advised to engage students in the teaching and learning activities below. These activities are designed to promote inquiry-based learning and cater to students with various learning styles.

Encourage students to:

TOPIC 1: UNITS OF LIFE

1. View instructional video clips about concepts such as osmosis, diffusion, and active transport to support and reinforce learning. Have students complete worksheets requiring them to answer related questions.
2. Work in small groups to construct models, for example, of different types of cells using plasticine or other materials found around the home or laboratory. Each group should display and explain their models to the class.
3. Draw and label simple diagrams showing the structure of unspecialised plant and animal cells.
4. Draw and interpret growth curves. Students should be able to extrapolate to predict height/rate of growth. Have students discuss their predictions with a peer and then make a presentation to the class.

TOPIC 2: REPRODUCTION AND GROWTH IN PLANTS

5. Participate in field trips and work in peers or small groups to engage in at least one of the following activities and creatively report on their findings/observations either orally or in writing. Have students:
 - (a) Visit nearby farms to study common agricultural practices with emphasis on the production of crops;
 - (b) Visit organic farms to observe their operations. Have students construct a table showing how organic farming differs from conventional farming;
 - (c) Visit a central Agricultural Institution/Ministry of Agriculture and work in pairs to observe how budding, grafting, cutting and tissue culture are executed;
 - (d) Observe strip farming, greenhouse farming and hydroponics farming and record the processes involved as well as the advantages and disadvantages of each type; and,
 - (e) In situations where a physical visit cannot be facilitated, virtual simulations are encouraged.
6. Work in small groups to analyse soil profiles and determine the nature of the soil. Have students share their findings with other groups. A guided teacher led discussion should follow.

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

7. *Review and analyse results of soil tests pertaining to fertility. Have students participate in a guided teacher led discussion around their conclusions.*
8. *Construct models demonstrating terracing, contour ploughing, and planting of catch crops to prevent soil erosion. This could be done as an individual or small group activity. Have students set up a display to showcase and explain their models.*

TOPIC 3: REPRODUCTION AND GROWTH IN ANIMALS

9. *Participate in guest lectures from resource persons, for example, health professionals engaging them in interactive sessions on topics of interest such as the use of contraceptives. Have students review guided questions before the lecture and complete worksheets after the lecture.*
10. *Observe teacher demonstrations using models/charts of human reproductive systems or watch guided videos introducing the concept. Have students locate and identify structures and label the human reproductive systems using print or online interactive resources. Have them share responses with each other or be guided in discussing what was observed in the videos.*
11. *Collect data on growth in humans, draw growth curves and interpret data. For example, measuring the heights of students of different ages within their class or school and draw bar charts using the data. Have students mount their graphs in the classroom.*
12. *Work in groups to analyse graphical representations of data showing effects of overpopulation. Students should then be engaged in a guided discussion regarding their analysis.*

TOPIC 4: TRANSPORT SYSTEMS

13. *Watch videos/simulations demonstrating the structures of a mammalian heart. Have students work in small groups to draw or construct and present a model of a mammalian heart.*
14. *Watch videos introducing concepts of how the heart pumps blood to the rest of the body. Have students review guided questions before watching the videos. Students should then be guided to discuss their observation.*
15. *Observe scenarios/situations with emphasis on transport systems in plants and animals. Have students work with a peer to discuss whether the transport systems in plants and animals can be compared to the transportation of goods in everyday life. Each pair should be asked to share their points in the class discussion.*

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

16. *Observe teacher demonstrations/simulations or watch guided videos on the concepts below. Have students record and discuss their observations. The concepts include:*
- (a) *Capillary action, use of herbaceous plant in coloured water;*
 - (b) *Matching blood types using food colouring; and,*
 - (c) *Prepared slides of blood cells. Students should make simple drawings of white and red blood cells.*
17. *Work in groups to create brochures, videos, podcasts or participate in a debate identifying and explaining issues that impact quality of life, for example, Rh factor and blood transfusion. Students should showcase their products in a class session.*

TOPIC 5: EXCRETION

18. *Construct models, for example, the kidney, the lungs, and the skin. Have students display their models along with explanatory notes.*
19. *Observe teacher demonstrations using cross section diagrams/models of the skin and kidneys of human beings. Have students examine the models and label the diagrams using annotations. This can be done as a peer/share activity after which selected persons/peers can share with the class.*
20. *Participate in a field trip to a hospital to observe the use of dialysis machines, attend lectures by resource persons, observe simulations or watch guided videos explaining how a dialysis machine works. Have students discuss their observations with a peer or complete a prescribed report template.*

TOPIC 6: SENSE ORGANS AND COORDINATION

21. *Observe teacher demonstrations incorporating the use of diagrams/models of the eye, ear, brain, and neurones of human beings in their lesson. Have students complete a worksheet requiring them to label related diagrams.*
22. *Watch videos introduce, support, or reinforce concepts of how the eyes, ears and brain work. Have students creatively present to the class their understanding of how these organs work.*
23. *Observe teacher demonstrations of simple reflexes for example, knee jerk. Have students record and discuss in groups their observations.*
24. *Observe and discuss diagrams showing the positions of the endocrine glands. For example, a think-pair-share in which students are allowed to discuss with a peer and then share with the class.*
25. *Complete worksheets or quiz on the functions of endocrine glands. Have students complete this activity to support lessons as a culminating activity or as in-class assessment.*
26. *Observe presentations made by resource persons on eye defects and colour blindness. Have students complete a follow-up activity requiring them to answer questions based on their observations.*

MODULE 1: ORGANISMS AND LIFE PROCESSES (cont'd)

TOPIC 7: HEALTH

27. *Complete and creatively present the findings from a group research project on communicable diseases to the class. For example, HIV Aids, Covid-19.*
28. *Participate by observing guest lectures from health professionals on issues related to health such as immunity, vaccination, and drugs. Have students prepare and ask questions or seek clarifications on the selected topic at the end of the lecture.*
29. *Construct posters, videos, podcasts, or brochures that depict critical issues or principles covered in topics, for example, food preservation and pest control. These can be showcased as a class activity and graded.*
30. *Participate in guided activities requiring them to examine case studies, for example, on topics that relate to health and hygiene. Have students discuss their analysis in a think-pair-share-square activity and then have the reporter share with the class.*
31. *Work in small groups to investigate and creatively present findings to the class on the effectiveness of different food preservation methods.*
32. *Participate in class debates on issues that impact the quality of life such as food contamination and environmental wastes.*
33. *Complete accessible online activities to reinforce concepts taught. For example, Microorganisms activity may be accessed using this link:
https://legacy.e-bug.eu/junior_pack.aspx?cc=eng&ss=2&t=Introduction%20to%20Microbes*
34. *Review and interpret health data by investigating the number of persons in their country who suffer from diabetes/hypertension. Guided questions can be given for example, what are the causes, incidence rates and treatments available in your area? Have students complete a worksheet.*

♦ **MODULE 2: ENERGY**

SKILLS AND ABILITIES: KC, UK and XS

DURATION: Minimum of 45 hours

CREDIT WEIGHTING: 1

GENERAL OBJECTIVES

On completion of this module, students should:

1. appreciate the importance of energy in everyday life;
2. understand the methods involved in the transfer of energy;
3. appreciate the inter-conversion and conversion of mass energy;
4. *understand the role photosynthesis plays in the transfer of energy;*
5. *understand the importance of energy in nutrition and respiration;*
6. *understand the impact of fossil fuels and suitable alternatives on the environment;*
7. appreciate the importance of electrical energy in everyday life;
8. understand the occurrence of accidents, hazardous situations and safety measures used in their prevention;
9. *understand the concept of heat transfer;*
10. understand the need for appropriate physical conditions in the home and workplace; *and,*
11. develop *collaborative*, investigative and problem-solving skills.

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
1. TOPIC 1: CONSERVATION OF ENERGY			
Students should be able to:			
1.1 explain the concept of energy;	(a) Energy as an ability to produce a change/do work.		
	(b) Types and forms of energy.		
	(c) Unit of energy (Joules).		
1.2 discuss the inter-conversion and conservation of mass energy;	(a) Inter-conversion of energy for example:		
	(i) in the sun;		

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 1: CONSERVATION OF ENERGY (cont'd)

Students should be able to:

(ii) *nuclear reactors; and,*

(iii) *electrical devices.*

(b) *Methods used to save energy supply to vehicles.*

(c) *Effects of the internal combustion engine on the environment. Mention how they can be reduced.*

Refer to Module 2, Topic 1, SO 3 – Photosynthesis.

Refer to Module 2, Topic 2, SO 2 – Alternative energy sources.

Refer to Module 3, Topic 4, SO 9 – Efficiency of machines.

1.3 *examine the role of photosynthesis in the conversion of energy; and,*

(a) *Definition.*

(b) *Identification of substrate, conditions and products; word and chemical equations.*

(c) *Importance of:*

(i) *substrates (carbon dioxide and water);*

(ii) *conditions (light, chlorophyll) (photo chemical reactions); and,*

(iii) *products (oxygen, glucose).*

Conduct experiments to establish conditions for photosynthesis (light and chlorophyll), tests for starch as a product of photosynthesis.

Mention that glucose can be converted to other substances.

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 1: CONSERVATION OF ENERGY (cont'd)

Students should be able to:

- (d) Awareness that light energy can be converted to chemical energy (glucose).

Refer to Module 1, Topic 5, SO 3 – Excretion in plants.

Refer to Module 2, Topic 1, SO 2 – Inter-conversion of energy.

- | | | |
|--|---|--|
| 1.4 analyse the transfer of energy in the environment. | <p>(a) Producers, consumers (primary and secondary) decomposers, habitat, herbivores, carnivores, omnivores, population, community, ecosystem.</p> <p>(b) Food chains and food webs, trophic levels.</p> <p>(c) Ecological pyramids:</p> <p style="margin-left: 40px;">(i) pyramid of numbers (species numbers);</p> <p style="margin-left: 40px;">(ii) pyramid of biomass; and,</p> <p style="margin-left: 40px;">(iii) pyramid of energy (energy efficiency).</p> | <p><i>Observe plants and animals in a nearby area or on the school grounds and classify them as producers, consumers, decomposers, herbivores, carnivores. Construct simple food chains and food webs in terrestrial and aquatic environments.</i></p> |
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Make mention of ecological balance for environmental sustainability.

Refer to Module 3, Topic 3, SO 5 – Water pollution.

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
2. TOPIC 2: ENERGY IN LIFE PROCESSES			
Students should be able to:			
2.1 examine the importance of food as a source of energy;	(a) <i>The Caribbean food groups, their sources and functions of their main nutrients.</i> (i) <i>carbohydrates;</i> (ii) <i>proteins;</i> (iii) <i>fats;</i> (iv) <i>vitamins (A, B₁₂, C, D); and,</i> (v) <i>minerals (calcium, iron, iodine).</i> (b) <i>Importance of fibre and water to health.</i> (c) <i>Definition of a balanced diet.</i> (d) <i>Balanced diet related to age, gender, occupation/sport.</i> (e) <i>Food additives (artificial food colouring, salt, MSG, artificial sweeteners) and their effect on health.</i> (f) <i>Protein-energy malnutrition (PEM) deficiency disease, diabetes and obesity.</i> (g) <i>Importance of food as a source of energy.</i>	<i>Conduct food tests – use local foods.</i> <i>Conduct experiments to measure energy value of food.</i>	
2.2 examine the process of digestion in humans;	(a) <i>Mechanical and chemical digestion.</i>	<i>Conduct experiments to show the effects of temperature and pH on enzymes.</i>	

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 2: ENERGY IN LIFE PROCESSES (cont'd)

Students should be able to:

- (b) *Definition and role of enzymes (activation energy).*
- (c) *Role of bile.*
 - (i) *Enzymes active at different stages (salivary, amylase, pepsin, renin, pancreatic, lipase, pancreatic amylase, trypsin, maltase, lactase, sucrose, galactase), substrates and products.*
 - (ii) *Effects of temperature and pH on enzymes.*
 - (iii) *Absorption.*
 - (iv) *Assimilation.*
 - (v) *Egestion.*

Refer to Module 1, Topic 5, SO 1 – Difference between excretion and egestion.

2.3	<i>relate the structure of the teeth to their function in human digestion;</i>	(a) <i>Types of teeth and dental formula.</i> (b) <i>Structural and functional adaptation of the different types of teeth.</i> (c) <i>The role of mechanical digestion to increase surface area of food and diet.</i> (d) <i>Importance of the care of the teeth (brushing, flossing, rinsing, diet, visit to the dentist).</i>	<i>Investigate the external structure of a tooth and draw a tooth specimen.</i>
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MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 2: ENERGY IN LIFE PROCESSES (cont'd)

Students should be able to:

2.4	<i>evaluate the importance of respiration in providing energy to organisms;</i>	(a)	<i>Definition.</i>	<i>Conduct experiments to show the release of energy and carbon dioxide by organisms.</i>
			<i>Substrate and products, word and chemical equation.</i>	
		(b)	<i>Site of respiration.</i>	
		(c)	<i>Importance of energy release.</i>	
2.5	<i>distinguish between aerobic and anaerobic respiration;</i>	(d)	<i>Energy related to type of substrate.</i>	
2.6	<i>examine the mechanism of breathing;</i>	(a)	<i>Compare amounts of energy produced, products and use.</i>	
		(b)	<i>Relevance of anaerobic respiration to sports and industries (bakeries/breweries).</i>	
		(c)	<i>Inhalation and exhalation, pressure and volume changes, role of ribcage and diaphragm.</i>	
		(d)	<i>Composition of inhaled and exhaled air; structure of the lungs.</i>	
		(c)	<i>Using kinetic energy to bring air into and out of the lungs.</i>	
		(d)	<i>Stages of Cardiopulmonary Resuscitation (CPR).</i>	

Refer to Module 2, Topic 4, SO 7 – First aid methods.

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 2: ENERGY IN LIFE PROCESSES (cont'd)			
Students should be able to:			
	Refer to Module 1, Topic 5, SO 2 – The excretory system.		
	Refer to Module 3, Topic 3, SO 10 – Effects of diving.		
2.7	explain the importance of gaseous exchange in organisms; and,	(a) Features common to respiratory surfaces: alveoli, stomata, fish gills. (b) Significance of gaseous exchange.	Drawings of the structures of stomata from prepared slides.
	Refer to Module 1, Topic 1, SO 1 – Diffusion.		
2.8	explain the effects of smoking on the gaseous exchange process.	(a) Importance of smoke free environments. (b) Effect of second- hand smoke on the gaseous exchange process. (Tobacco, marijuana, vaping – electronic cigarettes).	
	Refer to Module 3, Topic 8, SO1 – effect of air pollution.		
3.	TOPIC 3: FOSSIL FUELS AND ALTERNATIVE SOURCES OF ENERGY		
3.1	examine the use of fossil fuels; and,	(a) Types of fossil fuels: coal, natural gas, crude oil (petroleum derivatives - diesel, kerosene, gasoline, fuel oil). (b) Interconversion of energy using fossil fuels (internal combustion engine, and power plant). (c) Advantages of using fossil fuels.	

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 3: FOSSIL FUELS AND ALTERNATIVE SOURCES OF ENERGY (cont'd)

Students should be able to:

- (d) *Disadvantages of using fossil fuels.*
 - (i) *fossil fuels as a non-renewable and exhaustible resource.*
 - (ii) *environmental effects (acid rain, greenhouse effect).*
 - (iii) *the impact of emissions on air quality (sulphur dioxide, carbon dioxide, methane, carbon monoxide, lead).*

Refer to Module 3, Topic 7, SO 1 – Effects of air pollution.

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|-----|---|---|
| 3.2 | <i>appraise the extent to which alternative sources of energy can be used in the Caribbean.</i> | <ul style="list-style-type: none"> (a) <i>Description of alternative sources of energy: solar, wind, wave, biofuels (biogas, biomass, biodiesel, ethanol), geothermal, hydroelectric.</i> (b) <i>Uses of solar and wind energy: to include water heating, solar cells (photovoltaic cells), solar cookers/cooking.</i> (c) <i>Sustainability of the use of alternative sources of energy: environmental, economic, social.</i> |
|-----|---|---|

Refer to Module 2, Topic 1, SO 2 – Inter-conversion of energy.

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
4. TOPIC 4: ELECTRICITY AND LIGHTING			
Students should be able to:			
4.1 examine the use of conductors of electricity;	(a) Definition of conductors (good, semi and poor). (i) Definition of insulators. (ii) Use of insulators (poor conductors such as rubber and plastics in covering electrical wires and connections). Refer to Module 3, Topic 3, SO 3 – Generation of electricity.	Investigate to determine how well materials conduct electricity.	
4.2 examine the flow of electricity in circuits;	(a) Circuits: (i) electrical components of circuits and their symbols (ammeter, cell, lamps, resistors, switch, transformer, voltmeter and fuse); (ii) difference between series and parallel circuits; and, (iii) simple circuits (must be able to draw simple circuits). (b) Relationship between voltage (V), current (I) and resistance (R): (i) Units of electricity: Ampere, Volt, Watt and Ohm.	Set up circuits to show properties; draw diagrams of series and parallel circuits.	

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 4: ELECTRICITY AND LIGHTING (cont'd)			
Students should be able to:			
	(ii) Calculation of:	<i>Use ammeters and volt-meters to show how different resistances affect current.</i>	
	- Wattage Power (P), given voltage and current, use of formula $P = IV$.	<i>Calculate wattage, given voltage and current.</i>	
	- Voltage given current and resistance, use of formula $V = IR$ (for series circuit only).		
4.3	assess the consumption of electricity in the home;	(a) Energy consumption = power x time.	
		(b) The cost of using heating appliances (clothes iron, stoves) and non-heating appliances (radio, fluorescent bulbs, fans). Unit = 1 kWh.	
		(c) Costs contributing to electricity bills (meter rentals and fuel adjustments).	
4.4	discuss the safety features of electrical devices;	(a) Electrical devices:	
		(i) fuse; and,	
		(ii) wiring plug and flex.	
		(b) Colour code in wiring plug and choice of flex.	

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 4: ELECTRICITY AND LIGHTING (cont'd)

Students should be able to:

		(c) Dangers of overloading circuit (overheating of wire that may cause insulation to burn).	
		<i>Mention the use of thick wires as overhead cables and for heavy-duty appliances.</i>	
4.5	discuss energy conservation measures;	(a) Definition of energy conservation.	
		(b) Energy wastage in faulty electrical appliances. <i>Light Emitting Diode (LED), Liquid Crystal Display (LCD), Plasma.</i>	
		(c) <i>Methods of energy conservation such as turning off lights, unplugging appliances not in use, the use of occupancy sensors.</i>	
4.6	compare artificial sources of light;	(a) Artificial sources of light:	
		(i) fluorescent tubes;	
		(ii) LED bulbs;	
		(iii) compact fluorescent bulbs; and,	
		(iv) filament lamps.	
		(b) Comparison of sources in (a) using:	
		(i) shadow formation;	
		(ii) efficiency;	
		(iii) ease of brightness control; and,	
		(iv) similarity with daylight.	

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 4: ELECTRICITY AND LIGHTING (cont'd)

Students should be able to:

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| 4.7 | <i>discuss first aid methods for treating electrical accidents;</i> | <div style="display: flex; align-items: flex-start;"><div style="margin-right: 10px;">(a)</div><div><i>Description of methods to include principles of CPR.</i></div></div> <div style="display: flex; align-items: flex-start; margin-top: 10px;"><div style="margin-right: 10px;">(b)</div><div><i>Methods in relation to:</i><div style="margin-left: 20px;"><i>(i) electrical shock; and,</i>
<i>(ii) burns.</i></div></div></div> <div style="margin-top: 20px;"><i>Refer to Module 2, Topic 2, SO 6 – Mechanism of breathing.</i></div> |
| 4.8 | <i>discuss the hazards associated with electricity;</i> | <div style="display: flex; align-items: flex-start;"><div style="margin-right: 10px;">(a)</div><div><i>Hazards associated with powerlines to include:</i><div style="margin-left: 20px;"><i>(i) Illegal connections;</i>
<i>(ii) kite flying; and,</i>
<i>(iii) picking of fruits.</i></div></div></div> <div style="display: flex; align-items: flex-start; margin-top: 10px;"><div style="margin-right: 10px;">(b)</div><div><i>Mishandling of appliances to include:</i><div style="margin-left: 20px;"><i>(i) radios;</i>
<i>(ii) television sets;</i>
<i>(iii) computers; and,</i>
<i>(iv) microwaves.</i></div></div></div> <div style="margin-top: 20px;"><i>Refer to Module 2, Topic 4, SO 9 – Methods used in extinguishing fires.</i></div> |
| 4.9 | <i>discuss the various methods of extinguishing fires; and,</i> | <div style="display: flex; align-items: flex-start;"><div style="margin-right: 10px;">(a)</div><div><i>Elements required for fires (heat, fuel, oxygen).</i></div></div> |

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 4: ELECTRICITY AND LIGHTING (cont'd)

Students should be able to:

- (b) *Types of fires:*
 - (i) *Electrical;*
 - (ii) *Chemical; and,*
 - (iii) *Bush fires.*
- (c) *Types of fire extinguishers:*
 - (i) *Water;*
 - (ii) *Foam;*
 - (iii) *Carbon dioxide; and,*
 - (iv) *Powder.*
- (d) *Methods of extinguishing fires:*
 - (i) *Removing heat;*
 - (ii) *Removing oxygen; and,*
 - (iii) *Removing fuel source.*

*Refer to Module 2, Topic 4, SO 8
– Hazards associated with electricity.*

*Refer to Module 3, Topic 6, SO 1
– Chemicals are flammable, causing fires.*

*Refer to Module 1, Topic 2, SO 7 –
Bush fires result in soil erosion.*

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| 4.10 | <p><i>explain the appropriate use of conventional protective gear/wear.</i></p> | <p><i>Including gears/wears in the home, school and workplace. Protective clothing – gloves, goggles, and helmets.</i></p> |
|------|---|--|

*Refer to Module 3, Topic 7, SO 2 –
Community hygiene.*

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
5.	TOPIC 5: TEMPERATURE CONTROL AND VENTILATION		
Students should be able to:			
5.1	<i>examine the application of the methods of heat transfer;</i> <i>Description and use of each heat transfer methods.</i> (a) <i>Conduction (Use of conductors in the home).</i> (b) <i>Convection (Mention land and sea breezes).</i> (c) <i>Radiation (Heat from the sun and fire).</i> <i>Refer to Module 2, Topic 5, SO 1– Heat transfer.</i>	<i>Perform simple experiments to show the conduction and convection.</i>	
5.2	<i>outline the use of thermostats in household appliances;</i> (a) <i>Principles by which thermostats operate.</i> (b) <i>Functions of thermostats in electrical ovens, gas ovens, and electrical irons.</i>		
5.3	<i>compare the types of thermometers;</i> (a) <i>Definition of temperature and unit of measurement.</i> (b) <i>Principles by which thermometers work.</i> (c) <i>The use and temperature range of the types of thermometers:</i> (i) <i>clinical;</i> (ii) <i>laboratory; and,</i> (iii) <i>digital.</i> (d) <i>Advantages and disadvantages of alcohol and mercury thermometers.</i>		

MODULE 2: ENERGY (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 5: TEMPERATURE CONTROL AND VENTILATION (cont'd)

Students should be able to:

- 5.4 *explain temperature regulation in humans; and,*
- (a) Sweating.
Role of latent heat of vaporisation.
 - (b) Effect of changes in body temperature on metabolic rate.

Refer to Module 1, Topic 5, SO 2 – The excretory system.

- 5.5 *explain the need for proper ventilation.*
- (a) *The importance of ventilation to living organisms.*
 - (b) *Identification of features of buildings which promote ventilation.*
 - (c) *Types of ventilation (natural and mechanical including air conditioners and fans).*

Refer to Module 3, Topic 7, SO 1 – Air pollution.

MODULE 2: ENERGY (cont'd)

Suggested Teaching and Learning Activities

To facilitate students' attainment of the objectives of this Syllabus, teachers are advised to engage students in the teaching and learning activities below. These activities are designed to promote inquiry-based learning and cater to students with various learning styles.

Encourage students to:

TOPIC 1: CONSERVATION OF ENERGY

1. Complete graded activities, for example worksheets or quizzes:
 - (a) identifying the different forms of energy in everyday life. This should include both renewable and non-renewable energy sources.
 - (b) inter-conversion of energy.
2. Carry out a lab activity to investigate the importance of the conditions of photosynthesis (light, chlorophyll, carbon dioxide, water). Have students record their observations.
3. Visit websites to watch movies, play games and do quizzes and activities on topics such as food chains and photosynthesis to support and reinforce learning. For example, <https://www.brainpop.com/science/>.
4. Construct food chain, food webs and food pyramid in a terrestrial and aquatic environment. This can be done by using a worksheet or students could also participate in a field trip to gather data on flora and fauna. Students may be placed in groups and encouraged to use their gadgets to capture images and videos and use the information to prepare a presentation to be shared in the class.
5. Work with a peer to investigate the conductivity of different materials using a simple circuit. Have students document their observations.
6. Build circuits using parts from a flashlight (or other small device) to test which materials are conductors or insulators. Have students use materials (aluminium foil, rubber, plastic, wood, steel etc.) by putting them into the circuit to close the circuit. Have them observe if the material conducts electricity based on whether the bulb in the circuit lights up.
7. Conduct investigations and interpret data of results obtained, for example, on the use of electrical energy in the home/school. This can be done as a collaborative activity. Have students discuss their findings in a guided class discussion.
8. Visit websites to watch movies, play games and do quizzes and activities on topics such as electricity and lighting. For example, <https://www.brainpop.com/science/>.
9. Collect food labels to discern nutritive content and caloric value. Have students work with a partner to discuss their labels and how they can be used to help them to make healthy choices.

MODULE 2: ENERGY (cont'd)

TOPIC 2: ENERGY IN THE LIFE PROCESSES

10. Watch videos and observe models, or posters on how the digestive system works and the role of enzymes. Have student complete a worksheet requiring them to label and state the function of each part of the digestive system. They should also identify enzymes, digestive juices, and state acidity/alkalinity at each stage.
11. Collect and interpret data and do representations [mathematical] where necessary on concepts such as enzymes.
12. Participate in planned field trips to dental schools, dental offices or have a dentist serve as a resource person to reinforce content.
13. Visit sites and/or go on field trips to a brewery or bakery, to observe the use of fermentation. Teachers could also demonstrate fermentation by preparing yeast bread in the lab and encourage students to record and discuss their observation.
14. Work collaboratively to construct models. For example:
 - (a) Model lung to observe the mechanism of breathing.
 - (b) Model of the teeth.
15. Observe the features of fish gills (specimen). Have students draw the features.
16. Work collaboratively to research the usage of Ventilator machines. Have students complete a report and present their findings.
17. Participate in an interactive virtual field trip demonstrating gaseous exchange in humans. For example, <https://encounteredu.com/teacher-resources/google-expeditions-science-11-16-lesson-2-respiratory-system-gas-exchange-in-the-alveoli>.
18. Work collaboratively to create a public service announcement encouraging teenagers to quit smoking or not to start smoking and/or the effects of second-hand smoke. They should be encouraged to creatively present information using graphics, video, or skits.
19. Observe demonstrations from resource persons on how to perform CPR. Have students work with a peer to demonstrate what was observed or use pictorial representations to explain.
20. Work collaboratively to research the effects of smoking cigarettes, marijuana and/or vaping.
21. Work collaboratively to design posters or flyers that depict the effects of fossil fuels on the environment and on the quality of life.

MODULE 2: ENERGY (cont'd)

TOPIC 3: FOSSIL FUELS AND ALTERNATIVE SOURCES OF ENERGY

22. Watch videos depicting the effects of fossil fuel use on the environment and on the quality of life.
23. Participate by going on field trips to renewable energy projects in their country such as solar farms, wind farm, hydroelectric plant, geothermal plant, biogas plant to reinforce concept of alternative energy sources.
24. Work in small groups, to investigate the suitability of different forms of renewable energy in the Caribbean. Each group should present their findings to the class.
25. Work in small groups to design scaled models of the various renewable energy sources used in the Caribbean.
26. Participate in a debate on the suitability/sustainability of alternative energy use in the Caribbean.
27. Visit websites to learn about renewable energy technologies such as biomass, geothermal, hydrogen, hydropower, ocean, solar, wind, zero-energy buildings <https://www.nrel.gov/research/learning.html>.
28. Construct a solar cooker to demonstrate how the sun's energy can be used as an alternative source.

TOPIC 4: ELECTRICITY AND LIGHTING

29. Construct simple circuits both series and parallel using easily accessible materials and determine the effect on current by adding different resistors. Have students showcase their work.
30. Work in small groups to discuss relevant first aid methods that can be used to treat electrical accidents.
31. Complete activities reinforcing content taught. For example:
 - (a) Work out size of fuses for appliances.
 - (b) Use actual measurements on energy consumption (units on meter) from different appliances.
 - (c) Read both digital and analogue meters.
32. Work collaboratively to investigate the brightness of various filament lamps of different voltage and present their findings.

MODULE 2: ENERGY (cont'd)

33. *Participate by observing presentations made by resource persons, for example, firefighting personnel to engage students in interactive sessions on topics of interest, such as hazards associated with electricity in everyday activities, methods of extinguishing fires, protective gears to be used at home, school and workplace when handling electrical issues and safety features of electrical devices.*
34. *Participate by attending a field trip to a Fire Brigade station where they will observe or be engaged in activities on how to extinguish fires based on its origin (electrical, chemical).*
35. *Observe guest lectures by resource persons (nurse or doctor). Students could also visit the burn unit at a hospital where they would observe the necessary steps to treat electrical burns and shock. The Red Cross can also be invited where students are trained and certified in first aid.*

TOPIC 5: TEMPERATURE CONTROL AND VENTILLATION

36. *Observe, record and discuss demonstrations or simulations:*
 - (a) *of conduction and convection; and,*
 - (b) *to illustrate the principle of using a bimetallic strip. For example, <https://www.youtube.com/watch?v=9AWKkTPqrJE>.*
37. *Participate in activities that will show them how to use thermometers properly. Students should be supervised while using the apparatus. Have students reflect on the activity.*
38. *Watch instructional videos and/or observe a model of the skin to reinforce the concept of temperature regulation. Have students write a poem on the importance of temperature regulation.*
39. *Conduct an investigation on the effects of wind, temperature, and humidity of the area on evaporation and drying of materials. Have students work collaboratively to creatively present their findings.*
40. *Participate in a discussion on features of buildings which promote ventilation. Have students explain how this is applicable within their own context (home/school).*
41. *Participate in guest lectures by resource persons, for example, contractors or engineers to engage them in interactive sessions on topics of interest such as the importance of ventilation features in the homes and school environment.*

◆ **MODULE 3: OUR PLANET**

SKILLS AND ABILITIES: KC, UK and XS

DURATION: Minimum of 45 hours

CREDIT WEIGHTING: 1

GENERAL OBJECTIVES

On completion of this module, students should:

1. *understand the nature of the universe;*
2. *appreciate the solar system and the motions within it;*
3. *understand the rationale for space exploration;*
4. *explore the factors that contribute to weather systems;*
5. *understand the nature of forces;*
6. *explore the factors that lead to natural hazards;*
7. *appreciate the value of water and its importance to aquatic eco systems;*
8. *develop an appreciation of the nature of matter;*
9. *explore the uses and effects of chemicals on the environment;*
10. *appreciate the importance of proper sanitation; and,*
11. *develop collaborative, investigative and problem-solving skills.*

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
1. TOPIC 1: THE UNIVERSE AND OUR SOLAR SYSTEM			
Students should be able to:			
1.1 <i>identify the components of the universe;</i>	(a) <i>Galaxies to include the Milky way galaxy.</i> (b) <i>Solar system.</i> (c) <i>Asteroids.</i> (d) <i>Planets/Planetoids.</i> (e) <i>Meteorites/Meteoroids.</i> (f) <i>Comets.</i> (g) <i>Stars.</i>		

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 1: THE UNIVERSE AND OUR SOLAR SYSTEM (cont'd)			
Students should be able to:			
1.2	<i>explain how bodies stay in orbit;</i>	(a) <i>Gravitational pull.</i> (b) <i>Satellites (natural and artificial – geo stationary orbit).</i> <i>Refer to Module 3, Topic 4, SO 2 – Centripetal force.</i>	
1.3	<i>describe the solar system;</i>	(a) <i>Planets, elliptical shape, number of moons; size of planets, ecliptic orbits.</i> (b) <i>The sun, in relation to the planets (distance of planets from the sun).</i> (c) <i>Location of the earth in the solar system.</i>	
1.4	<i>discuss how earth is affected by other bodies; and,</i>	(a) <i>Day and night.</i> (b) <i>Concept of shadow formation.</i> <i>Refer to Module 2, Topic 4, SO 6.</i> (c) <i>Description of eclipses – Solar and Lunar.</i> (d) <i>Phases of the moon – new moon, first quarter, full moon, last quarter.</i> <i>Refer to Module 3, Topic 2, SO 3 – Tides.</i>	
1.5	<i>discuss human's exploration of the universe.</i>	<i>Including:</i> (a) <i>Reasons for exploration of the universe.</i>	

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 1: THE UNIVERSE AND OUR SOLAR SYSTEM (cont'd)

Students should be able to:

- (b) *Characteristics of outer space temperature, radiation, lack of oxygen and pressure, lack of gravity.*
- (c) *International space station.*
- (d) *James Webb and Hubble telescope.*
- (e) *Types of exploration of other planets (Mars, Jupiter; flyby, orbiter, rover (lander), humans).*

2. TOPIC 2: THE TERRESTRIAL ENVIRONMENT

- 2.1 *examine the effects of air masses in the Caribbean;*
- (a) *Types of air masses (maritime tropical, continental tropical, maritime polar, continental polar).*
 - (b) *Air masses affecting the Caribbean - the spread of Pollutants (radioactive fallout, volcanic dust, industrial waste, Sahara dust, landfill fumes).*
- Refer to Module 3, Topic 7, SO 1 – Air pollution.*
- (c) *Types of fronts (to include definition) and their effect on the weather (cold, warm, stationary, occluded).*

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 2: THE TERRESTRIAL ENVIRONMENT (cont'd)			
Students should be able to:			
2.2	examine the weather patterns in the Caribbean;	(a) Weather patterns (wet and dry seasons). (b) Definition of a cyclone – low pressure, wind speed, meeting of air masses. (c) Development of tropical depressions, tropical storms and hurricanes.	
2.3	examine the effects of tides; and,	(a) The types of tides - high, low, spring, neap tides. (b) The formation of tides. (c) The effects of tides (coastal erosion). Refer to Module 3, Topic 1, SO 4 - Eclipses. (d) Brief description of tidal waves. (e) Description and causes of a tsunami - underwater landslides, volcanic eruptions and earthquakes.	
2.4	explain the causes of the different types of volcanic eruptions.	(a) The types of volcanoes: (i) cinder cone or spatter cone; (ii) shield; and, (iii) composite or strato-volcano.	

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 2: THE TERRESTRIAL ENVIRONMENT (cont'd)			
Students should be able to:			
	(b) The ecological consequences of volcanoes in the long and short- term. Include Kick-em-Jenny underwater volcano off the coast of Grenada, <i>La Soufriere in St Vincent</i> .		
	(c) <i>The relationship between earthquakes and volcanoes.</i>		
	(d) <i>The function of the seismograph. The Richter scale. Significance of the numbers on the Richter scale.</i>		
3. TOPIC 3: WATER AND THE AQUATIC ENVIRONMENT			
3.1	discuss the properties of water;	(a) <i>Chemical properties of sea and fresh water.</i>	
		(b) <i>Physical properties of sea and fresh water.</i>	
		(c) Effects of sea and fresh water on aquatic life.	
3.2	distinguish between hard and soft water;	(a) <i>Characteristics of hard and soft water.</i>	Conduct experiments to determine degree of hardness of water.
		(b) <i>Advantages and disadvantages of hard and soft water.</i>	Conduct experiments to soften samples of hard water by:
			(i) <i>boiling;</i>
			(ii) <i>adding washing soda; and,</i>
			(iii) <i>distillation.</i>

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 3: WATER AND THE AQUATIC ENVIRONMENT (cont'd)

Students should be able to:

3.3	explain the uses of water;	(a) Role in life processes - <i>excretion, digestion, transport of substances, respiration.</i> (b) Uses in home - <i>drinking, cooking, washing, cleaning</i> (consider wastage and conservation). (c) <i>Agriculture including hydroponics, aquaculture, mariculture.</i> (d) <i>Recreational activities.</i> (e) <i>Firefighting.</i> (f) <i>Generation of electricity.</i>	<i>Perform calculations to determine water consumption over a 30-day period.</i>
3.4	describe the various methods used locally for fishing;	<i>Brief description of methods, including by hand; spears/harpoons; netting (trawling, long-lining); pots or traps; fish farming.</i>	
3.5	evaluate the effects of water pollution on aquatic life;	(a) Sources of pollution, for example, nitrates, phosphates, various pesticides, oil spills. (b) <i>Effects of pollution on aquatic life to include but not limited to eutrophication, fish kill, destruction of the coral reef, mangroves and wetlands.</i>	<i>Investigate the effects of the removal of oxygen from water.</i>

*Refer to Module 2, Topic 1, SO 4
Energy transfer.*

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 3: WATER AND THE AQUATIC ENVIRONMENT (cont'd)

Students should be able to:

3.6	<i>investigate methods of purifying water;</i>	(a) Sources of water. (b) <i>Methods of purifying water: the treatment of seawater for domestic use (desalination), boiling, filtration, chlorination, distillation and additives of alum and carbon.</i>	Conduct experiments to purify water.
		<i>Refer to Module 3, Topic 6, SO 5 – Separation techniques.</i>	
3.7	<i>determine conditions for flotation;</i>	(a) <i>Upthrust and density.</i> (b) Archimedes principle. (c) The Plimsoll line on boats and ships.	Conduct experiments to compare sinking and floating of similar materials in fresh and sea water. Simple measurements of densities.
		<i>Refer to Module 3, Topic 4, SO 3 – Maximum loading capacity.</i>	
3.8	<i>explain importance of navigational devices used at sea;</i>	(a) Compass, sonar, radar, GPS as devices. (b) <i>How the magnetic compass works.</i> (c) <i>Maritime safety standards.</i>	
3.9	identify water safety devices; and,	Life rafts and jackets, inflatable tubes.	

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 3: WATER AND THE AQUATIC ENVIRONMENT (cont'd)

Students should be able to:

- 3.10 discuss the *effects of diving on the human body.* *Effects of Scuba-diving and free-diving:*

- (a) Respiratory problems: damage to membrane due to high pressure.
- (b) *Decompression sickness:* The bends, nitrogen narcosis, *use of decompression chambers.*
- (c) *Embolism.*
- (d) *Baro trauma (ears, lungs).*

Refer to Module 2, Topic 2, SO 6 – Mechanism of breathing.

4. TOPIC 4: FORCES

- 4.1 investigate the principles of forces;
- (a) *Definition of force*
Use of formula $F = m \times a$. *Demonstrate friction on different surfaces using a ball.*
 - (b) *Types of forces – push and pull.*
 - (c) *Newton's third law – Action-reaction principle applied in space transport.*
 - (d) *Forces in air – the forward motion of jet aircrafts. Relationship between shape of wings of planes and birds and lift forces they experience while moving through air; the effects of wind speed and wind currents on the motion of aircraft.*
 - (e) *Friction – motion of vehicles, road surfaces and tyres.*

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 4: FORCES (cont'd)			
Students should be able to:			
4.2 describe gravity as a force;	(a) Definition – <i>gravitational force, and non-contact force.</i> (b) <i>Newton's first law – Centrifugal forces, centripetal forces (satellites).</i> <i>Refer to Module 3, Topic 4, SO 2 – Gravity.</i>		
4.3 determine the centre of gravity;	(a) <i>Definition of the centre of gravity.</i> (b) <i>The relationship between height of the centre of gravity of an object and its stability; the implications for stability on the loading of vehicles in relation to their centre of gravity; reasons for maximum loading capacity and tare.</i> <i>Refer to Module 3, Topic 3, SO 7 – Plimsoll line on boats.</i>	Use cardboard cutouts of triangles, rectangles, circles and irregular shapes to arrive at the approximate position of the centre of gravity of objects of different shapes.	
4.4 investigate equilibrium;	(a) <i>Types of equilibrium – stable, unstable, neutral.</i> (b) <i>Application of equilibrium in medicine (chemical balance in the body), chemistry (reversible chemical reaction), biology (homeostasis).</i> (c) <i>Moments of a force about a point is the product of the force and the perpendicular distance of its line of action from a point.</i> <i>Use of formula $M = F \times d$.</i> (d) <i>Conditions for equilibrium under parallel forces.</i>	<i>Investigate using rule suspended by a spring balance and kept horizontal by known suspended weights to show that:</i> (i) the sum of the forces in one direction must equal the sum in the opposite direction; and, (ii) the sum of the clockwise moments about a pivot must equal the sum of anti-clockwise moments.	

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 4: FORCES (cont'd)			
Students should be able to:			
4.5	examine the principle of momentum conservation;	(a) <i>Definition of momentum.</i> (b) <i>The law of conservation of momentum. Total momentum before the collision is equal to the total momentum after the collision.</i> (c) <i>Consider conservation of linear momentum (refer to vehicular collision, rocket engine, playing pool/billiards).</i> <i>Use of formula ($p = m \times v$) to do simple calculations.</i>	
4.6	explain the functions of simple machines;	(a) <i>Functions of simple machines:</i> (i) <i>they make work easier; and,</i> (ii) <i>they are used as force multipliers for convenience of application of a force.</i> (b) <i>Types of simple machines:</i> (i) <i>levers:</i> - <i>classes of levers (organising load, effort and fulcrum in three different ways);</i> - <i>compare the different classes of levers; and,</i> - <i>simple levers in the mammalian skeleton.</i>	

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 4: FORCES (cont'd)			
<i>Students should be able to:</i>			
	(ii) pulleys; and, (iii) inclined planes.		
	(c) Simple machines used in or associated with vehicles, for example, motor cars, push carts, draglines, bicycles.		
	<i>Refer to Module 3, Topic 4, SO 8 – Movement of the limbs in the skeleton.</i>		
4.7	relate the structure of the skeleton to its functions;	(a) Major bones of the human skeleton: Cranium, clavicle, scapula, vertebral column (cervical, thoracic, lumbar), humerus, radius, ulna, rib cage, sternum, pelvic girdle, femur, tibia, fibula. (b) Functions: Movement, protection, support, breathing, protection of blood vessels.	
4.8	explain how the skeletal muscles function in the movement of the limb; and,	(a) Joints in the human body: Hinge joint, fixed joint, ball, socket and gliding. (b) Role of antagonistic muscles.	
	<i>Refer to Module 3, Topic 4, SO 6 – Simple levers in the mammalian skeleton.</i>		
4.9	examine the efficiency of simple machines.	(a) The principle of energy efficiency and energy conversion. <i>Law of conservation of energy (energy is neither created nor destroyed it can be converted from one form to another).</i>	

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 4: FORCES (cont'd)

Students should be able to:

Use of the equation: work done = force x distance moved in the direction of the force.

- (b) *The principle of mechanical advantage (ratio of the load to the effort applied).*

Use of the equation: mechanical advantage = load ÷ effort.

- (c) *The factors that contribute to the inefficiencies of machines and ways of overcoming their influences. The motorcar, lawnmower, bicycle; factors such as rusting, corrosion and friction.*

Refer to Module 2, Topic 1, SO 2 – Inter-conversion of energy.

Refer to Module 3, Topic 5, SO 6 – Tarnishing of metals.

5. TOPIC 5: METALS AND NON-METALS

5.1 relate the physical properties of metals and non-metals to their uses;

- (a) Metals.
- (b) *Nonmetals to include plastics, wood, ceramics and textiles (natural and man-made).*
- (c) *Properties such as electrical conductivity, thermal conductivity, melting point, density, tensile strength.*
- (d) *Appropriateness of the materials used in sports/sporting equipment.*

Measuring the elasticity of specific substances.

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 5: METALS AND NON-METALS (cont'd)			
Students should be able to:			
5.2	<i>compare the reactivity of metals;</i>	Aluminium (Al); Copper (Cu); Iron (Fe); Tin (Sn); Silver (Ag); Zinc (Zn). <i>Refer to Module 3, Topic 6, SO 6 – Household appliances.</i>	Observe which metals react <i>with dilute acid</i> and which do not; Write simple word equations to show their reaction.
5.3	discuss the advantages and disadvantages of using cooking or canning utensils made of aluminium;	(a) <i>Advantages.</i> (b) <i>Disadvantages</i> (consideration of toxicity and corrosion).	
5.4	discuss the benefits of using alloys to make household items;	(a) <i>Alloys</i> and examples of alloys commonly found in the home and workplace - steel, brass, soft solder and <i>electroplating</i> . (b) <i>Benefits of using Alloys.</i>	
5.5	<i>examine the conditions which cause rusting; and,</i>	(a) <i>Factors which affect the rate of rusting:</i> (i) <i>sea spray;</i> (ii) <i>emission from industrial plants; and,</i> (iii) <i>climate (tropical).</i> (b) Tarnishing as a chemical process (oxidative). <i>Rusting in relation to iron and steel.</i> (c) <i>Word equation for rusting:</i> <i>Iron + Oxygen + Water → rust (Iron Oxide).</i>	<i>Investigate the conditions needed for rusting.</i>

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 5: METALS AND NON-METALS (cont'd)

Students should be able to:

- 5.6 discuss the methods used to reduce or prevent tarnishing of metals. *The scientific principles of the following methods used to prevent tarnishing for commercial and household purposes:*

- (a) *Painting.*
- (b) *Covering with oil or plastic.*
- (c) *Proper storage (moisture free).*
- (d) *Use of drying agents (desiccants).*
- (e) *Electroplating.*
- (f) *Galvanizing.*
- (g) *Alloy (steel).*

Refer to Module 3, Topic 6, SO 9 – Efficiency of machines.

6. TOPIC 6: HOUSEHOLD CHEMICALS

- 6.1 discuss the uses of common household chemicals;
- (a) *Water as the most common chemical in the home; water as a solvent used in many household chemicals.*
 - (b) *Household chemicals and their uses (chemical and household names and eco-friendly products):*
 - (i) *hard surface cleaners (disinfectant, ammonia, bleach, degreaser);*

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 6: HOUSEHOLD CHEMICALS (cont'd)

Students should be able to:

- (ii) *personal hygiene (bath soap, body wash, deodorant, shampoos, antiperspirants, conditioners, toothpaste, mouthwash);*
 - (iii) *laundry (soap, detergents, fabric softeners);*
 - (iv) *kitchen (dishwashing liquid, oven cleaner, salt, vinegar, baking powder, baking soda); and,*
 - (v) *health (antacid, antiseptic, pain killers, hand sanitizers, alcohol).*
- (c) *Safety symbols (alert to potential hazards) - corrosive, toxic, flammable, explosive, oxidizing, harmful or irritant, radioactive.*
- Refer to Module 2, Topic 4, SO 9 - Methods used in extinguishing fires.*
- (d) *The safe use of household chemicals.*
- (e) *The economic use of household chemicals (buying in bulk, using recommended quantities, long shelf life).*

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 6: HOUSEHOLD CHEMICALS (cont'd)			
Students should be able to:			
6.2	<i>examine the properties of acids, bases and salts;</i>	(a) The concept of pH <i>as it relates to the properties of acids, bases and salts.</i> (b) Classification of household chemicals into acids, bases and salts. (c) <i>Principle of neutralisation to include but not limited to:</i> (i) <i>agriculture;</i> (ii) <i>health;</i> (iii) <i>nutrition;</i> (iv) <i>cooking; and,</i> (v) <i>sanitation.</i>	Conduct simple investigations to determine the pH values of various household chemicals to determine whether they are acids, alkaline or neutral. Use pH paper. Conduct simple experiments on neutralisation using droppers.
6.3	<i>differentiate among the properties of the states of matter;</i>	(a) <i>Characteristics of solid liquid and gas:</i> (i) <i>arrangement of particles;</i> (ii) <i>shape and volume;</i> (iii) <i>forces of attraction; and,</i> (iv) <i>movement of particles.</i> (b) <i>Changes of state (processes to include deposition, freezing, condensation, sublimation, evaporation, and melting).</i>	Conduct simple experiments to demonstrate the changes of state of matter (solid liquid and gas).

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 6: HOUSEHOLD CHEMICALS (cont'd)

Students should be able to:

		(c) <i>Plasma (simple definition and example) of plasma (the fourth state of matter).</i>	
6.4	<i>examine the properties of mixtures;</i>	Mixtures: solutions, suspensions and colloids. (a) <i>Definition of the terms solutions, solute and solvent.</i> (b) <i>Differences between a solution, a suspension and a colloid.</i> (c) Classification of household chemicals in each category.	
6.5	<i>determine appropriate separation techniques;</i>	(a) <i>Difference between aqueous and non-aqueous solutions.</i> (b) <i>Application of separation methods to home and workplace to include but not limited to:</i> (i) <i>distillation (wine making);</i> (ii) <i>filtration (cooking);</i> (iii) <i>chromatography (bleeding colours in laundry);</i> (iv) <i>crystallization (honey to form sugar crystals, production of sugar from sugar cane); and,</i>	

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
TOPIC 6: HOUSEHOLD CHEMICALS (cont'd)			
Students should be able to:			
	(v) <i>evaporation (preparation of table salt).</i>		
	(c) <i>Desalination plants (distillation and reverse osmosis).</i>	<i>Plan and design experiments on stain removal - turpentine for paint; methylated spirit for glass; acetone for nail polish.</i>	
	<i>Refer to Module 3, Topic 3, SO 6 – Water purification.</i>	<i>Conduct simple experiment to separate salt from a mixture of salt and sand.</i>	
6.6	explain the effects of cleaning agents on household appliances; and,	(a) <i>Cleaning agents (scouring powders, detergents, rust removers, lime scale removers, oxidizing agents).</i>	
		(b) <i>Different reactions of cleaning agents: abrasive and chemical action.</i>	
		(c) <i>Effects on appliances made from Aluminum (Al), Copper (Cu), Iron (Fe), Tin (Sn), Silver (Ag), Zinc (Zn).</i>	
	<i>Refer to Module 3, Topic 5, SO 2 - Metals.</i>		
6.7	distinguish between soapy (soap) and soapless detergents.	(a) <i>Methods of production (soap – animal fats and oils; soapless – petroleum products).</i>	
		(b) <i>Scum formation.</i>	
		(c) <i>Advantages and disadvantages:</i>	
		(i) <i>biodegradable and non-biodegradable;</i>	

MODULE 3: OUR RUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 6: HOUSEHOLD CHEMICALS (cont'd)

Students should be able to:

- (ii) *pollution;*
- (iii) *irritants and non-irritants; and,*
- (iv) *production cost.*

7. TOPIC 7: POLLUTANTS AND ENVIRONMENT

- 7.1 *discuss the effects of air pollution;* *Effects:*
- (a) *Health: Allergies, lung cancer, asthma, other respiratory disorders, emphysema.*
 - (b) *Environmental: Acid rain, blocking stomata in plants.*
- Refer to Module 1, Topic 7, SO 4 - allergies.*
- Refer to Module 2, Topic 2, SO 8 – second-hand smoke.*
- Refer to Module 2, Topic 5, SO 5 - Ventilation.*
- Refer Module 2, Topic 3, SO 1 – Fossil fuels.*
- Refer to Module 3, Topic 2, SO 1 – Air masses.*
- 7.2 *justify the importance of community hygiene to the environment; and,*
- (a) *Types of waste: Domestic, industrial, biological, chemical and electronic waste. Bio-degradable and non-bio-degradable waste.*
 - (b) *Impact of improper waste disposal:*
 - (i) *Pollution of land and water (surface, ground and potable water);*

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 7: POLLUTANTS AND ENVIRONMENT (cont'd)

Students should be able to:

- | | | |
|-----|------------------------------|---|
| | | <ul style="list-style-type: none"> (ii) increase in pest population; (iii) prediction of their consequences; and, (iv) assessment of the effects of unsanitary conditions on the spread of pathogenic microorganisms and parasites such as worms. |
| | | <p>(c) <i>Recommended Practices: Proper disposal of solid waste (refuse, reduce, reuse, repair, refill, repurpose, recycle), composting, biogas production, adequate toilet and sewage disposal facilities (septic tanks, cesspools soak aways and sewage treatment plants), garbage collection and disposal.</i></p> |
| | | <p>(d) <i>Benefits: Prevention of infections, aesthetic value.</i></p> |
| 7.3 | discuss the use of plastics. | <p>(a) <i>Advantages of using plastics:</i></p> <ul style="list-style-type: none"> (i) <i>ease of production (manufacture and costs);</i> (ii) <i>durability; and,</i> (iii) <i>use of plastics in industry (medical, construction).</i> |

MODULE 3: OUR PLANET (cont'd)

SPECIFIC OBJECTIVES	EXPLANATORY NOTES	SUGGESTED ACTIVITIES	PRACTICAL
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TOPIC 7: POLLUTANTS AND ENVIRONMENT (*cont'd*)

Students should be able to:

- (b) *Negative effects on the environment:*
 - (i) *burning of plastics – release of toxic gases (dioxins) and respiratory illnesses;*
 - (ii) *length of time taken to degrade; and,*
 - (iii) *microplastics in the marine environment.*
- (c) *Benefits of recycling.*

Suggested Teaching and Learning Activities

To facilitate students' attainment of the objectives of this Syllabus, teachers are advised to engage students in the teaching and learning activities below. These activities are designed to promote inquiry-based learning and cater to students with various learning styles.

Encourage students to:

TOPIC 1: THE UNIVERSE AND OUR SOLAR SYSTEM

1. *Watch videos to reinforce the concepts of the solar system and the bodies within.*
2. *Work in small groups to construct and display models:*
 - (a) *of the solar system using recyclable materials; and,*
 - (b) *to demonstrate the effects of the moon on the earth, for example, day and night and types of eclipses.*
3. *Conduct simple lab activities using light sources and screens to show shadow formation. Have students record their observation.*
4. *Observe the moon over a 30-day period noting changes in shape and percentage of light emitted. Engage students in a guided discussion on their findings.*

MODULE 3: OUR PLANET (cont'd)

5. Visit the NASA website and view photographs of outer space from the James Webb and Hubble space telescope. They should also be encouraged to access virtual tours.
6. Participate as a member of a group to debate issues that impact human life. For example, effects of space travel on astronauts.

TOPIC 2: THE TERRESTRIAL ENVIRONMENT

7. Watch videos to reinforce concepts on various topics, for example, cycles in nature, tidal waves, volcanic eruptions and climate change. Have students complete a worksheet at the end of the video presentations.
8. Use a website or weather app to observe and track weather systems across the Caribbean. Have students document and discuss their observations.
9. Attend field trips, where possible, to meteorological/weather stations to observe equipment used to monitor weather systems.
10. Work in small groups to conduct a study of plant and animal life on seashores or riverbanks with respect to tidal patterns. Have students creatively present their findings to the class.
11. Work in small groups to construct and present working models to show volcanic eruptions.
12. Work in small groups to research and demonstrate evacuation and safety procedures during an earthquake.

TOPIC 3: WATER AND THE AQUATIC ENVIRONMENT

13. Use boiling and melting points to compare the physical properties of salt water and fresh water.
14. Watch videos to reinforce the concepts about the properties of water, fishing, water safety and floatation. Have students complete a worksheet or engage a peer in a guided discussion on their observations.
15. Attend site visits and/or field trips to seashores and riverbanks to observe life in the aquatic environment and water treatment plants to observe processes involved in purifying water. Have students document their observations.
16. Participate in guest lectures from resource personnel on fishing and/or water pollution to reinforce learning.
17. Attend a field trip to a fisheries complex or fish farm to observe types of fishing gears, navigation devices and boats and engage fisheries officers, and fishermen in discourse on their observations.

MODULE 3: OUR PLANET (cont'd)

18. *Use simple apparatus to illustrate the concept floating for example, density blocks.*
19. *Work in small groups to construct a working model of a compass. Have students present their models.*
20. *Compare water usage of their school during the term and during vacation using water bills. They should then calculate water consumption over a 30-day period. From their observations, students are encouraged to propose ways in which water can be conserved.*
21. *Participate in lectures by resource persons to discuss effects of diving. In cases where this is not possible, students should be encouraged to conduct research on the effects of diving.*

TOPIC 4: FORCES

22. *Watch videos to reinforce concepts such as force, gravity, equilibrium, momentum, and machines. Have students complete a worksheet or discuss with a peer their observations.*
23. *Observe teacher led demonstrations on forces. Students should then discuss their observations with classmates. For example,*
 - (a) *Action-reaction principle, for example, releasing an inflated balloon, using a pair of spring balances, balloon rocket.*
 - (b) *Demonstrate by blowing over strips of paper held at one end.*
 - (c) *Demonstrate using paper aircraft models and fan.*
24. *Conduct investigations. For example:*
 - (a) *Dropping a heavy and a light book from the same height at the same time and observe if they land at the same time or not. Discussion should ensue about the leaning tower of Pisa; and,*
 - (b) *Determine if it is easier to balance a yardstick on your finger than a pencil or ruler.*
25. *Observe teacher demonstrations on the concept of gravity by:*
 - (a) *Throwing a ball up and observing its motion.*
 - (b) *Releasing objects attached to suspended spring/ not attached to anything.*
 - (c) *Using models to demonstrate how an object can escape the pull of gravity if given enough kinetic energy by whirling around the head a rubber band attached to a weak thread.*
26. *Locate the centre of gravity of items such as pencils, rulers and solids with regular shapes. Have students reflect on the activity.*

MODULE 3: OUR PLANET (cont'd)

27. *Observe demonstrations of simple machines (hammer, bottle opener, crowbar, scissors, nutcracker, wheelbarrow, fishing rod, tweezers as levers; pulleys, wheels, hydraulic press, screw) in use to reinforce theoretical concepts. Have students work with a partner to discuss their observations.*
28. *Complete worksheets requiring them to perform calculations on mechanical advantage with respect to simple machines.*
29. *Observe teacher demonstrations on:*
 - (a) *The momentum conservation principle using simple qualitative trolley experiments, or they may incorporate the use of worksheets for students to complete.*
 - (b) *The three types of equilibrium using a small ball, a concave/ convex dish, or a cone shaped object and a flat surface.*

TOPIC 5: METALS AND NON-METALS

30. *Watch videos to reinforce concepts such as properties of metals and nonmetals and alloys. For example:*
<https://www.youtube.com/watch?v=AJbe5THaNUU>
<https://www.youtube.com/watch?v=PHu8hQBtPcY>
<https://www.youtube.com/watch?v=Rc2JBp91V7o>
<https://www.youtube.com/watch?v=TGPPPFczOj0&t=15s>
<https://www.youtube.com/watch?v=KqUmNQD6m5Q>
31. *Visit websites and use online resources to reinforce learning on the properties of metals and nonmetals. For example:*
<https://www.nagwa.com/en/lessons/280172486318/>
<https://keslerscience.com/metals-nonmetals-and-metalloids-lesson-plan-a-complete-science-lesson-using-the-5e-method-of-instruction/>
32. *Conduct research on the advantages and disadvantages of using aluminium and alloys in household items. They should work in small groups to present their findings in a creative way. For example, using flyers, brochures, or presentations.*
33. *Watch videos to reinforce the concept of rusting for example,*
https://www.youtube.com/watch?v=jQoE_9x37mQ
34. *Watch teacher guided videos and simulations explaining the application of electrolysis in electroplating, galvanizing, and painting. Have students record and discuss their observations.*
35. *View pictures and/or samples of household items made from alloys. Have students observe the pictures/samples and engage them in guided discussions on the advantages of alloys.*

MODULE 3: OUR PLANET (cont'd)

36. Visit websites to watch movies, play games and do quizzes and activities on topics such as metals. For example, <https://www.brainpop.com/science/>. Have students complete a worksheet at the end of the activity.
37. Observe demonstrations in which nails are placed under different conditions. Students should then discuss the conditions causing rusting.

TOPIC 6: HOUSEHOLD CHEMICALS

38. Make poster, brochures or flyers showing safety symbols and explaining the dangers of chemicals used in everyday life. Have students display their work.
39. Conduct a research on the functions of common household chemicals. Have students classify each into acid, bases, or salt.
40. Conduct a research on the use of salts in everyday life, for example, preservatives, controlling pests, medicines. This can be done as a collaborative or individual activity requiring students to creatively represent their findings.
41. Watch videos on concepts such as acid, bases, salts and principle of neutralization:
<https://www.youtube.com/watch?v=Ne8xQ4LGLf0>
<https://www.youtube.com/watch?v=mkAnvRIeATE>
<https://www.youtube.com/watch?v=vt8fB3MFzLk>
<https://www.youtube.com/watch?v=IBjwMcHUyBY>
<https://www.youtube.com/watch?v=RmnT9jwX4qQ>
42. Work in small groups to construct a 2D or 3D model of the states of matter. Have students use the models to explain the properties and changes of states of matter.
43. Watch video clips on plasma to enhance students learning. For example, <https://www.youtube.com/watch?v=94tReSbyPYc&t=3s>. Have students complete a worksheet at the end of the activity.
44. Observe teacher demonstrations showing the simple preparation and use of solutions, suspensions, and colloids to compare the characteristics. Have students record and discuss their observations.
45. Watch videos and/or observe teacher demonstration on concepts such as separation techniques to enhance learning. For example:
https://www.youtube.com/watch?v=vi_SJBnXmHo
<https://www.youtube.com/watch?v=eQlnHr9q6lo>
<https://www.youtube.com/watch?v=TdJ57SQ6GAQ>
<https://www.youtube.com/watch?v=mEXzzIDCkoo>
46. Observe teacher demonstrations on the effects of cleaning agents. For example, experiments on stain removal such as applying bicarbonate of soda on fruit stains; borax for fruit, wine and tea stains; and the removal of rust marks on clothing. As well as the use of stain removal pens and teeth whiteners. Have students record and discuss their observations.

MODULE 3: OUR PLANET (cont'd)

47. View websites to watch movies, play games and do quizzes and activities to reinforce learning on topics such as compound and mixture. For example, <https://www.brainpop.com/science/>. Have students complete a worksheet.
48. Engage in group activities to assess the impact of organic compounds used in everyday life on human health, society and the environment, for example, food additives, pharmaceuticals, detergents.

TOPIC 7: POLLUTANTS AND THE ENVIRONMENT

49. Attend guest lectures by resource persons to reinforce concepts. For example, lectures by health professionals to engage students in interactive sessions on topics of interest such as issues related to health and sanitation.
50. Attend field trips to landfills and dumpsites to enhance learning on the importance of community hygiene. Have students document their observations.
51. Examine case studies, for example, on topics that relate to health and hygiene. Students should then be engaged in a guided discussion on their analysis. Have them present their own recommendations to treat with any issues identified.
52. Work in small groups to create posters or flyers depicting the advantages and negative effects of plastics on the environment. Have students display their work.
53. Watch the video "Losing Paradise" <https://www.youtube.com/watch?v=vCanbznET3Y>. Ask students to write and creatively present a report on how all stakeholders can tackle environmental issues [lack of recycling or accumulation of plastic waste].

◆ GUIDELINES FOR SCHOOL-BASED ASSESSMENT

RATIONALE

The School-Based Assessment (SBA) is an integral part of student assessment in the course covered by this syllabus. It is intended to assist students in developing certain knowledge, skills and attributes that are critical to the subject. The activities for the School-Based Assessment are linked to the “Suggested Practical Activities” and should form part of the learning activities to enable the student to achieve the objectives of the syllabus. *Students are encouraged to work in groups of no more than six. However, when practical work is done in groups, the candidates must write up his or her own report.*

During the course of study in the subject, students obtain marks for the competence they develop and demonstrate in undertaking the SBA assignments. These marks contribute to the final marks and grades that are awarded to students for their performance in the examination.

The guidelines provided in this syllabus for selecting appropriate tasks are intended to assist teachers and students in selecting assignments that are valid for the purpose of SBA. These guidelines are also intended to assist teachers in awarding marks according to the degree of achievement in the SBA component of the course. In order to ensure that the scores awarded by teachers are not out of line with **CXC®** standards, the Council undertakes the moderation of a sample of the SBA assignments marked by teachers.

School-Based Assessment provides an opportunity to individualise a part of the curriculum to meet the needs of students. It facilitates feedback to the students at various stages of experience. This helps to build the self-confidence, and critical thinking and problem-solving skills of the students as they proceed with their studies. School-Based Assessment further facilitates the development of essential communicative, investigative, and practical skills that allow students to function more effectively in their chosen vocation. School-Based Assessment, therefore, makes a significant and unique contribution to the development of relevant skills of the students. It also provides an instrument for *assessing* them and rewarding them for their achievements.

The general Aims of this syllabus can only be achieved by using a practical approach. Skills that are not being assessed at a particular time should, therefore, not be neglected. Note also, that not all practical *activities* are used for assessment. Students should be given the opportunity to develop their skills and to feel free to ask for assistance without penalty.

MODERATION OF SCHOOL-BASED ASSESSMENT

The reliability (consistency) of the marks awarded by teachers on the School-Based Assessment is an important characteristic of high-quality assessment. To assist in this process, the Council undertakes on-site moderation of the School-Based Assessment conducted by visiting external Moderators.

*During the Term 2 of the year of the candidate’s first sitting, the Moderator will make a visit to the school/centre. **All School-Based Assessment Record of marks must be submitted online using the SBA data capture module of the Online Registration System (ORS) before the moderator visits.** Teachers must make available to the Moderator **ALL** SBAs (practical activities and investigative project) and support documents (Mark Schemes, Score Sheet from ORS, Sample sheet from ORS and Moderation of SBA forms). **Teachers are NOT required to submit to CXC® samples of candidates’ work, unless specifically requested to do so by the Council.***

The Moderator will remark the skills, and investigation reports for a sample of five candidates identified in the sample sheet from the ORS.

*Teachers' marks may be adjusted as a result of the moderation. The Moderator may remark the assignments of additional candidates. Where the total number of candidates is five or fewer, the Moderator will remark **ALL**.*

*The Moderator will submit the Assessment Sheets, moderation of SBA Sample and the moderation reports to the Local Registrar by April 30 of the year of the examination. A copy of the Assessment Sheets and candidates' work must be retained by the school for three months after the examination results are published by **CXC**®.*

*School-Based Assessment Record Sheets are available online via the **CXC**®'s website www.cxc.org.*

PROCEDURES FOR CONDUCTING SBA

SBA assessments should be made in the context of normal practical coursework exercises. It is expected that the exercises would be designed to provide authentic learning experiences. Assessments should only be made after candidates have been taught the skills and given enough opportunity to develop them. Although **CXC**® will only moderate *EACH skill TWICE*, teachers are reminded that there is no upper limit to the number of assessments **that should be conducted during the course of normal teaching**. Teachers are encouraged to allow candidates to conduct more labs than required for moderation to help candidates to **develop** their skills and to enhance the teaching and learning process with realistic experiences. Candidates should be encouraged to do corrections so that misconceptions will not persist.

Each skill must be assessed at **least two times** over the two-year period, *with the exception of Drawing which must be assessed once*. When selecting the SBA activities, *no one drawing can be used to record candidates score for Drawing and ORR for the purpose of moderation*. If the candidate was credited for the drawing as a part of the ORR a separate practical activity is needed for drawing.

When assessing certain skills, especially those which require on-the-spot observation or involve looking at several behaviours or criteria, teachers are advised to select not more than two skills in any activity. The practical exercises selected for assessment should make adequate demands on the candidates and the skills assessed should be appropriate for the exercises done. For the assessment of written work, the practical activity selected should be one that can be completed in the time allotted for the class and **it is recommended that the notebooks should be collected at the end of the period**. When the skill being assessed is ORR, candidates are required to complete their own write up even if the teacher provided the method/experimental Procedure.

1. In preparation for an SBA practical, the teacher should:

- (a) select tasks related to a given syllabus objective. These tasks may be chosen from the "Suggested Practical Activities" and should fit in with the normal work being done in that class;
- (b) list the materials including quantities and equipment that will be needed for each student;
- (c) carry out the experiment beforehand, if possible, to ascertain the suitability of materials, and the kind of results (observations, readings) which will be obtained, noting especially any unusual or unexpected results;

- (d) list the steps which will be required by the candidates in performing the experiment. From this it will be clear to the teacher how the candidates should be arranged in the laboratory, whether any sharing of equipment or materials is necessary, the skills which can be assessed from the practical, and the instructions to be given;
- (e) list the skills that may be assessed (for example, observation/recording/reporting, analysis and interpretation);
- (f) *ensure the practical is recorded using the stipulated format (Aim/Purpose, Apparatus and Materials, Method/Experimental Procedure, Results and Observation, Discussion, Conclusion);*
- (g) select the skills to be assessed on this occasion. Skills other than those required for that year should also be included for teaching purposes; and,
- (h) *review* the criteria for assessing each skill.

2. The teacher should carry out the assessment and record the marks.

This is *one of* the most critical *steps* in the assessment process. For a teacher to produce marks that are reliable, the marking must be consistent for all candidates and the marks should reflect the standard of performance at the level. The teacher must be able to justify the marks, and this occurs when there is a fixed set of conditions, factors or criteria for which the teacher looks. Marks should be submitted electronically to **CXC®** on a yearly basis on the SBA form provided. The forms should be dispatched through the Local Registrar to reach **CXC®** by the *given deadline as communicated by the Local Registrar each examination year.*

ASSESSMENT OF PRACTICAL SKILLS

School-Based Assessment will assess skills under the profiles Experimental Skills and Use of Knowledge (Analysis and Interpretation only).

The assessment will be conducted during Terms 1-5 of the two-year period following the programme indicated in the Table below.

SBA SKILLS TO BE ASSESSED FOR CXC® MODERATION

PROFILE	SKILLS	FIRST ASSESSMENT		SECOND ASSESSMENT		TOTAL
		NO. OF TIMES SKILLS TO BE ASSESSED	MARKS	NO. OF TIMES SKILLS TO BE ASSESSED	MARKS	
XS	Manipulation/ Measurement	1	10	1	10	20
	Observation Recording and Reporting	1	10	1	10	20
	<i>Drawing</i>	1	10	-	-	10
	Planning and Designing	1	10	1*	10	20
UK	Analysis and Interpretation	1	10	1*	20	30
	TOTAL	5	50	4	50	100

The investigative project is assessed for two skills, Planning and Design and Analysis and Interpretation. Students are encouraged to work in groups **of no more than six**. However, when practical work is done in groups, the candidates must still write up his or her own report.

Teachers are encouraged to provide candidates with multiple opportunities to develop their skills. The score selected for the SBA must be the candidate's **BEST** performance (an average of multiple activities is **NOT** to be used when calculating the final SBA mark).

CRITERIA FOR THE ASSESSMENT OF EACH SBA SKILL

The syllabus is grounded in the philosophy and methodology of all science disciplines. The teaching strategies that are recommended for its delivery are dictated by the scientist's approach to a task. A problem to be identified will be examined in the light of available evidence and suggestions or hypothesis as to its solution formulated. These will then be tested by repeated practical observations, modified or discarded as necessary until a hypothesis that does offer a solution is found.

The history of scientific thought shows that new ideas replace old ones that were previously accepted as factual. Students must be made to realise that no solution is final and infallible since modifications are continually made in light of new knowledge and technology.

Practical Activities

Teachers **must** use the criteria below to mark the SBAs. Teachers are **not** allowed to create their own marking criteria.

OBSERVATION/RECORDING/REPORTING [ORR] (10 marks)

A. REPORTING	4
ORGANISATION	2
Sections Named – Aim/Purpose, Apparatus and Materials, Method/Experimental Procedure, Results and Observation, Discussion, Conclusion.	
<i>All sections present in correct sequence.</i>	1
<i>Relevant information under each heading.</i>	1
(refers to the sections the candidates present even if all are not present)	
METHOD	2
<i>The method is logically sequenced (candidate presents a step-by-step procedure).</i>	1
<i>The method is written in the past tense using passive voice.</i>	1
B. RECORDING	6
<i>The model for recording observations is appropriate.</i>	1
<i>The observations recorded are accurate.</i>	1
<i>Award 4 marks if the table, graph, diagram and/or prose data have no flaws.</i>	4
<i>Award 3 marks if ONE area (a-c/a-d/a-e/a-c) has a flaw.</i>	
<i>Award 2 marks if TWO areas have flaws.</i>	
<i>Award 1 mark if THREE to FIVE areas have flaws.</i>	

Tables (Numerical)

- (a) *Columns headings correctly labelled (Physical quantity in heading, unit is stated in heading and abbreviations/symbols given for each heading).*
- (b) *Title (The title is at the top of the table, appropriate (aligned to content) and is underlined).*
- (c) *The table is enclosed.*

Tables (Non-Numerical)

- (a) *Column headings correctly labelled.*
- (b) *Details of data presented aligned to requirements for the experiment.*
- (c) *Title (The title is at the top of the table, appropriate (aligned to content) and is underlined).*
- (d) *Enclosed table.*

Graphs

- (a) *Axes appropriately drawn and labelled. Both axes (x and y) are correctly labelled with appropriate variables (where possible units must be included).*
- (b) *Appropriate scales used (at least 75% of the graph paper used).*
- (c) *ALL points are accurately plotted.*
- (d) *Smooth curve or smooth line of best fit drawn.*
- (e) *Graph title clearly displayed at the top of the graph sheet.*

Prose/other

- (a) *Attention to kinds of data.*
- (b) *Attention to details of data.*
- (c) *Details of data presented aligned to requirements for the experiment.*

Diagram (shading, three dimensional and free hand drawings are unacceptable).

- (a) *Clarity – clean continuous lines of even thickness.*
- (b) *Accuracy – faithfulness of reproduction.*
- (c) *Labelling/Labelling Lines (neat, drawn with a ruler, straight and do not cross and structure is clearly identified).*
- (d) *Title (The title is below the diagram and is appropriate).*

ANALYSIS AND INTERPRETATION [A/I] (10 marks)

A. DISCUSSION	8
INTERPRETATION/ANALYSIS OF RESULTS	5
Candidate correctly identifies trends/patterns/relationships.	1
There are no inaccuracies in the analysis (to include calculations).	1
Candidate explanation or interpretation is linked to theoretical concepts.	1
Candidate explanation or interpretation is linked to results presented.	1
Candidate explanation or interpretation is linked to the aim/hypothesis stated.	1
LIMITATION/SOURCES OF ERRORS/PRECAUTIONS	3
Candidates correctly explain at least ONE limitation. (anything that may impact the experiment that is outside of the control of the candidate)	1
Candidates correctly explain at least ONE source of error. (anything that was done incorrectly that resulted in inaccuracies)	1
Candidates correctly explain at least ONE precaution. (any measure taken to minimise negative impacts to experimental results)	1
B. REFLECTION	
Candidate presents either the relevance between the experiment and real life (self, society, or environment) AND/OR the Impact of knowledge gained from experiment on self.	1
C. CONCLUSION	
The conclusion is accurate and related to the aim.	1

DRAWING (D) (10 marks)

A. CLARITY	2
Clean continuous lines of even thickness (no shading).	1
Reasonable size (a large drawing is produced).	1
B. ACCURACY	4
Representative	
Award 2 marks if ALL structures/components typical of the specimen/apparatus are included.	2
Award 1 mark if only 75% of the structures/components typical of the specimen/apparatus are included.	
Faithfulness of reproduction of specimen or apparatus.	2
Award 2 marks if structures/components are accurately represented.	
Award 1 mark if more than TWO structures/components are inaccurately represented.	
C. LABELLING	2
Label lines are straight, do not cross and drawn with a ruler.	1
ALL structures/components are correctly named and/or annotated.	1
D. TITLE	2
The title is appropriate (must include the name of the structure or organism and the view from which the drawing was made). <u>The title must be placed below the drawing, written in caps, and underlined.</u>	1
The title includes correct magnification (candidates must show calculation).	1

PLANNING AND DESIGN (P/D) (10 marks)

A. HYPOTHESIS	2
Clearly stated.	1
Testable.	1
B. AIM	1
Related to hypothesis.	1
C. MATERIALS AND APPARATUS	1
Appropriate materials and apparatus suitable to test the hypothesis.	1
D. METHOD	3
The method is logically sequenced (candidate presents a clear and step-by-step procedure suitable to test the hypothesis).	1
The method is written in instructional language (command or imperative verb).	1
At least one manipulated or responding variable stated (must be relevant to hypothesis and can be stated under a heading or seen within the method).	1
E. CONTROLLED VARIABLE	1
Controlled variable stated (must be relevant to hypothesis).	1
F. EXPECTED RESULTS	1
Reasonable and linked with method.	1
G. ASSUMPTIONS/PRECAUTIONS/POSSIBLE SOURCES OF ERRORS	1
Any one stated.	1

INVESTIGATIVE PROJECT

REPORTING FORMAT OF INVESTIGATION

PART A THE PROPOSAL (Planning and Design)

Statement of the Problem – Can be an observation, a problem.

Hypothesis

Aim – Should be related to the hypothesis.

Materials and Apparatus

Method – Should also include variables.

Assumptions/Precautions

Expected Results

PART B THE IMPLEMENTATION (Analysis and Interpretation)

Method – Linked to Part A (change of tense).

Results

Discussion – Explanations/Interpretations/Trends.

Limitations

Reflections

Conclusion

CRITERIA FOR ASSESSING INVESTIGATIVE PROJECT

A. PLANNING AND DESIGN

HYPOTHESIS		2
- Clearly stated.	1	
- Testable.	1	
AIM		1
- Related to hypothesis.	1	
MATERIALS AND APPARATUS		1
- Appropriate materials and apparatus.	1	
METHOD		3
- The method is logically sequenced (candidate presents clear and step-by-step procedure suitable to test the hypothesis).	1	
- The method is written in instructional language (command or imperative verb).	1	
At least one manipulated or responding variable stated (must be relevant to the hypothesis and can be stated under a heading or seen within the method).	1	
CONTROLLED VARIABLE		1
- Controlled variable stated.	1	
EXPECTED RESULTS		1
- Reasonable (linked with method).	1	
ASSUMPTIONS/PRECAUTIONS/POSSIBLE SOURCES OF ERRORS		1
- Any one stated.	1	
TOTAL		(10)

B. ANALYSIS AND INTERPRETATION

METHOD		1
- Linked to Proposal, Change of tense.		
RESULTS		4
-The model for recording data is appropriate.	1	
-The model for recording data (table, graph, diagram and/or prose data) has less than 2 flaws.	1	
- Accuracy of data.	2	
Award 2 marks if fully accurate.		
Award 1 mark if partially accurate.		
DISCUSSION		5
- Explanation.	2	
Award 2 marks if the development of points is thorough.		
Award 1 mark if the development of points is partial.		

- Interpretation	2	
Award 2 marks if the interpretation is fully supported by the candidate's data.		
Award 1 mark if the interpretation is partially supported by candidate's data.		
- Trends stated.	1	
LIMITATIONS		3
- Sources of error identified.	1	
- Precautions stated.	1	
- Limitation stated.	1	
REFLECTIONS		5
- Relevance between the experiment and real life (self, society or environment).	1	
- Impact of knowledge gained from experiment on self.	1	
- Justification for any adjustment made during experiment.	1	
- Communication of information.	2	
Award 2 marks for use of appropriate scientific language, grammar and clarity of expression all of the time.		
Award 1 mark if there are errors.		
CONCLUSION		2
- Stated.	1	
- Related to the aim.	1	
TOTAL		(20)

EXEMPLAR OF INVESTIGATIVE PRACTICAL

EXEMPLAR 1

PART A – THE PROPOSAL

Observation: Some textile materials fade or lose colour after being washed repeatedly. In some cases, these fabrics including cotton, wool and linen, may be reused to make mats, cushions or other household decorations. How can these fabrics be made more brightly and creatively coloured? Which types of fabrics are best for dyeing with a selected natural dye that can be made from readily available plant materials?

Hypothesis: The woollen samples will have a more intense (brighter) colour when treated with the selected natural dye and a mordant, when compared with the linen fabric samples.

Aim: To determine which of the three (3) strategies is best for providing brighter coloured fabrics after dyeing. To determine which type of fabric, cotton, wool or linen, produces a more intense colour when organic dyes are used with sodium chloride as a mordant.

Materials/Apparatus

Fabrics for dyeing: cotton, woollen and linen materials (6 x 6 cm² size samples)

1. Onion skin (6)
2. Beakers (4)
3. Filter paper
4. Filter funnel
5. Tripod and gauze
6. Glass rod
7. Scissors
8. Measuring cylinders
9. Tongs
10. Stopwatch
11. Potassium alumina sulfate
12. Balance

Method

This method may require 5 days.

1. Prepare your fabric samples, two samples per type of material: Cotton, woollen and linen (6 x 6 cm² size samples each).
2. Weigh the fabric samples to be dyed.
3. In tap water, dissolve the alum (10% of mass of fabric to be dyed) in 300 cm³ water.
4. Boil the fabric for about 45-60 minutes then leave overnight to cool.
5. Drain the fabric and leave in a dark area for 3 days.
6. Label three beakers, C for cotton, W for wool and L for linen and place a sample of each fabric type into the corresponding beaker.
7. Prepare a sample of onion skin dye by soaking the onion skins in water overnight, then boiling for about 5 minutes in 500 cm³ water until the water becomes coloured).
8. Using the filter paper and funnel, filter the dye mixture.

9. Pour 50 cm³ of filtrate on to the fabric (cotton samples) in the beaker (C) and leave for 5 minutes of boiling or until the colour appears in the fabric.
10. Carefully remove the sample fabric pieces with the tongs then rinse in the beaker until the water appears clean.
11. Thoroughly rinse using distilled water and allow the sample to air dry for a day.
12. Repeat steps 9-11 above with samples L and W.
13. Compare the colour of each piece of fabric.
14. Repeat this procedure and compare your results. Record your results in the table relative to the colour of the dye solution (For example, Yellow with greater intensity of dye; Yellow with less intensity of dye).

Precautions: Exercise all precautionary procedures regarding heating. Ensure proper filtering techniques are used.

Variables

Controlled: Size of the materials, type of dye used, volume of each solution used, time allotted to selected portions of the procedure.

Manipulated: Type of fabric used.

Responding: The intensity of the colour.

Expected Result

Strategy:

The fabric with the most intense colouration after the procedure will be considered the best fabric to use. If the woollen fabric is the one with the most intense colouration, then the hypothesis will be accepted. If not, then the hypothesis will be rejected.

PART B – THE IMPLEMENTATION

Introduction

Some textile materials lose some of their colour after being washed repeatedly. In some cases, these materials including cotton, wool and linen, may be reused to make mats, cushions or other household decorations. How can these materials be made more brightly and creatively coloured? Which materials are best for dyeing with a selected natural dye that can be made from readily available plant materials?

The success of a dye on a fabric is highly dependent on the nature of the dye as well as on the nature of the fabric. An acidic dye tends to work best on protein-based fabrics such as wool or silk. The fibre reactive dye is generally better at staining non-protein-based fabrics such as cotton and linen.

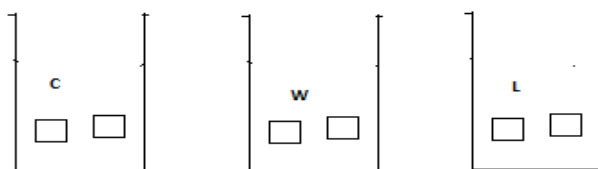
In this experiment the relationship between the type of fabric and organic dye will be explored so as to offer an explanation to the observation made.

Some dyeing processes may require a mordant. The mordant allows for the fixing of the colour from the dye mixture into the fabric. Where a mordant is not essential, for dyeing to occur, the appearance of the outcome colour may be affected.

Method

Two phases are in this process: (a) Mordanting the fabric and (b) Dyeing.

1. Prepare your fabric samples, two samples per type of material: Cotton, woollen and linen (6 x 6 cm² size samples each).
2. Weigh the fabric samples to be dyed.
3. In tap water, dissolve the alum (10% of mass of fabric to be dyed) in 300 cm³ water.
4. Boil the fabric for about 45-60 minutes then leave overnight to cool.
5. Drain the fabric and leave in a dark area for 3 days.
6. Label three beakers, C for cotton, W for wool and L for linen and place a sample of each fabric type into the corresponding beaker.



Beakers with fabric samples

7. Prepare a sample of onion skin dye by soaking the onion skins in water overnight, then boiling for about 5 minutes in 500 cm³ water until the water becomes coloured.
8. Using the filter paper and funnel, filter the dye mixture.
9. Pour 50 cm³ of filtrate on to the fabric (cotton samples) in the beaker (C) and leave for 5 minutes of boiling or until the colour appears in the fabric.
10. Carefully remove the sample fabric pieces with the tongs then rinse in the beaker until the water appears clean.
11. Thoroughly rinse using distilled water and allow the sample to air dry for a day.
12. Repeat steps 9–11 above with samples L and W.
13. Compare the colour of each piece of fabric.
14. Repeat this procedure and compare your results. Record your results in the table relative to the colour of the dye solution (For example, Yellow with greater intensity of dye; Yellow with less intensity of dye).

Precaution: Exercise all precautionary procedures regarding heating of substances in the ball. Ensure proper filtering techniques are used.

Results

The table below shows the colour of dye seen on each fabric type investigated.

Type Of Fabric	Colour Description Due To Dyeing	
	Trial 1	Trial 2
Cotton		
Wool		
Linen		

Discussion

From this experiment it was noted that the cotton fabric had the brightest colour due to dyeing. This goes to show that the organic dye being fibre reactive may be used to stain these types of non-protein types of fabric. This is further assisted by the fact that the linen was also better stained than the wool.

The fact that organic dyes show covalent type bonding within its structure makes them similar in nature to the non-protein fabrics.

Clothing made from such materials will take a longer time to lose their colour as they were better stained in the first place.

Limitation

Every effort was taken to ensure that the experiment was error free. However, there was one limitation. The concentration of the dye could not have been strong enough to properly stain the fabric in order to give a definitive colour distinction.

Reflection

This experiment has taught me that the probability of clothing keeping their brightness is dependent on the type of fabric which makes it up and the nature of the dye used to colour it.

The experiment was carried out as designed.

Conclusion

Durability of colouring on fabric is dependent on the compatibility of the fabric and the dye used to stain the material.

EXEMPLAR 2

PART A: THE PROPOSAL

Observation: Farmers often choose larger seeds for propagating/replanting crops while discarding or rejecting smaller seeds. When asked, the typical farmer would say that bigger seeds give bigger crops. Is this only a myth? Will there be significant growth differences in crops propagated with larger or smaller seeds?

Hypothesis: Crops propagated from larger seeds will grow more than those propagated from smaller seeds.

Aim: To determine whether crops grown from larger seeds will grow more than crops grown from smaller seeds.

Materials/Apparatus

1. 8 peanuts with masses **greater than or equal to 5g**
2. 8 peanuts with masses **less than or equal to 3g**
3. 16 pots or planting bags (same size)
4. Loam soil
5. Hand trowel
6. Measuring cylinder
7. Tap water
8. Labelling tape
9. Permanent marker
10. Wheelbarrow
11. 30 cm ruler
12. Scale (that measures in grams)

Variables

Manipulated Variable: Size of seeds.

Responding Variable: Growth (length/height and number of leaves).

Controlled Variables: Equal amounts of water, soil and sunlight. Same type of soil. Same time/duration.

Method

1. Collect loam soil in the wheelbarrow.
2. Use the hand trowel to mix the soil thoroughly while still in the wheelbarrow.
3. Three-quarters ($\frac{3}{4}$) fill each pot/bag with the thoroughly mixed soil.
4. Examine the 16 shelled peanuts for defects (bites, scratches). Discard and replace samples where necessary.

5. Group peanuts into two groups according to size. The 8 peanuts that are $\geq 5\text{g}$ will form the **L group** and those $\leq 3\text{g}$ will form the **S group**.
6. Place **each** of the 8 large peanuts in separate pots. Each peanut should be covered with approximately 1 cm of soil. Use the tape and marker to label these eight pots '**L-1**' to '**L-8**'.
7. Repeat step 6 but with the smaller peanuts and label these pots '**S-1**' to '**S-8**'.
8. Use the measuring cylinder to gently add 40cm^3 of tap water to each pot. Repeat the watering process so that each pot is watered twice per day at approximately the same time daily.
9. Once the seeds have germinated, place all pots in open sunlight and continue to water twice per day.
10. Allow the experiment to proceed for 6 weeks.

DATA CAPTURE

1. Height

- (a) Record the heights of each crop after each week (every 7 days). Tabulate the results for six weeks as shown below.

TABLE 1: HEIGHTS OF PLANTS

	<i>Heights in cm of Plants Grown from Seeds $\geq 5\text{g}$</i>								<i>Heights in cm of Plants Grown from Seeds $\leq 3\text{g}$</i>							
	L-1	L-2	L-3	L-4	L-5	L-6	L-7	L-8	S-1	S-2	S-3	S-4	S-5	S-6	S-7	S-8
1																
2																
3																
4																
5																
6																

- (b) Calculate the average heights for plants grown from peanuts $\geq 5\text{g}$ for each week. Do the same for plants grown from seeds $\leq 3\text{g}$. Record the results in Table 2. **Round off values to one decimal place.**

TABLE 2: AVERAGE HEIGHTS

WEEK	Average Heights in cm of Plants Grown from Seeds ≥ 5 g (Large)	Average Heights in cm of Plants Grown from Seeds ≤ 3 g (Small)
1		
2		
3		
4		
5		
6		

- (c) Using weeks and average heights from Table 2, draw an appropriate graph to represent the growth of plants grown from larger seeds and those grown from smaller seeds. **Represent this data on ONE graph.**

2. Number of Leaves

- (a) Record the number of **fully opened leaves** of each crop each week (every 7 days) after sowing the seeds. Tabulate the results for 6 weeks as shown below.

TABLE 3: NUMBER OF FULLY OPENED LEAVES

	Number of Leaves on Plants Grown from Seeds ≥ 5 g								Number of Leaves on Plants Grown from Seeds ≤ 3 g							
WEEK	L-1	L-2	L-3	L-4	L-5	L-6	L-7	L-8	S-1	S-2	S-3	S-4	S-5	S-6	S-7	S-8
1																
2																
3																
4																
5																
6																

- (b) Calculate the average number of leaves found on plants grown from peanuts ≥ 5 g for each week. Do the same for plants grown from seeds ≤ 3 g. Tabulate the results in the table below. **Round off averages to the nearest whole number.**

TABLE 4: AVERAGE NUMBER OF LEAVES

WEEK	Average Number of leaves Found on Plants Grown from Seeds ≥ 5 g	Average Number of leaves Found on Plants Grown from Seeds ≤ 3 g
1		
2		
3		
4		
5		
6		

- (c) Using the weeks and average number of leaves from Table 4, draw an appropriate graph to represent the growth of plants grown from larger seeds and those grown from smaller seeds. **Represent this data on ONE graph. The Graph MUST be of a different type from the one drawn for height.**

Precautions: Handle the hand trowel with care. Take extreme care not to break the week-old seedlings when measuring heights.

Assumption: Seeds were free from pest and parasites.

Expected Results

1. The plants grown from the larger seeds (≥ 5 g) should grow taller than those grown from smaller seeds (≤ 3 g) after a six-week period.
2. The plants grown from the larger seeds ≥ 5 g should have more leaves than those grown from smaller seeds ≤ 3 g after a six-week period.

PART B: THE IMPLEMENTATION

Introduction

Growth may be defined as a permanent increase in size. It is often measured using units of length or by counting numbers of structures.

Farmers normally plant larger seeds with the assumption that larger crops will be produced. This experiment investigates whether or not the size of seeds used for propagation makes a difference in the growth of crops. Height differences for plants grown with small seeds will be compared with the heights of those from larger seeds. In addition, the number of leaves present will be counted as a function of growth over a period of time.

Method

1. Loam soil was collected in the wheelbarrow.
2. The hand trowel was used to thoroughly mix the soil while it was still in the wheelbarrow.

- Each pot was filled up to three quarters with the thoroughly mixed soil.
- The 16 peanuts were examined for defects (bites, scratches). Defective samples were discarded and replaced where necessary.
- Peanuts were grouped into two groups according to size. Eight peanuts with masses ≥ 5 g were piled together and considered the **L group** while eight peanuts with masses ≤ 3 g were piled together as the **S group**.
- Each of the eight large peanuts was placed in separate pots. Each peanut was covered with approximately 1 cm of soil. The tape and marker were used to label these eight pots '**L-1**' to '**L-8**'.
- Step 6 was repeated with the smaller peanuts and the pots were labelled '**S-1**' to '**S-8**'.
- The measuring cylinder was used to gently add 40 cm³ of tap water to each pot. The watering process was repeated so that each pot was watered twice per day at approximately the same time daily.
- After the seeds germinated, they were placed in open sunlight and were watered twice per day.
- The experiment continued for six weeks.

Results

1. Height

TABLE 1: HEIGHT OF PLANTS TAKEN OVER A SIX-WEEK PERIOD

WEEK	Heights in cm of Plants Grown from Seeds ≥ 5 g								Heights in cm of Plants Grown from Seeds ≤ 3 g							
	L-1	L-2	L-3	L-4	L-5	L-6	L-7	L-8	S-1	S-2	S-3	S-4	S-5	S-6	S-7	S-8
1	2	1.5	2	3	3	2.5	2	3	1	1	2	1	2	2	1.5	1.5
2	3	2	3	4	4	3	3	3	2	2	2	2	3	2	2.5	2
3	4	3	4	5	5	3.5	3.5	5	3	3	3	4	5	4	4	5
4	6	6	5	6	7	6	5	6	5	5	5	5.5	7	6	5	6
5	8	7	6	8	9	7	7	8	6	6	7	7	8	7	7	7
6	10	9	8	9	11	10	9	11	7	8	7	8	9	8	8	8

TABLE 2: AVERAGE HEIGHT OF PLANTS TAKEN OVER A SIX-WEEK PERIOD

WEEK	Average Heights in cm of Plants Grown from Seeds ≥ 5 g (Large)	Average Heights in cm of Plants Grown from Seeds ≤ 3 g (Small)
1	2.4	1.5
2	3.1	2.2
3	4.1	3.9
4	5.9	5.7
5	7.5	6.9
6	9.6	7.9

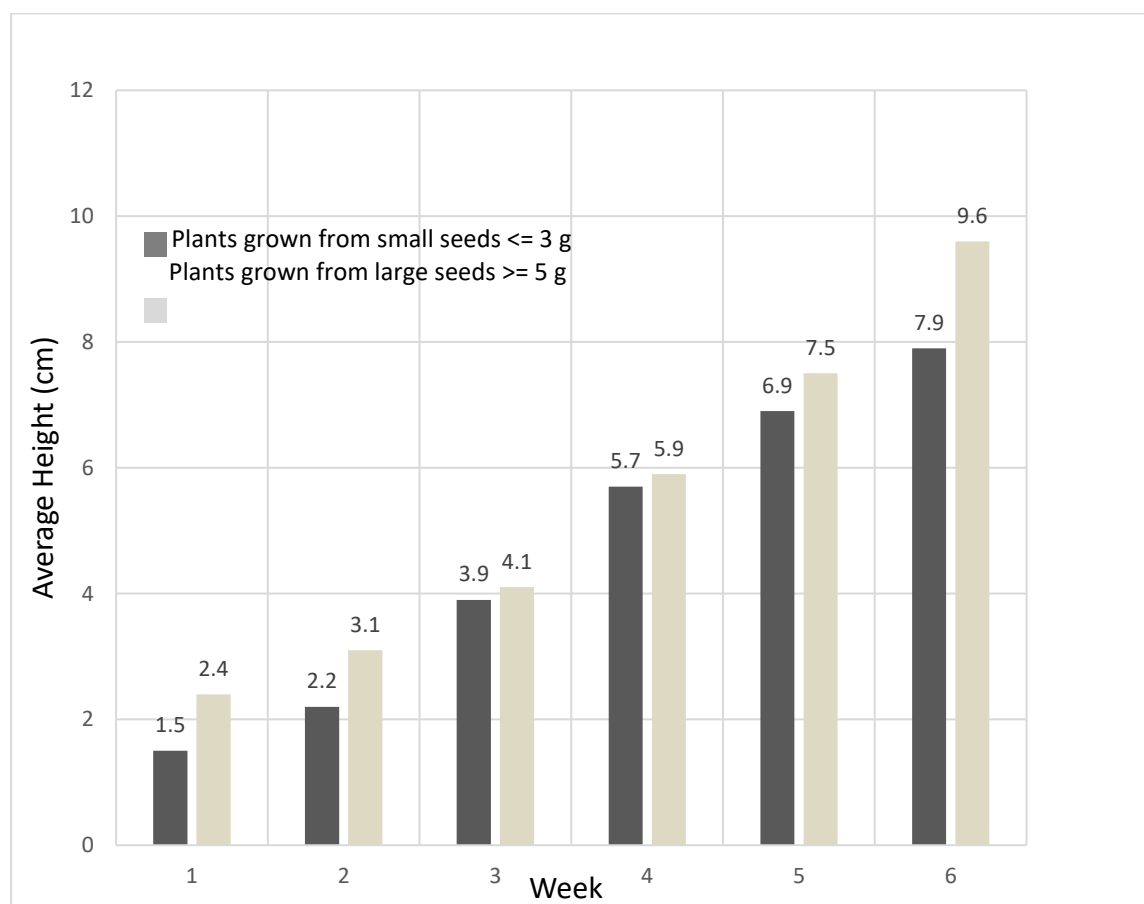


Figure 1: Average Height for Small Seeds and Large Seeds over a Six-week Period

2. Number of Leaves

TABLE 3: NUMBER OF LEAVES ON PLANTS TAKEN OVER A SIX-WEEK PERIOD

	<i>Number of Leaves on Plants Grown from Seeds ≥ 5g</i>								<i>Number of Leaves on Plants Grown from Seeds ≤ 3g</i>							
WEEK	L-1	L-2	L-3	L-4	L-5	L-6	L-7	L-8	S-1	S-2	S-3	S-4	S-5	S-6	S-7	S-8
1	3	3	3	3	2	2	2	3	2	3	2	2	2	3	2	3
2	4	6	6	6	5	6	5	6	4	4	4	3	4	4	5	4
3	8	10	12	10	10	11	10	10	6	6	6	5	6	6	7	6
4	12	14	15	16	15	16	13	13	10	11	10	10	11	9	9	8
5	16	18	18	19	18	21	17	17	12	13	13	14	13	12	13	12
6	20	25	25	27	26	25	23	24	16	17	18	20	19	17	18	16

TABLE 4: AVERAGE NUMBER OF LEAVES ON PLANTS TAKEN OVER A SIX-WEEK PERIOD

WEEK	<i>Average Number of Leaves Found on Plants Grown from Seeds ≥ 5 g</i>	<i>Average Number of Leaves Found on Plants Grown from Seeds ≤ 3 g</i>
1	3	2
2	6	4
3	10	6
4	14	10
5	18	13
6	23	18

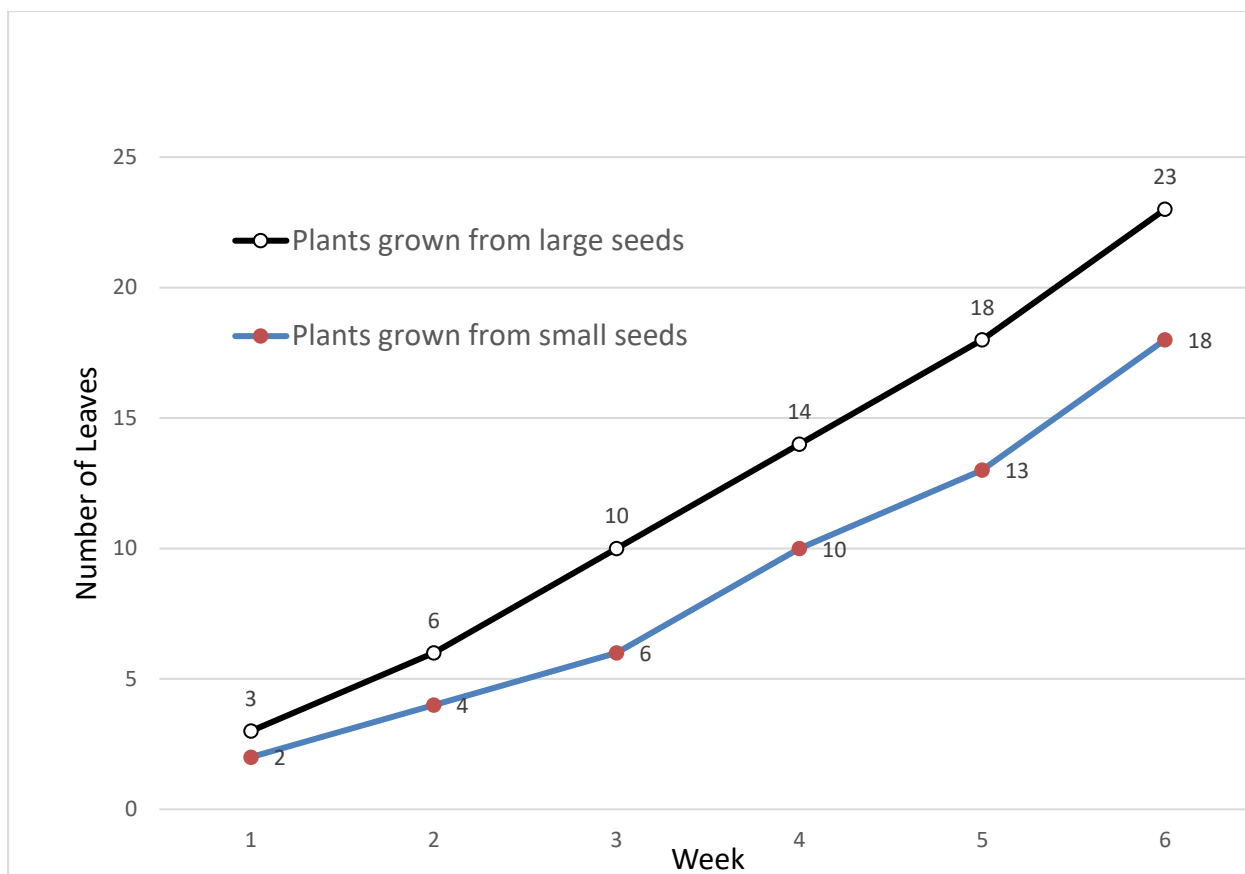


Figure 2: Average Number of Leaves Found on Plants Grown from Small Seeds and Large Seeds over a Six-Week Period

Discussion

Growth is a permanent increase in size. Growth in plants originate from areas called meristems found in shoots or buds. The cells in these areas divide by a process called mitosis where each successive cell is identical to the original cell with respect to the number of chromosomes.

In this experiment, one would readily observe that plants grown from larger seeds tend to grow taller than those grown from smaller seeds (9.6 cm for larger seeds and 7.9 cm for smaller seeds) at the end of week six. The height of a plant is significant as taller plants have the advantage of trapping sunlight more efficiently as they outgrow competing weeds. This translates to an increase in the rate of photosynthesis. This is important to farmers as greater yields are expected when photosynthesis rates are higher.

Also, crops grown from larger seeds on average had more leaves at the end of six weeks (23 leaves) than crops grown from smaller seeds (18 leaves). This represents a difference of 27.8 per cent. (See Tables 3 and 4 and Graph 2).

Having leaves is advantageous to a plant. Leaves are the sites for photosynthesis. With more leaves, the rate of photosynthesis increases and so does the amount of food storage. This will result in more and/or bigger vegetables and fruits.

Greater numbers of leaves also provide better ground cover which reduces evaporation of ground water. Water is essential for photosynthesis. Therefore, more water which is needed for photosynthesis is available to plants with more leaves.

The fact that the growth rate was consistently greater in plants grown from larger seeds may speak to genetic influences and not environmental factors.

Limitations

- Defects to internal structures could not be ascertained.
- Pest and parasites may have unknowingly attacked samples causing inaccurate results.
- No attention was given to the types or varieties of peanuts.
- Some varieties naturally grow larger so results could be skewed if more than one variety was in the sample.

Reflection

This investigation has given me an appreciation for traditional farming in my country. The daunting task of helping relatives sort out larger seeds for planting was always a meaningless exercise to me until I did this investigation. I now understand their strategy. It means that less fertilisers could be used to boost growth because of the fact that larger seeds naturally grow larger. This investigation also revealed to me that plants grown from larger seeds produce more leaves so I now understand how we can produce more food in my country by simply selecting larger seeds for planting.

Conclusion

Peanut plants produce bigger crops when larger seeds are used for propagation as opposed to when smaller seeds are used.

General Notes to the Teacher/Student

1. Figures in this exemplar are not to be taken literally. Figures were deliberately not drawn to scale to discourage plagiarism. Your actual experimental data will differ.
2. This investigation could be modified to determine the relationship between the propagating seed size and fruit production/yields.
3. Other species of seeds may be substituted in this investigation. Depending on the species of plant seeds used, there would not be any significant growth differences or the conclusion stated above may be completely reversed.

Safety

Teachers should observe all the following safety precautions before conducting laboratory work:

1. Investigations involving human blood and other fresh human material (for example, cheek cell, and saliva) should NOT be conducted.
2. Extreme care should be taken when handling live animals. Wild rodents should not be handled since they pass on disease by biting or through their urine. These diseases include leptospirosis.
3. A fire extinguisher or fire blanket must be readily accessible. Both teacher and student should know how to use them. The extinguisher purchased should be appropriate for a biology laboratory.

4. **A first aid kit should be kept in the laboratory and should be checked regularly.**
5. **Corrosive** solutions and inflammable solvents (for example, concentrated acids, alcohols) should be clearly labelled as such and handled with great care and should be locked away when not in use.
6. Candidates should know the correct way to light and use a Bunsen burner. Flints rather than matches are safer to use.
7. Electrical equipment and fittings should be regularly checked and serviced. Electrical outlets should be properly labelled (example 110v and 220v).
8. A laboratory safety manual should be available.
9. All safety precautions should be maintained regarding field trips.

Audio-Visual Aids

The dynamic nature of Integrated Science requires the teacher to make use of a variety of resource materials as teaching aids. Audio-visual aids are particularly useful to reinforce and deepen understanding.

Teachers are encouraged to use the following aids:

1. Film projectors
2. Slide projectors
3. Overhead projectors
4. Videotape machines (VCR)
5. Tape recorders (Cassette)
6. CD-ROM and other interactive media
7. Multi-media projector
8. Camcorders
9. Digital cameras

Cost might prohibit departmental ownership but hardware may be kept in a common pool for use within a school or among a group of schools.

Sources or resource materials include:

1. Overseas information services, for example, USIS, UNESCO, and High Commissions;
2. Government ministries;
3. The media; and,
4. The Internet.

VALIDITY AND RELIABILITY OF TEACHERS MARKS

The reliability of marks awarded is a significant factor in SBA and has far-reaching implications for the candidate's final grade. Teachers are asked to note the following:

1. The criteria for assessing a skill should be clearly identified. A mark scheme must be submitted with the sample of books sent for moderation. Failure to do this could result in the candidates being unavoidably penalised.
2. The relationship between the SBA marks in the practical workbooks and those submitted to **CXC**® on the SBA forms must be clearly shown. It is important that the marks awarded reflect the degree of mastery of the skills assessed.
3. Workbooks should contain all practical work and those exercises used for SBA marks should be clearly identified.
4. The standard of marking must be consistent, hence the need for a mark scheme.
5. Collaboration among teachers especially in the same centre is urged to minimise the discrepancy in the standard of assessment between teachers.

RECORD-KEEPING

Each candidate is required to keep a practical workbook containing all practicals done over the two-year period prior to the examination. Those assessed for **CXC**® will be used to determine the standard of marking by the teacher. A mark scheme must be sent with each set of books. All practicals should be dated and an index made by the candidates of the practicals done. Those assessed for **CXC**® should be clearly indicated along with the marks awarded for each skill.

Candidates' workbooks should be durable and neatness should be encouraged. The pages should be numbered and all exercises should be dated. The workbook should contain a contents page providing the following information concerning the practicals:

1. page number;
2. date;
3. aim of practical;
4. an indication by an asterisk, of which practicals were assessed for **CXC**®; and,
5. the skills assessed.

Teachers

An example of the teacher's records follows:

Recording Marks for SBA

TEACHER'S MARK BOOK

The total figures from the teacher's mark book will be transferred to the **CXC**[®] School-Based Assessment Form and the latter will be submitted to **CXC**[®] by the given deadline as communicated by the Local Registrar each examination year.

Note that no special assessment exercises need to be planned. The teachers will, as is customary, be recording periodic "marks" for all students. The difference is that, since these "marks" will now contribute to an assessment external to the school, they need to be more carefully arranged to clearly stated criteria.

*Teachers are encouraged to provide candidates with multiple opportunities to show their skill and select the BEST performance for the SBA, **not** the average of multiple activities.*

The Record Card

The SBA Record Book will show each candidate's mark for each skill/quality at the end of each year. *Teachers are required to select and record the candidate's BEST performance for the SBA, not the average of multiple activities.* The Record Card should be completed in duplicate. The original of the Card is to be submitted to **CXC**[®] and the copy retained by the school.

SBA Record Card should be dispatched through the Local Registrar to reach **CXC**[®] by the given deadline as communicated by the Local Registrar each examination year.

A sample of the Record Card is included in Appendix 1 to this syllabus. An example of the teacher's completed record card follows:

NAMES	Year 1						Year 2					Grand Total	Remarks
	ORR	DR	MM	AI	PD	Total YR1	ORR	MM	AI	PD	Total YR1		
	10	10	10	10	10	50	10	10	20	10	50	100	
Allen, Veronica	6	2	8	6	5	27	8	10	14	7	39	66	
Williams, Ann	4	7	6	7	8	32	4	9	18	9	40	72	
Cuthbert, Bryan	5	3	9	3	7	27	5	7	16	6	34	61	
Moore, Jason	9	10	0	5	5	29	9	8	14	6	37	66	
Worte, Stewart	3	9	3	4	4	23	6	5	10	4	25	48	

Teachers will also be expected to supply to **CXC**[®] a record of tasks set for School-Based Assessment.

◆ RESOURCES

The following is a list of books, which may be used for **CXC®**'s Integrated Science syllabus. This list is neither exhaustive nor prescriptive but indicates some possible sources which teachers and students may use as appropriate.

Atwaroo-Ali, L.	<i>CXC® Biology</i> , Oxford: Macmillan Publishers Limited, 2003.
Avison, J., Henry, D., and Neeranjan, D.	<i>Physics for CSEC® 2nd Revised Ed.</i> , London: @Oxford University Press, 2014.
Chung-Harris, T.	<i>Integrated Science for CSEC® Examinations 2nd Ed.</i> , Oxford: Macmillan Caribbean, 2010.
Farley, A., and Trotz, C.	<i>CXC® Physics</i> , Oxford: Macmillan Education, 2007.
Gadd, P.	<i>CXC® Human and Social Biology</i> , Oxford: Macmillan Education, 2007.
Kirby, P., Madhosingh, L., and Morrison, K.	<i>Biology for CSEC®</i> , London: Nelson Thornes Limited, 2008.
Koh, A., Hong, C., and Jai, L. M.	<i>CSEC® Integrated Science</i> , Trinidad and Tobago: Caribbean Educational Publishers, 2008.
McMonagie, D., and Anning, P.	<i>Integrated Science for CSEC® 2nd Ed.</i> , Oxford, 2016.
Potter, S. and Oliver, R.	<i>Integrated Science for CSEC®</i> , London: Pearson Longman Publishing, 2003.
Ragoobirsingh, D.	<i>Longman Human and Social Biology for CSEC®</i> , London: Pearson Education Limited, 2007.
Taylor, M., and Chung-Harris, T.	<i>CXC® Chemistry</i> , Oxford: Macmillan Education, 2006.
Tindale, A., and Murray, J.	<i>Chemistry for CSEC® 2nd Ed.</i> , London: Oxford University Press, 2014.

◆ GLOSSARY OF EXAMINATION TERMS

WORD/TERM	DEFINITION/MEANING	NOTES
annotate	Add a brief note to a label.	Simple phrase or a few words only.
apply	Use knowledge/principles to solve problems.	Make inferences/ conclusions.
appraise	To judge the quality or worth of.	
assess	Present reasons for the importance of particular structures relationships or processes.	Compare the advantages and disadvantages or the merits and demerits of a particular relationship or process.
calculate	Arrive at the solution to a numerical problem.	Steps should be shown; units must be included.
classify	Divide into groups according to observable characteristics.	
comment	State opinion or view with supporting reasons.	
compare	State similarities and differences.	An explanation of the significance of each similarity and difference stated may be required for comparisons which are other than structural.
construct	Use a specific format to make and/or draw a graph, histogram, pie chart or other representation using data or material provided or drawn from practical investigations, build (for example, a model), draw scale diagram.	Such representations should normally bear a title, appropriate headings and legend.

WORD/TERM	DEFINITION/MEANING	NOTES
deduce	Make a logical connection between two or more pieces of information; use data to arrive at a conclusion.	
define	State concisely the meaning of a word or term.	This should include the defining equation/formula where relevant.
demonstrate	Show clearly by giving proof or evidence; direct attention to.	
derive	To deduce; determine or extract from data by a set of logical steps some relationship, formula or result.	This relationship may be general or specific.
describe	Provide detailed factual information of the appearance or arrangement of a specific structure or the sequence of a specific process.	Descriptions may be in words, drawings or diagrams or any appropriate combination. Drawings or diagrams should be annotated to show appropriate detail where necessary.
determine	Find the value of a physical quantity.	
design	Plan, and present with appropriate practical detail.	Where hypotheses are stated or when tests are to be conducted, possible outcomes should be clearly stated and/or the way in which data will be analyzed and presented.
develop	Expand or elaborate an idea or argument with supporting reasons.	
differentiate/distinguish (between/among)	State or explain briefly those differences between or among items which can be used to define the items or place them into separate categories.	

WORD/TERM	DEFINITION/MEANING	NOTES
discuss	Present reasoned arguments; consider points both for and against; explain the relative merits of a case.	
draw	Make a line representation from specimens or apparatus that shows an accurate relationship between the parts.	In case of drawings from specimens, the magnification must always be stated. A diagram is a simplified representation showing the relationship between components.
estimate	Make an approximate quantitative judgment.	
evaluate	Weigh evidence and make judgments based on given criteria.	The use of logical supporting reasons for a particular point of view is more important than the view held; usually both sides of an argument should be considered.
explain	Give <i>reasoned</i> account for.	
find	Locate a feature or obtain as from a graph.	
formulate	To express in a formula or in a systematic manner.	
identify	Name or point out specific components or features.	
illustrate	Show clearly by using appropriate examples or diagrams, sketches.	

WORD/TERM	DEFINITION/MEANING	NOTES
justify	To prove a statement or claim true.	
label	Add names to identify structures or parts indicated by pointers.	
list	Itemise without detail.	
measure	Take accurate quantitative readings using appropriate instrument.	
name	Give only the name of.	No additional information is required.
note	Write down observations.	
observe	Pay attention to details which characterise a specimen, reaction or change taking place; to examine and note scientifically.	Observations may involve all the senses and/or extensions of them, but would normally exclude the sense of taste.
<i>outline</i>	<i>Give details/steps.</i>	
plan	Prepare to conduct an exercise.	
predict	Use information provided to arrive at a likely conclusion or suggest a possible outcome.	
record	Write an accurate description of the full range of observations made during a given procedure.	This includes the values for any variable being investigated where appropriate recorded data may be depicted in graphs, histograms or tables.
relate	Show connections between; explain how one set of facts or data depend on others or are determined by them.	
sketch	Make a simple freehand diagram showing relevant proportions and any important details.	

WORD/TERM	DEFINITION/MEANING	NOTES
state	Provide factual information in concise terms, omitting explanation.	
suggest	Offer an explanation deduced from information or previous knowledge.	No correct or incorrect solution is presumed but suggestions must be acceptable within the limits of scientific knowledge.
suggest an hypothesis	Provide a generalisation which offers a likely explanation for a set of data or observations.	
test	To find out by following set procedures.	

◆ NOTE TO TEACHERS

MEASUREMENT

The SI system is used in this syllabus and will be used in all examination papers. Common multiples and sub-multiples of base units (for example, kilometres, centimetres and millimetres) will also be used.

SCHOOL-BASED ASSESSMENT

Preparing the Candidate

During Term 1 of the two-year period, teachers should ensure that the candidates are familiar with the assessment criteria and the mark scheme. Involving the candidates in practice assessments might accomplish this.

The teacher should also ensure during the first term that all candidates use their practical notebooks to record the relevant activities and that such records are made in a systematic way.

Assessing ‘Manipulation/Measurement’ and ‘Observation’

In assessing ‘Manipulation/Measurement’ and ‘Observation’ the teacher should ensure that the candidate has had at least two prior experiences in manipulating/measuring or observing with the apparatus or in making other observations for recording, before the candidate is assessed on these criteria.

Manipulation and Measurement [M/M]

1. *Use of basic laboratory equipment with competence and skill.*
 - (a) *Handle selected measuring devices – balance, thermometer, measuring cylinder, burette, syringe, watch/clock or any timing device, voltmeter, ammeter, reagent bottles, Bunsen burner.*
 - (b) *Makes accurate reading.*
2. *Mastering of laboratory techniques – simple distillation, heating of solids and liquids in test tubes, detection of gases, filtration, constructing simple series and parallel electrical circuits.*
3.
 - (a) *Prepare biological materials for observation or investigation.*
 - (a) *Handle living things with care.*

Sample of Teachers’ Records

The following three pages are samples of the Record Card and Record Book.

CARIBBEAN EXAMINATIONS COUNCIL

SCHOOL-BASED ASSESSMENT IN INTEGRATED SCIENCE

NAME OF SCHOOL: _____

SCHOOL CODE: _____

YEAR OF FINAL EXAMINATION: _____

NAME OF TEACHER: _____

COUNTRY: _____

CANDIDATES NUMBERS	CANDIDATES NAMES	FIRST ASSESSMENT						SECOND ASSESSMENT					GRAND TOTAL 100	REMARKS
		O/R/R	Dr	M/M	A/I	P/D	TOTAL YEAR 1	O/R/R	M/M	A/I	P/D	TOTAL YEAR 2		
		10	10	10	10	10	50	10	10	20	10	50		

TEACHER'S SIGNATURE: _____

PRINCIPAL'S NAME: _____

DATE: _____

PRINCIPAL'S SIGNATURE: _____

◆ SOME GUIDELINES CONCERNING PRACTICAL WORK

It is a syllabus requirement that practical work be done from all *three* sections of the syllabus.

Each *practical* write-up should reflect the candidate's own work and analysis. When practical work is done in groups, the candidates must still write up his or her own report.

KINDS OF PRACTICAL WORK

Practical work usually falls into three broad categories that sometimes overlap. The categories are described below.

Practical Exercises

These are the types that are most often done. They are usually done to help students develop certain practical skills or gain insights into scientific concepts.

Investigations/Information Gathering

In this kind of work, students use their skills to investigate a problem or to find out about a certain phenomenon *or* area of interest. Investigations are best done in areas in which students are interested. There is much scope for planning and designing in this kind of experiment.

Technology

Students may also be interested in using their knowledge of science in making simple devices or in solving simple problems. Emphasis is on using readily available materials (even discards) and appropriate techniques of a very simple nature. Devices constructed should usually be tested by the student and performance data recorded and evaluated.

In writing up practical exercises, candidates must be encouraged to discuss the relevance of their work and be made aware of the limitations of their methods and conclusions.

◆ SUGGESTED PRACTICAL ACTIVITIES

MODULE 1

TOPIC I: UNITS OF LIFE

1. Conduct simple investigations to demonstrate the processes of diffusion and osmosis.
2. Examine prepared slides of cells under a microscope and make simple drawings.

TOPIC II: REPRODUCTION AND GROWTH IN PLANTS

3. Examine and draw storage organs including corms, bulbs, rhizomes, runners, and cuttings.
4. Draw and label cross section of various types of flowers.
5. Draw and label internal and external views of seeds and fruits.
6. Plot graph of plant growth at regular intervals of one week and extrapolate to predict height at future time.
7. Conduct sedimentation tests in order to make inferences about the amount of clay and sand in the soil. Draw a labelled diagram of a soil profile.
8. Conduct soil tests to determine the percentage of air, pH of soils, drainage, and water retention.

TOPIC IV: TRANSPORT SYSTEMS

9. Conduct experiments to investigate diffusion using agar cubes of different sizes to show how surface area/volume ratio affects total diffusion.
10. Conduct experiments to investigate factors which affect the rate of transpiration.

TOPIC VI: SENSE ORGANS AND COORDINATION

11. Conduct simple investigations using convex and concave lenses.
12. Conduct simple investigations on pitch and loudness.

TOPIC VII: HEALTH

13. Investigate the effect of exercise on the pulse rate.
14. Investigate growth of mould on bread under different conditions.
15. Investigate one of the methods for preserving food.

MODULE 2**TOPIC 1: CONSERVATION OF ENERGY**

1. *Conduct experiments to establish conditions for photosynthesis (light and chlorophyll), tests for starch as a product of photosynthesis.*
2. *Observe plants and animals in a nearby area or on the school grounds and classify them as producers, consumers, decomposers, herbivores, carnivores. Construct simple food chains and food webs in terrestrial and aquatic environments.*

TOPIC 2: ENERGY IN LIFE PROCESSES

3. *Conduct food tests – use local foods.*
4. *Conduct experiments to measure energy value of food.*
5. *Conduct experiments to show the effects of temperature and pH on enzymes.*
6. *Investigate the external structure of a tooth and draw a tooth specimen.*
7. *Conduct experiments to show the release of energy and carbon dioxide by organisms.*
8. *Drawings of the structures of stomata from prepared slides.*

TOPIC 4: ELECTRICITY AND LIGHTING

9. *Investigation to determine how well materials conduct electricity.*
10. *Set up circuits to show properties; draw diagrams of series and parallel circuits.*
11. *Use ammeters and voltmeters to show how different resistances affect current.*
12. *Calculate wattage, given voltage and current.*

TOPIC 5: TEMPERATURE CONTROL AND VENTILLATION

13. *Perform simple experiments to show the conduction and convection.*

MODULE 3**TOPIC 3: WATER AND THE AQUATIC ENVIRONMENT**

1. *Use boiling and melting points to compare the physical properties of salt water and fresh water.*
2. *Conduct experiments to determine degree of hardness of water.*

3. *Conduct experiments to soften samples of hard water by:*
 - (a) *boiling;*
 - (b) *adding washing soda; and,*
 - (c) *distillation.*
4. *Investigate the effects of the removal of oxygen from water.*
5. *Conduct experiments to purify water.*
6. *Conduct experiments to compare sinking and floating of similar materials in fresh and sea water. Simple measurements of densities.*

TOPIC 4: FORCES

7. *Demonstrate friction on different surfaces using a ball.*
8. *Use cardboard cutouts of triangles, rectangles, circles and irregular shapes to arrive at the approximate position of the center of gravity of objects of different shapes.*
9. *Investigate using rule suspended by a spring balance and kept horizontal by known suspended weights to show that:*
 - (a) *the sum of the forces in one direction must equal the sum in the opposite direction; and,*
 - (b) *the sum of the clockwise moments about a pivot must equal the sum of anti-clockwise moments.*

TOPIC 5: METALS AND NON-METALS

10. *Measuring the elasticity of specific substances.*
11. *Observe which metals react with dilute acid and which do not; Write simple word equations to show their reaction.*
12. *Investigate the conditions needed for rusting.*

TOPIC 6: HOUSEHOLD CHEMICALS

13. *Conduct simple investigations to determine the pH values of various household chemicals to determine whether they are acids, alkaline or neutral. Use pH paper.*
14. *Conduct simple experiments on neutralisation using droppers.*
15. *Plan and design experiments on stain removal - turpentine for paint; methylated spirit for glass; acetone for nail polish.*
16. *Conduct a simple experiment to separate salt from a mixture of salt and sand.*

◆ SUGGESTED CHEMICALS/MATERIALS LIST

Acetone	Sodium Sulphate
Agar	Starch
Agar, Nutrient	Steel wool
Aluminium foil	Sucrose
Ammonia solution	Sulphuric Acid
Benedict's solution	Turpentine
Bicarbonate indicator solution	Universal indicator paper
Cobalt Chloride	Universal indicator solution
Calcium Carbonate, precipitated	Zinc (granulated)
Charcoal powder	
Chloroform	
Copper, thick wire/strings/ turnings	
Copper Sulphate	
Crude oil	
Ethanol	
Ethanoic (acetic) acid	
Formaldehyde solution	
Glucose	
Hydrochloric Acid (dilute)	
Hydrogen Peroxide (20 volume)	
Iodine	
Iron filings	
Iron Nails	
Lead foil	
Litmus paper, blue	
Litmus paper, red	
Magnesium ribbon	
Manganese Dioxide	
Methylated spirit	
Nitric Acid	
Phenolphthalein	
Potassium Iodide/Sodium Iodide	
Potassium Nitrate	
Potassium Permanganate	
Pyrogallol 40% w/v	
Silver Chloride/nitrate	
Sodium Carbonate	
Sodium Carbonate hydrated (washing soda)	
Sodium Chloride	
Sodium Hydrogen Carbonate (baking soda)	
Sodium Hydroxide (caustic soda)	

◆ SUGGESTED EQUIPMENT LIST

Abrasives	Mirrors, plane (concave, $f = 15\text{ cm}$; convex, $f = 15\text{ cm}$)
Ammeters	<i>Multimedia projectors</i>
Aquaria	Needles, dissecting
Balances (range 1 kg, sensitivity 0.1 g)	Nets for collecting specimens
Balances, spring (10N, 100N)	*Oscilloscope
Beakers, 250 cm ³ (graduated)	Paper, chromatography
Beakers, 400 cm ³ /500 cm ³ (graduated)	Paper, filter
Bell jars with bungs (solid, one hole, two holes)	Pipettes
Borers, cork	Plugs, 3-pin
Bottles, dropping	Poster board (for displaying charts and articles)
Bottles, reagent, assorted	Potometres
Boxes, ray	Power packs (main or batteries) low voltage d.c
Brass	Press, plant
Bronze	Prism, triangular and rectangular
Buckets, plastic, with covers	Pulleys (single, stepped, block and tackle)
Burners, Bunsen or alcohol	Pumps, filter
Box Camera	Quadrats
Carbon microphone	Racks, test tube
Cardboard (for making charts)	Resistors (assorted 1ohm up to 1000 ohm at 1W rating)
Clock (or stopwatch)	Ripple tanks (with accessories for demonstrating rectilinear propagation plane and curved reflection, refraction, diffraction)
Compasses, magnetic	Rules, metre/half metre
<i>Computer</i>	Scalpels/razor blades/knives/scissors
Coverslips	Shelves, beehive
Crocodile clips	Skeleton, mammalian, complete
Crucibles with lids	Slides, microscope (plain)
Cylinders, measuring, assorted	Slides, prepared
	• Leaf, T.S; • Human Blood smear; and, • Dicot root, T.S.
Desiccators	Sockets, lamp
Dishes, petri, glass	Solar system, model of
Droppers, teat	Solder
<i>E – beam</i>	Sonometer (commercial or improvised – a guitar can work)
Ear, model of	Stands, retort with clamps

SUGGESTED EQUIPMENT LIST (Cont'd)

Eye, model of	Stands, tripod (heights must be suitable for use with Bunsen burners)
<i>Flash Drives</i>	Switches
Flasks, conical 250 ml	Telephone, earpiece and mouthpiece
Forceps	Thermometers -10°C - 110°C
Funnels, filter	Tongs, crucible
Fuses household	Trays, sorting
Heart, model of	Trolleys
Hi-fi equipment data (catalogues of)	Test Tubes (assorted sizes)
Holders, lens (convex)	Tubes, Y-piece connectors
Holders, mirror	Capillary Tubing
Holders, test tube	Glass Tubing (assorted lengths)
Jars, gas with cover plates	Rubber Tubing
Jars, with plastic screw top lids	Vertebrae (different types)
Lamps, low voltage	Voltmeter (d.c. dual range 0 - 5v, 0 - 15v)
Lenses, concave cylindrical	White metal
Lenses, concave spherical	Electrical Wire (flex) colour coded, connecting
Lenses, convex cylindrical	Wire gauzes with insulated centres
Lenses, convex spherical (f=5 cm, f = 30 cm)	Nichrome Wire (assorted) 1056 ohm m-1 – 156 ohm -1
Lenses, hand, large (x 6 or more)	
Lungs, bell jar model of	
Magnets, bar	
Masses, sets of (10, 100, 200, 500, 1000g)	
*Metre, joule	
Microscope, light, Magnification x 300	

Items with an asterisk (*) need not be bought but may be borrowed for the relevant lesson.

Western Zone Office

4 September 2025

CARIBBEAN EXAMINATIONS COUNCIL

**Caribbean Secondary Education Certificate®
CSEC®**



Integrated Science

Specimen Papers and Mark Schemes/Keys

Specimen Papers, Mark Schemes and Keys:

- Paper 01
- Paper 02
- Paper 032



TEST CODE **01230010**

SPECIMEN 2025

CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

INTEGRATED SCIENCE

Paper 01 – General Proficiency

1 hour 15 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

Benzene dissolves stains caused by

- (A) tar
- (B) fruit
- (C) paint
- (D) starch

Sample Answer



The best answer to this item is “tar”, so (A) has been shaded.

5. If you want to change your answer, erase it completely before you fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. Which of the following products of respiration is used during active transport?
- (A) Water
 - (B) Energy
 - (C) Oxygen
 - (D) Carbon dioxide
2. A farmer decides to propagate a crop using cuttings instead of seeds. Which of the following would NOT be an advantage of using cuttings?
- (A) Growth is more rapid.
 - (B) Larger food reserves are available.
 - (C) A larger number of offspring could be produced.
 - (D) The new plants would be better than the original plants.
3. Which of the following is a complete list of conditions necessary for germination?
- (A) Glucose, oxygen and water
 - (B) Temperature, oxygen and water
 - (C) Temperature, glucose and oxygen
 - (D) Temperature, carbon dioxide and water

4. Which of the following BEST describes sandy soil and clay?

	Sandy Soil	Clay
(A)	Large particles, poor drainage, excellent water retention	Small particles, good drainage, poor water retention
(B)	Large particles, good drainage, poor water retention	Small particles, poor drainage, excellent water retention
(C)	Small particles, poor drainage, poor water retention	Large particles, good drainage, excellent water retention
(D)	Small particles, good drainage, excellent water retentions	Large particles, poor drainage, poor water retention

5. Both the atria and the ventricles relax
- (A) before diastole
 - (B) during diastole
 - (C) before atrial systole
 - (D) during atrial systole

GO ON TO THE NEXT PAGE

6. Plants lose water through the stomata of leaves in the process known as
- (A) Osmosis
 - (B) Diffusion
 - (C) Respiration
 - (D) Transpiration
7. The phloem is responsible for the transport of
- I. sucrose
 - II. starch
 - III. water
- (A) I and II
 - (B) I and III
 - (C) II and III
 - (D) I, II and III
8. Which of the following term refers to the process of artificially cleaning the blood?
- (A) Dialysis
 - (B) Reabsorption
 - (C) Ultrafiltration
 - (D) Osmoregulation
9. Which of the following part of the plant stores waste products and is shed when not required?
- (A) Bud
 - (B) Stems
 - (C) Flower
 - (D) Leaves
10. Which of the following is excreted by the skin?
- (A) Water and salt
 - (B) Water vapour and urea
 - (C) Urea and carbon dioxide
 - (D) Carbon dioxide and water vapour
11. In which part of the kidney is urine present?
- (A) Ureter
 - (B) Tubules
 - (C) Arteries
 - (D) Nephrons
12. Which of the following structures controls the amount of light entering the eye?
- (A) Cornea
 - (B) Sclera
 - (C) Lens
 - (D) Iris
13. Which of the following structure of the ear collects sound waves and directs them down the external auditory passage?
- (A) Pinna
 - (B) Cochlea
 - (C) Oval window
 - (D) Round window
14. Which of the following BEST defines the condition where a person can see near objects clearly but is unable to see distant objects clearly.
- (A) Myopia
 - (B) Glaucoma
 - (C) Astigmatism
 - (D) Hypermetropia

GO ON TO THE NEXT PAGE

15. Which of the following statements is NOT true about microbes?

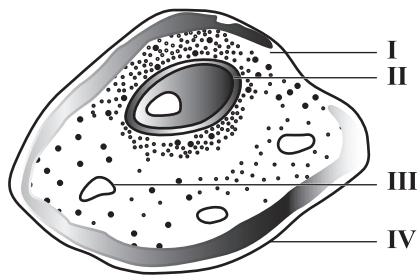
- (A) They aid in the spoilage of food
- (B) They are a source of disease trans-formation
- (C) They act as a barrier on the skin and are a key part of our immune system
- (D) They contribute to soil fertility by the fixation of nitrogen in the nitrogen cycle

18. Which of the following types of bacteria found in the soil are BEST for improving soil fertility?

- I. Nitrifying bacteria
- II. Denitrifying bacteria
- III. Nitrogen-fixing bacteria

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

Item 16 refers to the following diagrams.



16. Which part of the cell is responsible for holding genetic information?

- (A) I
- (B) II
- (C) III
- (D) IV

17. Which of the following sequences represents the correct order of the reproductive processes in sexual reproduction in plants?

- I. Pollination
- II. Germination
- III. Flower production
- IV. Fertilization

- (A) I, II, III, IV
- (B) II, III, I, IV
- (C) III, I, IV, II
- (D) IV, III, II, I

19. Which of the following BEST shows the process of accommodation in the eyes?

- (A) Ciliary muscle bends light to focus on cornea, retina contract/relax to change shape of lens.
- (B) Cornea bends light to focus on cornea, conjunctiva contract/relax to change shape of lens.
- (C) Cornea bends light to focus on retina, ciliary muscles contract/relax to change shape of lens.
- (D) Retina bends light to focus on cornea, ciliary muscles contract/relax to change shape of lens.

20. Which of the following are ways in which a person can acquire the antibodies which gives immunity to a particular disease?

- I. Vaccination
- II. Infection by disease
- III. A baby feeding on breast milk

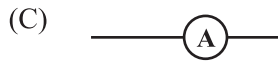
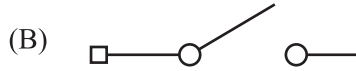
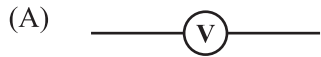
- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

GO ON TO THE NEXT PAGE

21. Energy is BEST defined as the ability to
- (A) do work
 - (B) move an object
 - (C) change liquids to gases
 - (D) convert from potential to kinetic energy
22. Which of the following organisms is typically the MOST numerous in a food chain?
- (A) Producer
 - (B) Herbivores
 - (C) Carnivores
 - (D) Omnivores
23. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$, is the balanced chemical equation for
- (A) metabolism
 - (B) fermentation
 - (C) photosynthesis
 - (D) aerobic respiration
24. The movement of digestive products and water across the villi and into the underlying blood and lacteal vessels is called
- (A) egestion
 - (B) digestion
 - (C) absorption
 - (D) assimilation
25. Loss of muscle mass, stunted height and an enlarged belly are symptoms of
- (A) Obesity
 - (B) Diabetes
 - (C) Marasmus
 - (D) Kwashiorkor
26. The absorption of iron in the small intestine is aided by
- (A) Vitamin A
 - (B) Vitamin B
 - (C) Vitamin C
 - (D) Vitamin D
27. Which of the following is a non-renewable resource?
- (A) Coal
 - (B) Solar
 - (C) Wind
 - (D) Geothermal
28. Which of the following gases is NOT emitted by a car engine?
- (A) Oxygen
 - (B) Methane
 - (C) Carbon dioxide
 - (D) Nitrogen dioxide
29. Geothermal energy is obtained from
- (A) wind
 - (B) waves
 - (C) hot rock
 - (D) running water
30. Which of the following is considered as a good insulator?
- (A) Tin
 - (B) Iron
 - (C) Steel
 - (D) Plastic

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31. Which of the following is the symbol for a switch?



32. Which of the following occurs in the body when the surrounding temperature rises?

- (A) Shivering
- (B) Sweating stops
- (C) Body hair becomes erect.
- (D) The metabolic rate decreases.

33. Which of the following is correct for both alcohol thermometer and mercury thermometer?

	Alcohol Thermometer	Mercury Thermometer
(A)	Alcohol is highly toxic	Mercury is safe
(B)	Suitable for high temperatures	Suitable for low temperatures
(C)	Alcohol must be dyed	Mercury does not have to be dyed
(D)	Only measures low temperatures to -37°C	Can measure temperatures as low as -115°C

34. Which of the following are the three elements of the fire triangle?

- (A) Paper, heat, and air
- (B) Heat, fuel, and oxygen
- (C) Matches, paper, and air
- (D) Spark, petrol, and oxygen

35. Which of the following is NOT usually an outcome of being electrocuted?

- (A) Cardiac arrest
- (B) Cuts on the limbs
- (C) Burns on the body
- (D) Difficulty breathing

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36. The power in a circuit of 20 V and 10 A is

- (A) 0.5 W
- (B) 2 W
- (C) 10 W
- (D) 200 W

37. Which of the following observations is caused by conduction?

- (A) A handheld in front of a fire is warmed.
- (B) A breeze moving from the sea to the land during the day.
- (C) Heat energy moving through a vacuum from the sun to the earth.
- (D) The handle of a spoon gets hot when placed in a pot of hot water.

Item 38 refers to the following table which shows a list of alternative types of energy, their source and method of harnessing.

Type of Energy	Source	Method of harnessing
Wind	Wind	Wind turbines on high ground, near coast or in the sea
Wave	Waves at sea	Floats at sea
Solar	Sun	Photovoltaic solar panels at homes and offices
Hydroelectric	Water dams	Turbines at dams in mountainous areas

38. Which alternative source of energy would be MOST feasible to replace fossil fuel as a source of energy across the Caribbean?

- (A) Solar
- (B) Wind
- (C) Wave
- (D) Hydroelectric

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39. Which of the following types of energy is electrical energy converted to when a television is switched on?
- I. Heat
 - II. Light
 - III. Sound
- (A) I and II only
(B) II and III only
(C) I and III only
(D) I, II and III
40. The energy consumption of an 800-watt refrigerator which was operating for 420 minutes is
- (A) 5.6 kWh
(B) 3 360 kWh
(C) 5 600 kWh
(D) 336 000 kWh
41. Which of the following air masses are found in the Caribbean?
- (A) Maritime polar
(B) Continental polar
(C) Maritime tropical
(D) Continental tropical
42. The number of high tides and low tides around the Earth each day is
- (A) 1
(B) 2
(C) 3
(D) 4
43. Which of the following map symbols represents an occluded front?
- (A) 
- (B) 
- (C) 
- (D) 
44. The movement of the Earth around the sun is called a/an
- (A) eclipse
(B) rotation
(C) revolution
(D) gravitational force
45. Which of the following changes in the weather occur during the passage of a warm front?
- (A) The frontal speed is fast
(B) There are intense showers
(C) The weather becomes cooler
(D) The clouds formed are horizontal
46. Which of the following are causes of water pollution?
- I. Pesticides
 - II. Oil Spills
 - III. Fertilizers
- (A) I, II and III
(B) I, II and IV
(C) II, III and IV
(D) I, III and IV

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47. Which of the following statements about the characteristics of non-metals is TRUE?
- (A) Low melting points
 - (B) High tensile strengths
 - (C) Good conductor of electricity
 - (D) Excellent thermal conductivity
48. Which of the following is the correct order of reactivity for the following materials, from lowest reactivity to highest reactivity?
- (A) Copper, iron, tin, silver
 - (B) Silver, iron, copper, tin
 - (C) Silver, copper, tin, iron
 - (D) Tin, silver, copper, iron
49. Mrs Gray bought some canned fruit which was passed its expiry date. Even though the cans were intact, why shouldn't she use the fruit?
- (A) Oxygen might have gotten into the can
 - (B) Bacteria levels in the can may be high
 - (C) The inside of the can may have rusted
 - (D) The quality of the fruit may have deteriorated
50. Which of the following BEST describes the purpose of disinfectants?
- (A) Purify solutions
 - (B) Mask bad smell
 - (C) Discolour solutions
 - (D) Kill microorganisms
51. Permanent hardness of water is caused by the presence of
- (A) sodium chloride
 - (B) sodium sulphate
 - (C) calcium sulphate
 - (D) calcium carbonate
52. Which of the following is NOT a consequence of acid rain?
- (A) An increase in pH of soil and water
 - (B) The death of aquatic plants and animals
 - (C) A change in the balance of soil chemistry
 - (D) The destruction of statues and corrosion of metals
53. The centre of gravity of an asymmetrical object will be
- (A) exactly at its edge
 - (B) exactly at its centre
 - (C) closer to where most of its mass is
 - (D) further from where most of its mass is
54. The unit of measurement for force is
- (A) Watt
 - (B) Joules
 - (C) Ampere
 - (D) Newton
55. Which of the following methods is BEST suited to capture fish for a small gathering?
- (A) Netting
 - (B) Trawling
 - (C) Rod and line
 - (D) Fish by hand

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56. Darren made himself a cricket bat with the grain of the wood running horizontally and carefully waxed it. It was a rainy day when Darren first used the bat and it cracked in two when the ball hit it. Which of the following reasons BEST explains why?

(A) The wood was too strong
(B) The wood was not elastic enough
(C) The wood absorbed too much moisture.
(D) Too much force was transmitted to the Darren.

57. Tim left a bottle of medicine and after placing it on the table. After an hour he noticed that particles were settled at the bottom. Which of the following type of mixture BEST describes the mixture Tim's observation?

(A) Alloy
(B) Colloid
(C) Solution
(D) Suspension

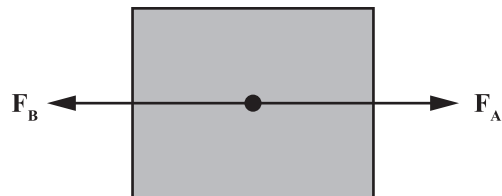
58. Bee stings are acidic. Which of the following household items can be used to neutralize a bee sting?

(A) Water
(B) Vinegar
(C) Lime juice
(D) Baking soda

59. What would be the mechanical advantage of a lever that requires an effort of 40 N to lift a load of 160 N?

(A) 4
(B) 2.5
(C) 0.4
(D) 0.25

Item 60 refers to the following diagram which shows a box being acted on by two equal but opposite forces.



60. The box will eventually
- (A) topple over
(B) remain stationary
(C) move in the direction of B
(D) move in the direction of A

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

Question	Key	Syllabus Objective	Profile	Question	Key	Syllabus Objective	Profile
1	B	1.1.1	KC	31	B	2.4.2	KC
2	D	1.2.1	KC	32	D	2.5.4	KC
3	B	1.2.4	KC	33	C	2.5.3	KC
4	B	1.2.6	KC	34	B	2.4.9	KC
5	B	1.4.2	KC	35	B	2.4.8	KC
6	D	1.4.1	KC	36	D	2.4.2	UK
7	B	1.4.2	KC	37	D	2.5.1	UK
8	A	1.5.2	KC	38	A	2.3.2	UK
9	D	1.5.3	KC	39	D	2.1.2	UK
10	A	1.5.2	KC	40	A	2.1.3	UK
11	A	1.5.2	KC	41	C	3.2.1	KC
12	D	1.6.2	KC	42	B	3.2.3	KC
13	A	1.6.4	KC	43	D	3.2.1	KC
14	A	1.6.3	KC	44	C	3.1.4	KC
15	C	1.7.3	KC	45	D	3.2.1	KC
16	B	1.1.2	UK	46	D	3.5.5	KC
17	C	1.2.3	UK	47	A	3.5.1	KC
18	B	1.2.3	UK	48	C	3.5.2	KC
19	C	1.6.2	UK	49	D	3.5.3	KC
20	B	1.7.3	UK	50	D	3.6.1	KC
21	A	2.1.1	KC	51	A	3.6.1	KC
22	A	2.1.4	KC	52	A	3.7.1	KC
23	D	2.2.4	KC	53	D	3.4.3	KC
24	C	2.2.2	KC	54	C	3.4.1	KC
25	D	2.2.1	KC	55	C	3.3.3	KC
26	C	2.2.1	KC	56	B	3.5.3	UK
27	A	2.3.1	KC	57	D	3.6.4	UK
28	A	2.3.1	KC	58	D	3.6.2	UK
29	C	2.3.1	KC	59	A	3.4.9	UK
30	D	2.4.1	KC	60	B	3.4.4	UK

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

**CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

**TABLE OF SPECIFICATION 2025
INTEGRATED SCIENCE
PAPER 02**

Question	Summary	Marks		
		KC	UK	XS
1	Topic/s: Reproduction and Growth in Plants and Excretion Specific Objectives: 1.2.2, 1.2.3, 1.5.1 and 1.5.2	7	8	5
2	Topic/s: Units of Life, Transport Systems and Health Specific Objectives: 1.1.1, 1.1.2, 1.4.3, 1.7.2 and 1.7.3	8	7	-
3	Topic/s: Energy in Life Processes and Electricity and Lighting Specific Objectives: 2.3.2, 2.4.2, 2.4.4	7	8	5
4	Topic/s: Conservation of Energy and Energy in Life Processes and Electricity and Lighting Specific Objectives: 2.1.9, 2.2.1, 2.4.6	8	7	-
5	Topic/s: The Terrestrial Environment and Water and the Aquatic Environment Specific Objectives: 3.2.2, 3.3.1, 3.3.3, 3.3.5	7	8	5
6	Topic/s: Metals and Non-metals, Household Chemicals and Pollutants and Environment Specific Objectives: 3.5.1, 3.6.4, 3.6.5 and 3.7.3	8	7	-
Total		45	45	15

SPECIMEN 2025



TEST CODE **01230020**

CARIBBEAN EXAMINATIONS COUNCIL

**CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

INTEGRATED SCIENCE

Paper 02 – General Proficiency

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This examination paper consists of THREE sections. Each section consists of TWO questions.
2. Answer ALL questions.
3. Write your answers in the spaces provided in this booklet.
4. Do NOT write in the margins.
5. You may use a silent, non-programmable calculator to answer questions.
6. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
7. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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01230020/CSEC/SPEC 2025

“*”Barcode Area*”
Sequential Bar Code

SECTION A

MODULE 1: ORGANISMS AND LIFE PROCESSES

Answer BOTH questions.

1. (a) Farmer Navid's main cash crops are yams and sweet potatoes which he rotates with tomatoes and then beans. The farmer also cultivates flowers and keeps honeybees.

- (i) Identify the term which BEST defines EACH of the following.

- The process by which pollen grains are transferred from anthers to receptive stigma.

.....
(1 mark)

- The process by which male and female gametes fuse.

.....
(1 mark)

- The process by which seeds begin to grow.

.....
(1 mark)

- (ii) State TWO advantages and ONE disadvantage of growing crops using cuttings as a method of vegetative propagation.

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(3 marks)

- (iii) Bee populations in the world are decreasing. Explain why this would be devastating news for farmer Navid.

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(3 marks)

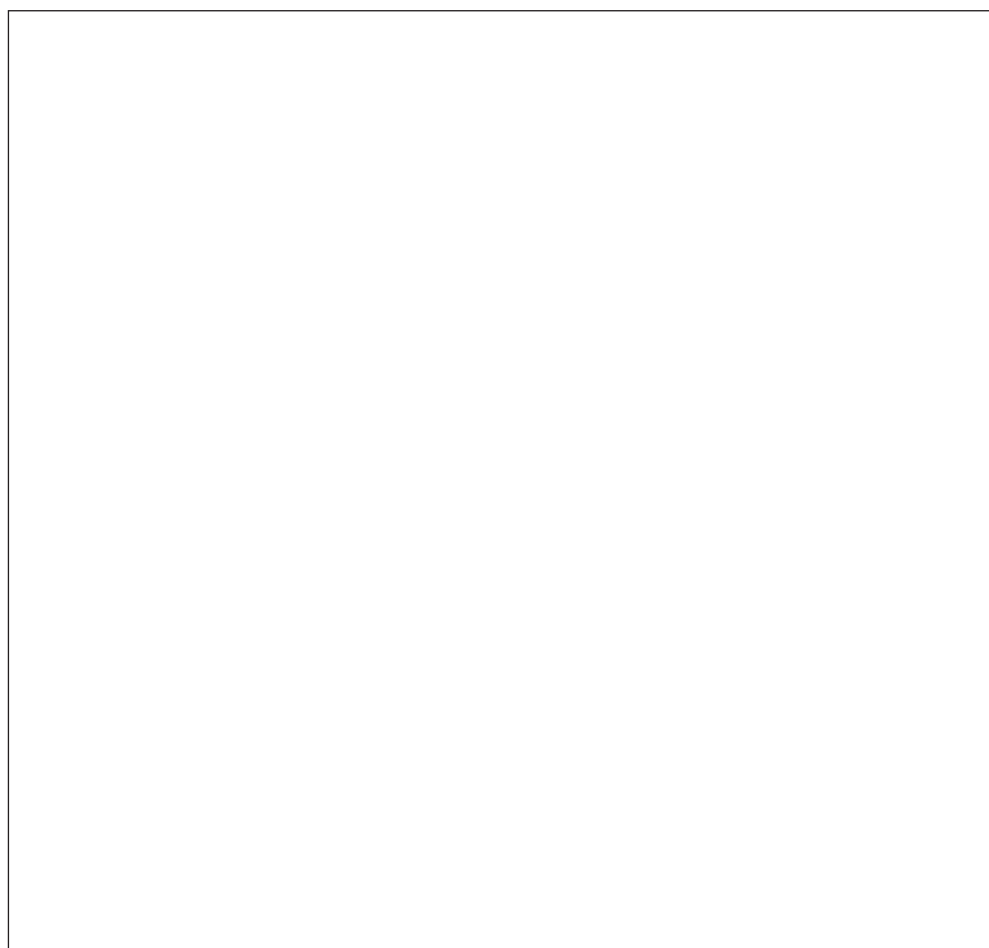
DO NOT WRITE IN THIS AREA

- (iv) Figure 1 is a photograph of a dissected flower. In the space provided, draw a simple diagram of the dissected flower.



Figure 1. Photograph of a dissected flower

Source: Unknown



(5 marks)

- (b) Roshaun drank a large glass of water at 12.30 pm. The salt concentration in his urine was recorded at thirty minutes intervals for one hundred and fifty minutes. The results obtained are shown in Table 1.

TABLE 1: SALT CONCENTRATION OF URINE

Time/minutes	30	60	90	120	150
Volume of urine/cm ³	20	50	70	40	30
Concentration of salt/ug	70	40	20	30	60

- (i) Define the term ‘excretion’.

.....

.....

(1 mark)

- (ii) Outline how the observations for volume of urine would change if Roshaun were to drink twice the amount of water.

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(2 marks)

- (iii) Explain the role of Anti-Diuretic Hormone (ADH) in regulating the water balance in the body if Roshaun were to drink twice the amount of water consumed.

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.....

(3 marks)

Total 20 marks

GO ON TO THE NEXT PAGE

2. (a) On landfills, fires can get started and thick clouds of smoke can be produced. The smoke affects the community which lives nearby.

- (i) Define the term 'diffusion'.

.....
.....

(1 mark)

- (ii) Outline how the process of diffusion would have contributed to the impact on the community.

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.....
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(2 marks)

- (iii) The process of diffusion can also occur in the cells of living organisms. Figure 2 shows the diagram of a cell in a living organism.

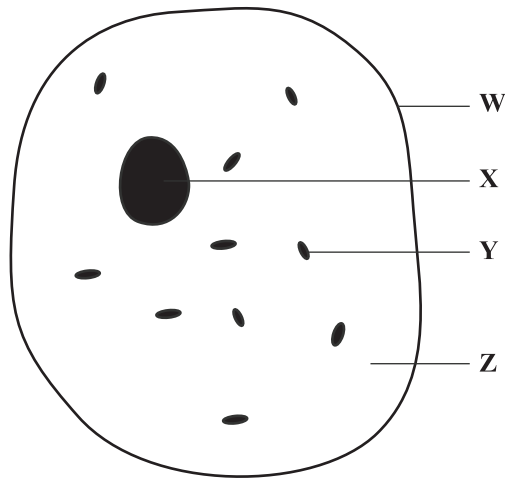


Figure 2. Diagram of a cell in a living organism

Identify the parts of the cell labelled W–Z on Figure 2.

W

X

Y

Z

(4 marks)

- (b) Mary's doctor was informed that her blood type is O-negative and she was concerned when her doctor mentioned negative.

- (i) Outline the meaning of the term 'O-negative' in terms of blood type.

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(2 marks)

- (ii) Mary's first child was born O-positive, explain why the doctors would be concern if her second foetus were O-positive as well.

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(3 marks)

(c) In 2020 there was a worldwide outbreak of the communicable disease, COVID–19.

(i) Define the term ‘communicable disease’.

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.....

(1 mark)

(ii) Outline how the principle of immunization may be used to control the spread of communicable diseases.

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.....

(2 marks)

Total 15 marks

SECTION B

MODULE 2: ENERGY

Answer BOTH questions.

3. (a) In the Caribbean, the major source of energy is provided by fossil fuels. However, there are many adverse effects associated with the use of fossil fuels.

- (i) State TWO advantages and TWO disadvantages of the use of solar energy as a replacement for the use of fossil fuels.

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(4 marks)

- (ii) List THREE alternative sources of energy which can be utilised within the Caribbean other than solar and hydroelectric energy.

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(3 marks)

- (b) Figure 3 shows current flowing through two circuits. Circuit A shows current flowing normally and Circuit B shows current flowing in a short circuit.

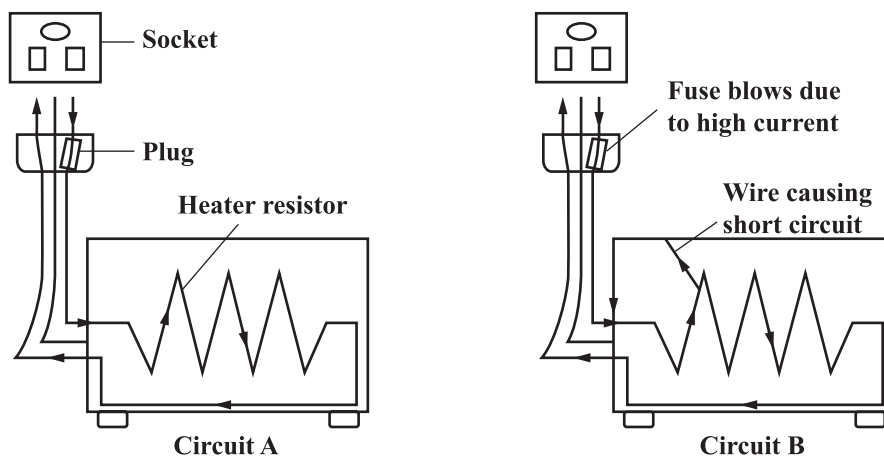


Figure 3. Diagram of Circuit A and Circuit B

- (i) Navin plugged an appliance into a power strip which had 5 other appliances plugged in. As soon as he plugged in the 6th appliance, the power in the house went out. Explain the role of the fuse in ensuring Navin's safety.

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(3 marks)

- (ii) State ONE possible danger to Navin if he uses the faulty appliance.

.....

(1 mark)

- (iii) Using the formula, $P = IV$, calculate the power in a circuit with a current of 1500 A and a voltage of 25 V. You MUST show your working.

(2 marks)

- (iv) Outline why thicker wires are used for overhead lines in power transmission.

.....

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(2 marks)

- (c) A group of students were tasked to carry out an experiment using the following: a cell, a lamp, connecting wires, crocodile clips and different materials such as wood, metal, plastic, salt crystals and pencil lead.

- (i) State a suitable hypothesis for the experiment.

.....
.....

(2 marks)

- (ii) State a suitable aim for the experiment.

.....
.....

(1 mark)

- (iii) State the manipulated variable.

.....

(1 mark)

- (iv) State the responding variable.

.....

(1 mark)

Total 20 marks

4. (a) Kendra went to a beach cookout on a moonlit night. She noticed that the breeze was flowing from the land to the sea rather than what she usually experienced during the day when the breeze came from the sea.

- (i) Distinguish between heat and temperature.

.....

.....

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(2 marks)

- (ii) List THREE types of heat transfer.

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(3 marks)

- (iii) Explain the method of heat transfer Kendra observed while on the beach.

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(3 marks)

- (b) Table 2 shows the energy requirements for some activities.

TABLE 2: THE ENERGY REQUIREMENTS FOR SOME ACTIVITIES

Activity	Energy Requirement (KJ)
Resting	60
Dancing	350
Bicycling	1000
Carpentry	230
Aerobic exercise	700

Source: <https://aapt.scitation.org/doi/10.1119/1.2343405>

- (i) Identify ALL the activities from Table 2 that will require a diet that is high in carbohydrates.

.....

.....

(1 mark)

- (ii) Outline why the data in Table 2 supports the saying that “there is no single diet that will work for everyone”.

.....

.....

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(2 marks)

- (c) Figure 4 shows a bar chart with the kilowatt-hour (kWh) of electricity consumed by of three different artificial sources of light (A, B, and C) over a 1000-hour period.

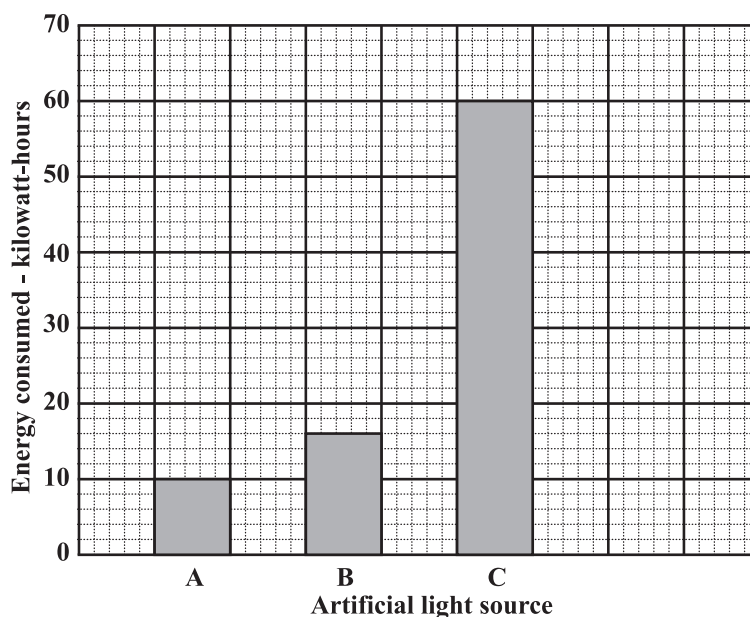


Figure 4. Diagram Circuit A and Circuit B

- (i) Identify which of the artificial light sources in Figure 4 is MOST likely an LED light bulb.

.....
(1 mark)

- (ii) List THREE other artificial light sources other than LED light bulbs.

.....
.....
.....
(3 marks)

Total 15 marks

SECTION C

MODULE 3: OUR PLANET

Answer BOTH questions.

5. (a) Water is essential for life due to its unique properties.

- (i) List THREE physical properties of water that makes water essential to life.

.....
.....
.....

(3 marks)

- (ii) State TWO uses of water.

.....
.....

(2 marks)

- (iii) State TWO sources of water pollution.

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.....

(2 marks)

- (iv) Commercial fishing provides a livelihood for many persons in the Caribbean.

Explain ONE negative impact of pollution of the ocean on commercial fishing.

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(3 marks)

- (b) Climate change is being experienced across the world. Table 3 shows the average, monthly temperature over a nine-month period in Caribbean.

**TABLE 3: TEMPERATURE READINGS IN THE CARIBBEAN
OVER NINE MONTHS**

Month	Temperature (°C)
Jan	23
Feb	25
Mar	27
Apr	32
May	35
Jun	37
Jul	38
Aug	39
Sept	40

- (i) On the grid provided **on page 19**, plot a line graph of month against temperature to represent the data in Table 3. You **MUST** connect each plotted point and label **EACH** axis. **(5 marks)**

A full-page sheet of graph paper featuring a uniform grid of small squares. The grid consists of 20 columns and 20 rows, creating a total of 400 small square units. The lines are thin and black, set against a white background. There are no margins, text, or other markings on the page.

- (ii) State ONE valid conclusion which can be made from the graph.

.....

.....

(1 mark)

- (iii) Discuss ONE possible consequence of the temperature trend shown in Table 3 on the hurricane season in the Caribbean.

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(4 marks)

Total 20 marks

6. (a) Ice-cream is one type of food that contains several food dyes.

(i) Define the term 'colloids'.

.....
.....
(1 mark)

(ii) State ONE difference between a solution and a suspension.

.....
.....
(1 mark)

(iii) State ONE suitable separation technique that Mr Kie could use to identify the different dyes in the ice-cream.

.....
.....
(1 mark)

(iv) Explain how the technique stated in (a) (iii) will separate the different dyes.

.....
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.....
(3 marks)

- (b) Food preparation involves the use of many utensils and items which are made of metal or plastic.

- (i) State TWO advantages and TWO disadvantages of using plastic.

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(4 marks)

- (ii) State TWO properties of metals.

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(2 marks)

- (iii) Explain why stainless steel may be considered superior to aluminium for making cooking pots.

.....

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(3 marks)

Total 15 marks

END OF TEST

“*”Barcode Area”*”
Sequential Bar Code

01230020/CSEC/SPEC 2025

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C A R I B B E A N E X A M I N A T I O N S C O U N C I L

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

INTEGRATED SCIENCE

SPECIMEN PAPER

PAPER 02 — GENERAL PROFICIENCY

KEY AND MARK SCHEME

2025

INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 1		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(a)	(i)	1.2.3	• Pollination • Fertilization • Germination 1 mark each	3		
	(ii)	1.2.2	Advantages - More nutrients are available for the new plant. - They are more likely to survive. - They have the identical characteristics of the parent plant, ensuring quality. Any two 1 mark each Disadvantages - The crop cannot be improved. - Overcrowding may occur. Any one 1 mark each	3		
	(iii)	1.2.3	Bees are necessary for the pollination of flowers (1) pollination is necessary for fertilization and fruit formation (1). Without pollination, farmer Navid would not have any tomatoes or beans (1). 3 marks for a full explanation 2 marks for a partial explanation 1 mark for a limited explanation		3	

INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 1		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
	(iv)	1.2.3	 Clarity (2 marks) - 1 mark for clean continuous lines of even thickness - 1 mark for reasonable size (large drawing is produced) Accuracy (3 marks) - 1 mark for at least 75% structures included - 2 marks for faithfulness of reproduction mark for proportion - o Award 2 marks if all structures are accurately represented - o Award 1 mark if more than TWO structures are <u>inaccurately</u> represented			5
(b)	(i)	1.5.1	Excretion is the process by which metabolic waste is removed from the body.	1		
	(ii)	1.5.2	The volume of urine should double that given in table 1 every 30-minute interval.		2	
	(iii)	1.5.2	ADH release will be suppressed (1) , and the permeability of the duct wall will not increase (1) . Therefore, excess water will be removed through the urine (1) . 3 marks for a full explanation 2 marks for a partial explanation 1 mark for a limited explanation		3	
			Total	7	8	5

INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 2		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(a)	(i)	1.1.1	Diffusion is the movement of particles from an area of low concentration to an area of high concentration.	1		
	(ii)	1.1.1	- Smoke moved from an area of high concentration, where the fire started in the landfill (1) to an area of low concentration in the community. (1) - There is a high concentration of smoke in the landfill and a low concentration in the community (1). Particles move from an area of high concentration to low concentration (1). 1 mark for identification of concentration in the different areas 1 mark for direction of movement		2	
	(iii)	1.1.2	W - Cell membrane X - Nucleus Y - Mitochondrion Z - Cytoplasm 1 mark each	4		
(b)	(i)	1.4.3	O refers to the blood group/ blood group with no antigen A or B on the red blood cell and antibodies A and B in the blood plasma (1). negative refers to the absence of the rhesus factor which is a type of antigen found in the red blood (1). 2 marks for a full outline 1 mark for a partial outline	2		
	(ii)	1.4.3	When she was pregnant with her first child her body may have produced anti-Rh antibodies (1). If she were to get pregnant a second time, with a rhesus positive baby, these antibodies will cross the placenta and attack the baby's red blood cells (1) causing anaemia and other health concerns for the baby (1). 3 marks for a full explanation 2 marks for a partial explanation 1 mark for a limited explanation		3	

INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 2		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(c)	(i)	1.7.2	Communicable disease is one that is spread from one person to another (through a variety of ways that include, contact with blood, body fluids, breathing etc.).	1		
	(ii)	1.7.3	Immunity is achieved by injecting small amounts of dead or weakened antigens in the body. This stimulates the body to produce antibodies which help to fight the infection. 2 marks for a full outline 1 mark for a partial outline		2	
			Total	8	7	

INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 3		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(a)	(i)	2.3.2	Advantages - solar energy is free. - solar energy is a clean form of fuel. - Lower electricity bills Any two, 1 mark each Disadvantages - High set up and storage cost - Weather dependency - Space requirements Any two, 1 mark each	4		
	(ii)	2.3.2	Alternate sources of energy - wind - hydroelectric - geothermal - biomass Any three, 1 mark each	3		
(b)	(i)	2.4.2	As soon as a short circuit occurs, current flows in the live and earth wires instead of in the live and neutral wires (1). This current is higher as most of the resistance is bypassed, causing the fuse to 'blow' and the circuit to break (1) If no earth wire was connected and someone then touches the case, they can receive an electrical shock (1). 3 marks for a full explanation 2 marks for a partial explanation 1 mark for a limited explanation		3	
	(ii)	2.4.2	Electrical Shock		1	
	(iii)	2.4.2	$P = IV$ $= 1500 \times 25$ $= 37,500 \text{ W}$ 1 mark for correct substitution 1 mark for correct answer		2	
	(iv)	2.4.4	Thick wires have less resistance (1) and therefore provide less chance of overheating /destroying insulation/ and causing electrical fires (1). 2 marks for a full outline 1 mark for a partial outline		2	

INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 3		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(c)	(i)	2.4.2	Hypothesis - If the material is a conductor, the bulb will light up. If the material is an insulator the bulb will remain off. - wood, plastic and salt crystals are expected to be insulators while metal and pencil lead are expected to be conductors. 1 mark for clearly stated 1 mark for testable			2
	(ii)	2.4.2	- To investigate how different materials affect the flow of electricity in a circuit. - To investigate which materials are conductors and which are insulators. 1 mark for aim related to the hypothesis			1
	(iii)	2.4.2	The various materials used - wood, metal etc			1
	(iv)	2.4.2	The bulb lighting up or not			1
Total				7	8	5

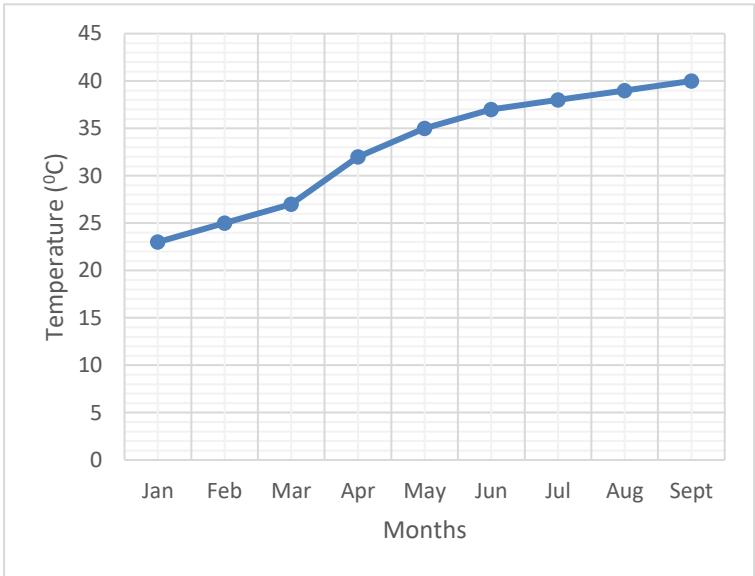
INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 4		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(a)	(i)	2.1.9	Heat is a form of energy (1) whereas temperature measures how hot an object is (1) . 1 mark for each part as stated	2		
	(ii)	2.1.9	- Convection - Conduction - Radiation 1 mark each	3		
	(iii)	2.1.9	With convection warm air rises and cool air descends. With the decrease temperature at night as the earth surfaces cool, the warm air blows towards the sea which is different from during the day when the land is hotter, and the sea breeze blows towards the land. 3 marks for a full explanation 2 marks for a partial explanation 1 mark for a limited explanation		3	
(b)	(i)	2.2.1	A diet high in carbohydrates is required for bicycling and aerobic exercise. (Both activities are required)		1	
	(ii)	2.2.1	Based on the data in the table, each activity has a different energy requirement (1) so the diet would be determined by the activities a person is engaged in (1) . 2 marks for a full outline 1 mark for a partial outline		2	
(c)	(i)	2.4.6	Bar A		1	
	(iii)	2.4.6	- Filament lamp (incandescent bulbs) - Compact fluorescent bulbs (CFL) - Fluorescent tubes 1 mark each	3		
			Total	8	7	

INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 5		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(a)	(i)	3.3.1	Properties of Water - Boils at 100 degrees Celsius, - freezes at 0 degrees Celsius, - density of water at 4 degrees Celsius is less than that at 0 degree Celsius, therefore ice floats, - high specific heat capacity, - excellent capillary action. Any three, 1 mark each	3		
	(ii)	3.3.3	Uses of Water - Drinking - Washing - Watering plants - Recreation - Cooking Any two, 1 mark each	2		
	(iii)	3.3.5	Sources of Pollution - Sewage - Soil - Plastics - Industrial waste - Pesticides - Fertilizers - Oil spills - Any other suitable source Any two, 1 mark each	2		
	(iv)	3.3.5	Pollution introduces harmful substances which are ingested by fishes (1) this can lead to death of fishes which depletes the fish stock (1) as well as contamination of seafood making it unsafe to consume. (1) 3 marks for a full explanation 2 marks for a partial explanation 1 mark for a limited explanation		3	

INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 5		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(b)	(i)	3.2.2	 - 2 marks correct Plotting of points. 2 marks for 7-9 points plotted accurately 1 mark for 4-6 points plotted accurately - 1 mark for <u>x and y</u> axes labelled (with units where possible). - 1 mark for using an appropriate scale (at least 75% of graph paper used). - 1 mark for smooth line connecting all points.			5
	(ii)	3.2.2	As the months go along the temperature increases.		1	
	(iii)	3.2.2	Higher temperatures in the months which coincide with hurricane season (1) results in warmer water in the Atlantic Ocean/Caribbean Sea (1) this leads to an increase in the frequency (1) and intensity of hurricanes/tropical cyclones (1). 4 marks for a full discussion 3 marks for a satisfactory discussion 2 marks for a partial discussion 1 mark for a limited discussion		4	
Total				7	8	5

INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 6		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(a)	(i)	3.6.4	A mixture with small particles that are spread evenly.	1		
	(ii)	3.6.4	- A solution is homogenous. - A suspension is heterogenous. - Particles dissolve in a solution - Particles settle in a suspension Any one, 1 mark each	1		
	(iii)	3.6.5	Paper chromatography		1	
	(iv)	3.6.5	Since ice-cream contains several food dyes which are pigments (1), paper chromatography is suitable as it involves separating a mixture of pigments(solutes) using a suitable solvent (1), in which all the different pigments will move at different rates (1). 3 marks for a full explanation 2 marks for a partial explanation 1 mark for a limited explanation		3	
(b)	(i)	3.7.3	Advantages - Plastic is easy to produce. - Plastic is durable - Plastic is readily accessible, inert/ does not react with other substances like food/drink Any two, 1 mark each Disadvantages - burning releases toxins in the environment, - length of time to degrade. - Harmful to marine life. Any two, 1 mark each	4		
	(ii)	3.5.1	- Metals conduct heat well. - Metals conduct electricity. - Metals are malleable. - Metals are ductile. Any three, 1 mark each	2		

INTEGRATED SCIENCE
PAPER 02
KEY AND MARK SCHEME

Question 6		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
	(iii)	3.5.1	Stainless steels as are less prone to corrosion than aluminium pots (1) which means that they will last for much longer periods (1) making them a cost-effective option (1). 1 mark for comparing to aluminium. 2 marks for developing the response. 1 mark each for relevant supporting points Candidate may also explain any one of the following characteristics. - Stainless steels can withstand high heat. - Stainless steels are highly durable. - they are quick and easy to clean.		3	
			Total	8	7	

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

**CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

**TABLE OF SPECIFICATION 2025
INTEGRATED SCIENCE
PAPER 032**

Question	Summary	Marks		
		KC	UK	XS
1	Topic/s: Household Chemicals and The Terrestrial Environment Specific Objectives: 3.6.2 and 2.2.7	-	5	20
2	Topic/s: Reproduction and Growth in Animals Specific Objectives: 1.3.7	-	5	10
3	Topic/s: Water and the Aquatic Environment Specific Objectives: 3.3.2	-	-	10
Total		-	10	40

SPECIMEN 2025



TEST CODE **01230032**

CARIBBEAN EXAMINATIONS COUNCIL

**CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

INTEGRATED SCIENCE

Paper 032 – Alternative to School-Based Assessment

General Proficiency

2 hours

In addition to the 2 hours, candidates are allowed 10 minutes to read through the entire paper. Writing may begin during the 10-minute period.

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. Answer ALL questions.
2. Use this answer booklet when responding to the questions. For EACH question, write your answers in the spaces provided and return the answer booklet at the end of the examination.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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01230032/CSEC/SPEC 2025

“*”Barcode Area”
Sequential Bar Code

1. You are provided with the following apparatus and material to carry out an investigation to determine the pH of a sample of household items.

It is recommended that you read through the entire procedure before you begin the investigation.

Quantity	Item
3	Hibiscus flower
1	Beaker
1	Glass rod
50 ml	Rubbing alcohol
1	Stopwatch
1	Egg tray
2.5 ml	Samples I–V
1	Medicine dropper
1	pH chart

Procedure

Preparation of pH indicator

- Step 1: Remove the petals from two hibiscus flower, cut them up into small pieces and place them in a beaker.
- Step 2: Add rubbing alcohol until the petals are covered.
- Step 3: Stir the mixture.
- Step 4: Wait 15 minutes for the indicator to be ready. You should complete Step 5 while waiting.

Testing pH of household chemicals

- Step 5: In the space provided at (a) **on page 3**, construct a table suitable for recording the results of the investigation.
- Step 6: Add a sample of each household chemical to the egg tray.
- Step 7: Add 3 drops of universal indicator solution to each sample in the egg tray.
- Step 8: Record your observations in a table.

GO ON TO THE NEXT PAGE

- (a) In the space below, construct a table to record your results. You MUST include a suitable title for your table.

(10 marks)

- (b) State whether EACH sample tested is an acid, an alkali or neutral.

Sample I

Sample II

Sample III

Sample IV

Sample V

(5 marks)

- (c) Figure 1 shows a specimen of the epidermal layer of a leaf showing the stomata.

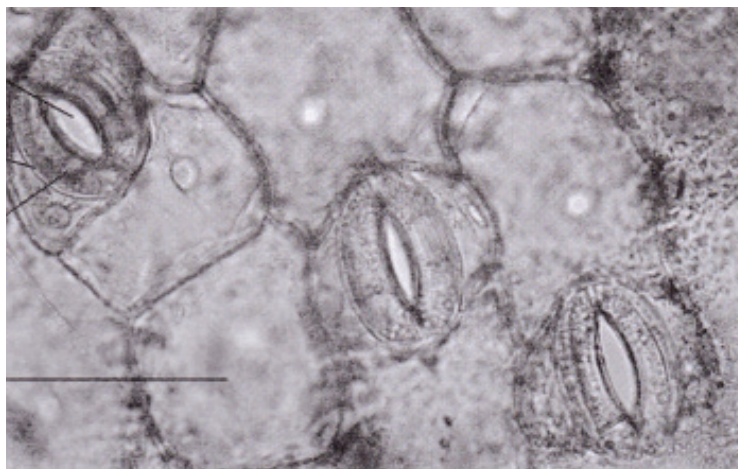
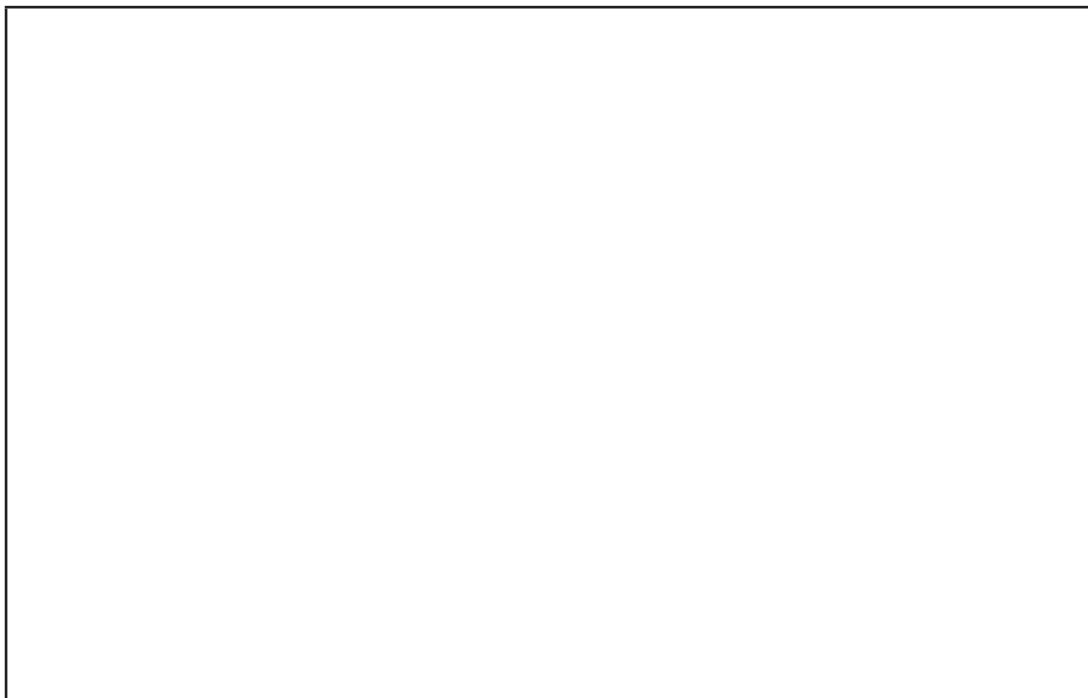


Figure 1. Specimen of the epidermal layer of a leaf

Source: <https://www.cbsetuts.com/ncert-class-10-science-lab-manual-stomata/>

In the space provided below, draw a diagram of the specimen shown in Figure 1. Your diagram **MUST** include a title (with magnification) and at least ONE label to clearly identify ONE stoma.



(10 marks)

Total 25 marks

GO ON TO THE NEXT PAGE

2. A student collected data on the average height of boys and girls aged 12–17 at a school. The result of the investigation is presented in Table 1 below.

TABLE 1: AVERAGE HEIGHT VS AGE OF GIRLS AND BOYS

Age	Average Height of Boys/cm	Average Height of Girls/cm
12	155	156
13	160	158
14	164	161
15	173	163
17	176	166

- (a) On the grid provided **on page 7**, plot a graph to represent the data in Table 1. You must draw a line of best fit through the data points and label the axes. **(10 marks)**

- (b) Using your graph predict the average height of boys and girls at 16. You **MUST** show the point on your graph.

Boys

Girls.....

(2 marks)

- (c) At what age was the greatest increase in height for. You must use the graph plotted and the information predicted in (b).

Boys

Girls.....

(2 marks)

- (d) State what happens to the growth rate of boys between 12 and 14 years of age.

.....

.....

(1 mark)

Total 15 marks

GO ON TO THE NEXT PAGE

[illegible]

01230032/MJ/CSEC 2025

“*”Barcode Area”*”
Sequential Bar Code

3. You are provided with the following apparatus and materials to plan and design an investigation to determine the hardness of water.

Quantity	Item
50 mL	Distilled water
50 mL	River water
50 mL	Salt water
3	Test tube
10 mL	Liquid soap
1	Dropper
6	Labels
3	10 mL measuring cylinder

- (a) State a suitable aim hypothesis for the investigation.

(2 marks)

- (b) Outline a suitable procedure to test the hypothesis stated in (a). Your response MUST be sequentially arranged.

(5 marks)

GO ON TO THE NEXT PAGE

- (c) State ONE variable which must be controlled in the investigation.

.....

.....

(1 mark)

- (d) Outline your expected results for the investigation.

.....

.....

.....

.....

(2 marks)

Total 10 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

01230032/CSEC/SPEC 2025

“*”Barcode Area”*”
Sequential Bar Code

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

INTEGRATED SCIENCE

SPECIMEN PAPER

PAPER 032 – GENERAL PROFICIENCY

KEY AND MARK SCHEME

2025

INTEGRATED SCIENCE

PAPER 032

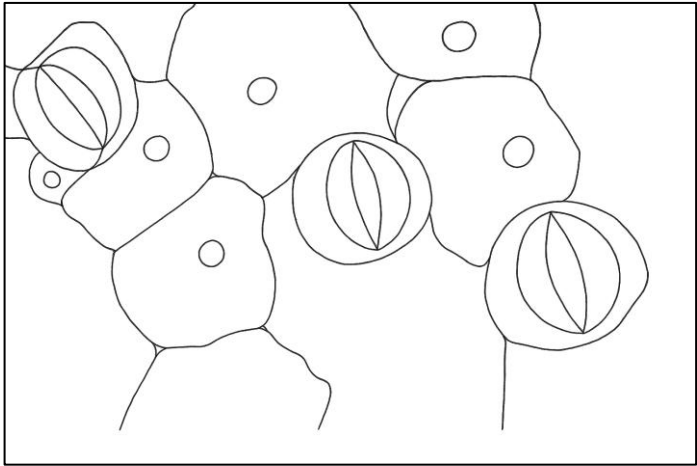
KEY AND MARK SCHEME

Question 1		Syll. Ref.	Possible Response Instructions or Comments	Marks														
				KC	UK	XS												
(a)		C.VI.2	<table border="1"><thead><tr><th>Samples</th><th>Observation</th></tr></thead><tbody><tr><td>Sample I (water)</td><td>Light Pink</td></tr><tr><td>Sample II (vinegar)</td><td>Deep Pink</td></tr><tr><td>Sample III (bleach)</td><td>Yellow</td></tr><tr><td>Sample IV (milk)</td><td>Pink</td></tr><tr><td>Sample V (baking soda)</td><td>Blue</td></tr></tbody></table> 1 mark for appropriate title 1 mark for enclosed table 1 mark for column heading "samples." 1 mark for column heading "observation". 1 mark for sample I-V shown in table. 5 marks for correct observations, 1 mark each	Samples	Observation	Sample I (water)	Light Pink	Sample II (vinegar)	Deep Pink	Sample III (bleach)	Yellow	Sample IV (milk)	Pink	Sample V (baking soda)	Blue			10
Samples	Observation																	
Sample I (water)	Light Pink																	
Sample II (vinegar)	Deep Pink																	
Sample III (bleach)	Yellow																	
Sample IV (milk)	Pink																	
Sample V (baking soda)	Blue																	
(b)		C.VI.2	<table border="1"><thead><tr><th>Samples</th><th>Observation</th></tr></thead><tbody><tr><td>Sample I (water)</td><td>Neutral</td></tr><tr><td>Sample II (vinegar)</td><td>Acid</td></tr><tr><td>Sample III (bleach)</td><td>Alkali</td></tr><tr><td>Sample IV (milk)</td><td>Acid</td></tr><tr><td>Sample V (baking soda)</td><td>Alkali</td></tr></tbody></table> 1 mark for each correct conclusion	Samples	Observation	Sample I (water)	Neutral	Sample II (vinegar)	Acid	Sample III (bleach)	Alkali	Sample IV (milk)	Acid	Sample V (baking soda)	Alkali		5	
Samples	Observation																	
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INTEGRATED SCIENCE

PAPER 032

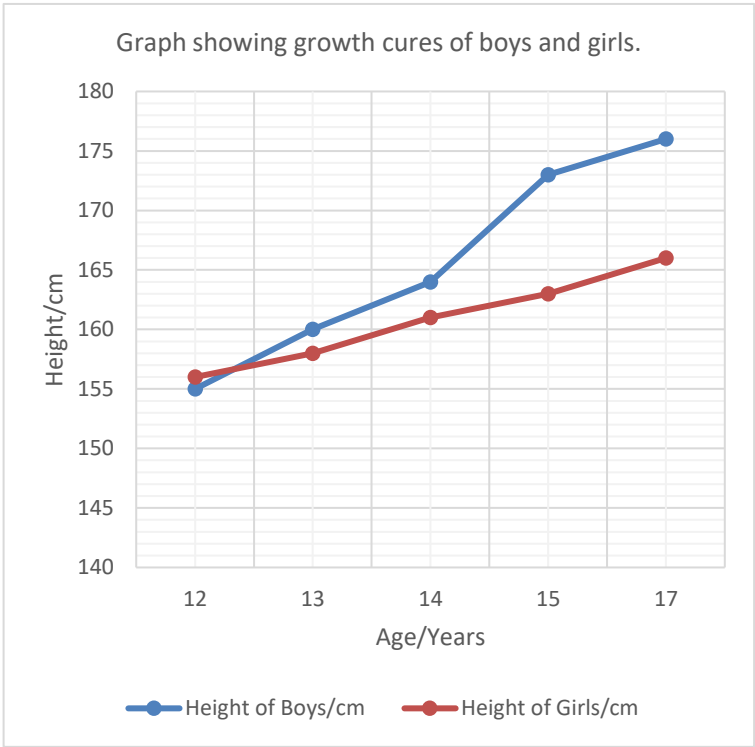
KEY AND MARK SCHEME

Question 1		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(c)		B.II.7	 Clarity (2 marks) 1 mark for clean continuous lines of even thickness (no evident shading) 1 mark for large drawing Accuracy (4 marks) Representative – 2 marks 2 marks if ALL structures are included. 1 mark if only 75% of the structure are included. Faithfulness of representation – 2 marks 2 marks if structures are accurately represented 1 mark if more than TWO structures/components are inaccurately represented. Labelling (2 marks) 1 mark for a straight label line 1 mark for correctly identifying a stoma. Title (2 marks) 1 mark for appropriate title. 1 mark for correct magnification.			10
Total					5	20

INTEGRATED SCIENCE

PAPER 032

KEY AND MARK SCHEME

Question 2		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(a)		A.III.7	Graph showing growth curves of boys and girls.  6 marks for correctly plotting for boys and girls. 3 marks each 3 marks – 5-6 points plotted accurately. 2 marks – 3-4 points plotted accurately. 1 mark – at least 2 points plotted accurately. 1 mark for x axes labelled (with units where possible). 1 mark for y axes labelled (with units where possible). 1 mark for using an appropriate scale (at least 75% of graph paper used) 1 mark for neat line/curve that connects all points.			10

INTEGRATED SCIENCE

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KEY AND MARK SCHEME

Question 2		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(b)		A.III.7	Boys 175 Girls 164 1 mark for correct each value using candidates' graph		2	
(c)		A.III.7	Boys 14-15 years Girls 13-14 years 1 mark each		2	
(d)		A.III.7	• Growth rate rapidly increases. • Boys reach their growth spurt. • The growth rate was more rapid than the girls. • The boys outgrew the girls. • The boys started going through puberty. Any one, 1 mark each		1	
			Total		5	10

INTEGRATED SCIENCE

PAPER 032

KEY AND MARK SCHEME

Question 3		Syll. Ref.	Possible Response Instructions or Comments	Marks		
				KC	UK	XS
(a)		C.III.2	Different sources of water have different levels of hardness. 1 mark for clearly stated. 1 mark for testable.			2
(b)		C.III.2	Procedure 1. Label test-tubes 2. Label measuring cylinders 3. Measure 30ml of each type of water in a different measuring cylinder. 4. Add water to labelled test tubes. 5. Add three drops of liquid detergent to each test-tube. 6. Shake each test-tube for about 1 minute of time. 7. Leave each to stand for the same time, then observe. 2 marks for logical sequence. 2 marks for no flaws with sequencing 1 mark for more than 2 flaws with sequencing 1 mark for linked to the hypothesis. 1 mark for instructional language 1 mark for including ALL apparatus/material listed.			5
(c)		C.III.2	- The amount of liquid soap used, - The volume of water used. Any one, 1 mark each			1
(d)		C.III.2	The sea water would be the hardest, followed by river water. The distilled water will be soft. 2 marks for a complete outline 1 mark for a partial outline			2
			Total			10



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